Machine Learning 2025

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1 What is Machine Learning?

1.1 What is Machine Learning?

Machine learning aims at programming a computer to learn from information (Data) to perform a task (Prediction, etc.) in an optimal way in terms of a performance measure (Loss Function).

1. Data availability: huge growth due to automation, digitalization, web applications.
2. Computing power: significant increase thanks to GPU computing, parallelization, etc.
3. The learning algorithms: theoretical developments of new statistical models and training methods.

1.2 The statistical framework

Types of variables:

- **Quantitative variables** take values in an ordered set, typically the real numbers (R);
- **Qualitative variables** (also called categorical or factor) take values in a finite set $G = G1, ..., Gq$ without ordering, e.g., Yes, No, $blue, red, green, yellow$, etc.

Supervised learning

We observe training data $(x_i, y_i), i = 1, ..., n$ where

- the inputs $xi \in R^p$ are called predictors, covariates or features;
- the outputs y_i are called responses (in this course we assume they are univariate).

The goal is to predict the (unknown) response y_0 for a new (known) predictor x_0.

Regression: the responses $y_i \in R$ are quantitative.

Classification: the responses $y_i \in G$ are qualitative.

Unsupervised learning

We only observe multivariate data $x_i \in R^p, i = 1, ..., n$ but without a particular response.

The goal is to summarize, understand, interpret and visualize the data.

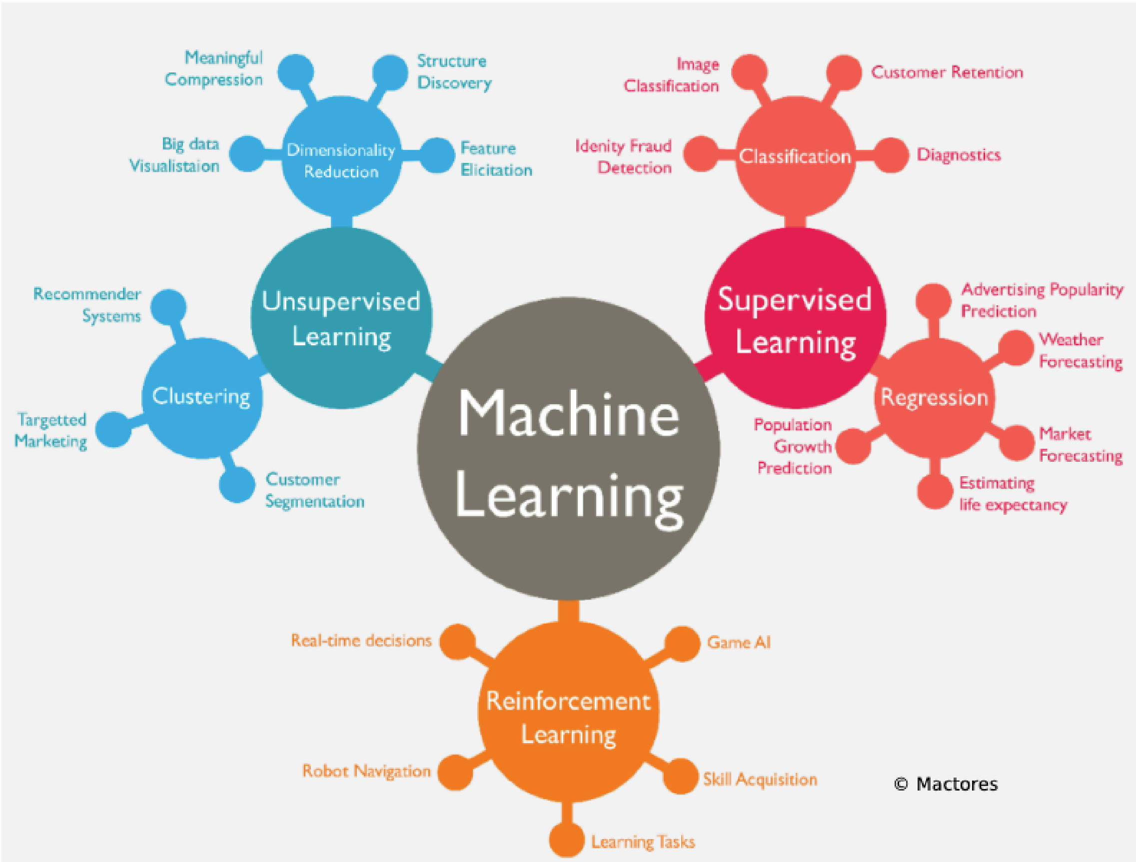


Figure 1: Machine Learning: an overview

1.3 Regression: simple linear model

- **Data**: Observations (x1, y1), . . . , (xn, yn)
- **Goal**: Predict unknown outcome y0 for new predictor x0.

1.4 Linear regression

$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$

- $\hat{y}_0$: This is the predicted value of the response variable y when the input is x_0. The “hat” (^) indicates it’s an estimate, not the true value.

- $\hat{\beta}_0$: This is the estimated intercept.
  It represents the predicted value of y when $x = 0$. It's called an *estimate* because it's computed from data and is an approximation of the true underlying intercept β_0.
- $\hat{\beta}_1$: This is the estimated slope (also an approximation). It tells us how much y is expected to increase (or decrease) when x increases by 1 unit.
- x_0: This is the value of the predictor (input) variable that we're using to make a prediction.

The numbers $\hat{\beta}_0, \hat{\beta}_1 \in \mathbb{R}$ are estimated model parameters.

1.4.1 What does $\hat{\beta}_0, \hat{\beta}_1 \in \mathbb{R}$ mean?

It says both $\hat{\beta}_0$ and $\hat{\beta}_1$ are **real numbers**, i.e., elements of the set of real numbers $\mathbb{R}$. In practical terms: they're just numerical values (like 2.3, -1.5, etc.) estimated from the data using statistical methods (usually least squares).

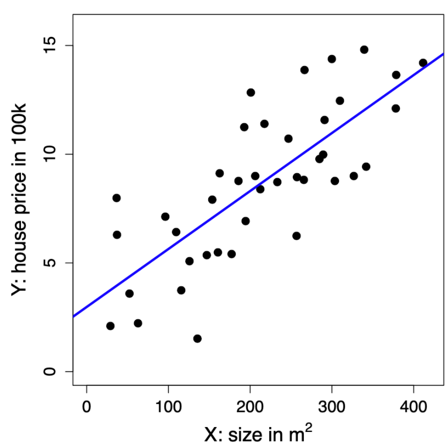


Figure 2: Regression: simple linear model

1.5 Regression: kNN method

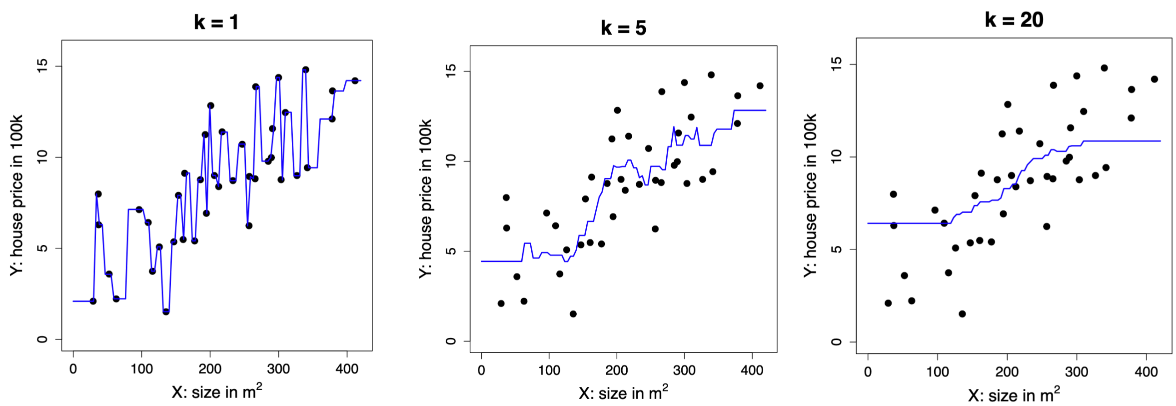
- **Data**: Observations $(x_1, y_1), \ldots, (x_n, y_n)$
- **Goal**: Predict unknown outcome y0 for new predictor x_0.

1.5.1 The k-Nearest Neighbor (kNN) Method

The prediction for a new input x_0 is given by:

$$\hat{y}_0 = \frac{1}{k} \sum_{i: x_i \in N_k(x_0)} y_i = \text{ave}\{y_i : x_i \in N_k(x_0)\}$$

- $N_k(x_0)$ refers to the k closest points x_i to x_0 in the training data.
- The number k is a **tuning parameter** that controls how many neighbours are used to make the prediction.



As we see in the figures above, the greater the number of neighbours, the smoother the curve since we are taking into account more information.

Business application: advertising

Medical application: body fat

1.6 Classification: Simple Linear Regression

- **Data:** Observations $(x_1, y_1), \ldots, (x_n, y_n)$, where $y_i \in G = \{0, 1\}$
- **Goal:** Predict unknown class y_0 for new predictor x_0.

Directly apply linear regression treating the classes as quantitative values 0 and 1.
The prediction is

$$\hat{y}_0 = \begin{cases} 1, & \text{if } \hat{\beta}_0 + \hat{\beta}^T x_0 \geq \frac{1}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

The numbers $\hat{\beta}_0, \hat{\beta}_1 \in \mathbb{R}$ are estimated model parameters.

The previous expression can be explained term-by-term as follows:

- $\hat{y}_0$: The predicted class label for a new input x_0. The hat (^) indicates it's an estimate from the model.
- **The piecewise notation** $\{\ldots$ means the prediction depends on a condition.
- $\hat{\beta}_0$: The estimated intercept of the linear regression model, a scalar.
- $\hat{\beta}^T x_0$: The dot product (transpose of $\hat{\beta}$ times x_0), representing the weighted sum of the input features.
- $\hat{\beta}_0 + \hat{\beta}^T x_0$: The linear combination of the intercept and weighted features, i.e., the model's predicted score.
- $\geq \frac{1}{2}$: The threshold. If the score is greater than or equal to 0.5, predict class 1; otherwise, 0.
- **1 and 0**: The two possible class labels (e.g., positive/negative or class 1/class 0).

In simple terms: You calculate a score from the model for x_0. If it's at least 0.5, you predict class 1; otherwise, class 0.

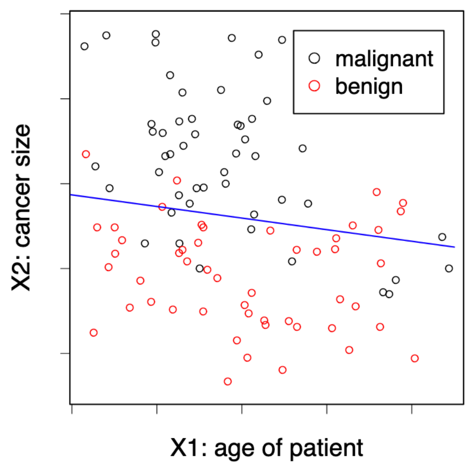


Figure 3: Classification: simple linear model

1.7 Classification: the kNN Method

- **Data:** Observations $(x_1, y_1), \ldots, (x_n, y_n)$, where $y_i \in G = \{0, 1\}$.
- **Goal:** Predict unknown class y_0 for new predictor x_0.

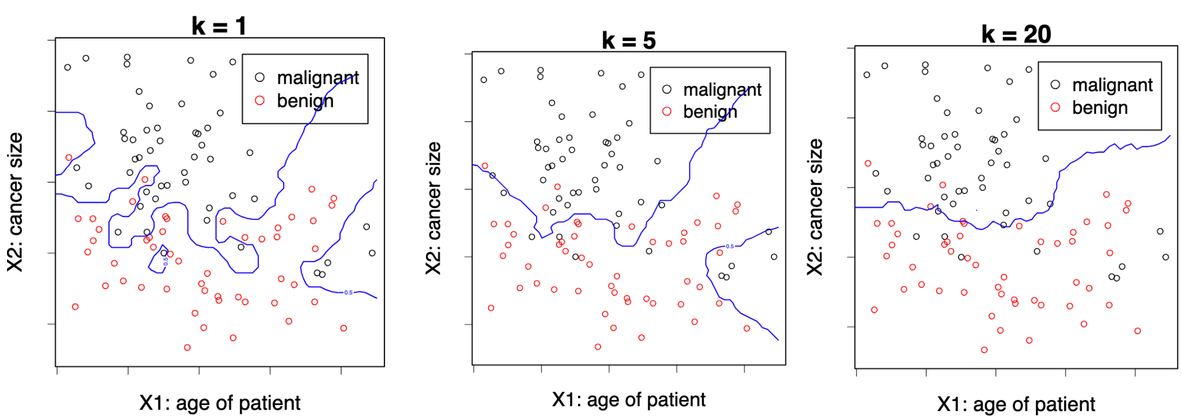
The k-Nearest-Neighbor (kNN) method predicts the majority class in the neighborhood:

$$\hat{y}_0 = \begin{cases} 1, & \text{if } \frac{1}{k}\sum_{i:x_i \in N_k(x_0)} 1\{y_i = 1\} > \frac{1}{2}, \\ 0, & \text{otherwise.} \end{cases}$$

Here, $N_k(x_0)$ are the k closest points x_i to x_0. The number k is a tuning parameter, meaning that it does not come from the data. We can choose which k would be better using Cross Validation, for example.

1.7.1 Explanation of terms:

- $\hat{y}_0$: The predicted class label for the new input x_0.
- $N_k(x_0)$: The set of the k closest neighbors to x_0 in the training data.
- $1\{y_i = 1\}$: An indicator function that equals 1 if the class label y_i is 1, and 0 otherwise.
- $\frac{1}{k}\sum_{i:x_i \in N_k(x_0)} 1\{y_i = 1\}$: The fraction of neighbours whose class is 1; effectively, the proportion of class 1 in the neighbourhood.
- $> \frac{1}{2}$: The decision threshold: if more than half the neighbours are class 1, predict 1; otherwise, predict 0.
- k: The tuning parameter controlling how many neighbours to consider.



Again, with low k, we tend to overfit the model to our training data.

1.7.2 Handwritten characters / digits recognition

Each pixel gives us the amount of darkness in a grey scale (0 to 9). We have 32 predictors (16 times 16, the size of the pictures). Given this predictors the system has to predict

1.8 What is the right model?

All models are wrong, some models are useful.

— George Box

In a sense its right since we would never find the model that has generated our data since Nature is too complex for this. However if they are flexible enough to capture the essence of what is happening in Nature.

Everything should be made as simple as possible. But not simpler.

— Albert Einstein

We want to create a model that represents reality but we don't want to make it too complicated, if we can make it simpler without decreasing the goodness of fit, then we need to go with the simpler. In deep learning this does not apply.

2 Regression Framework

2.1 Regression: Stochastic Framework

2.1.1 The stochastic (population) model

- We have a **quantitative response** Y and p **predictors** $X_1, \dots, X_p$.
- We assume there is a relationship between Y and $X = (X_1, \dots, X_p)$ given by

$$Y = f(X) + \varepsilon$$

  where ε is a random error term with $\mathbb{E}(\varepsilon) = 0$ (on average, errors cancel out and there is no systematic bias in the error) and $\text{Var}(\varepsilon) = \sigma^2$ (the variability of ε is constant, i.e., homoscedastic (it does not depend on X)).
- The function f is **fixed but unknown**[1] and represents the **systematic information** that X provides about Y.

Example Let: $Y =$ *house price* (in 100k) and $X =$ *size* (in m^2)

Then:

$$Y = 2 + 0.03X + \varepsilon$$

which can be interpreted as: house price $= 2 + 0.03 \times$ size $+$ error.

- This true relationship $f(X) = 2 + 0.03X$ is called the **population regression line**.

2.1.2 Random vectors and dimensionality

- (X, Y) is a $(p + 1)$-dimensional random vector. In this simulated example:

$$X \sim \mathcal{N}(0, \sigma_X^2), \quad Y = 2 + 0.03X + \mathcal{N}(0, \sigma_\varepsilon^2)$$

Understanding why it is $(p + 1)$-dimensional

In regression, X is not just one variable, it's a vector of predictors:

$$X = (X_1, X_2, \dots, X_p)$$

[1] It's fixed because the relationship between X and Y in the population exists, it's not random. In our housing example, there really is a true relationship between size and price. And it's unknown, we don't know what f actually is, that's what we're trying to estimate using data. When we run a regression, we build an estimate $\hat{f}(X)$ that approximates the true (but hidden) $f(X)$.

1. **What we mean by** (X, Y)

   - Each X_j represents one feature (e.g. size, number of rooms, location, etc.), and there are p such features in total.
   - The response variable Y is another random variable, i.e., the outcome we want to predict (e.g. house price).
   - So when we consider the pair (X, Y) together, we're grouping all predictors and the response into one joint random vector.

2. **Counting the dimensions**

   - X alone has p dimensions, because it contains p predictor variables.
   - Y adds one more dimension (the outcome variable). Hence, when combined:

   $$(X, Y) = (X_1, X_2, \dots, X_p, Y)$$

   - This vector has $p + 1$ components → therefore, it's a $(p + 1)$-dimensional random vector.

3. **Why it matters**

   - We call it a **random vector** because all its components (the predictors $X_1, \dots, X_p$ and the response Y) are random variables (they represent quantities that vary across individuals in the population, and whose values are not known in advance).
   - Their joint distribution $P(X_1, X_2, \dots, X_p, Y)$ describes how they vary together in the population.
   - The goal of regression is to understand the **conditional relationship** between them, i.e., how Y depends on X:

   $$\mathbb{E}[Y \mid X] = f(X)$$

   - That conditional expectation is exactly the **regression function** we try to estimate.

Example

At the **population level**, (X, Y) represents the theoretical random vector:

$$(X, Y) = (X_1, X_2, \dots, X_p, Y)$$

Each X_j is a random variable corresponding to one predictor, and Y is the random response variable.

At the **observation level**, we work with realised data points, in other words, specific numerical values drawn from this population. For observation i, we write:

$$(x_i, y_i) = (x_{i1}, x_{i2}, \dots, x_{ip}, y_i)$$

For example, if $p = 2$ (size and number of rooms) and Y is house price:

Observation	x_{i1} (size)	x_{i2} (rooms)	y_i (price)
1	120	4	6.8
2	85	3	5.1
3	150	5	8.2

For the first observation:

$$(x_1, y_1) = (x_{11}, x_{12}, y_1) = (120, 4, 6.8)$$

Thus, each (x_i, y_i) is one **realisation** of the $(p+1)$-dimensional random vector (X, Y).

2.2 Regression: The Statistical Framework

2.2.1 The Training Data

- Assume we are given n measurements $(x_1, y_1), \dots, (x_n, y_n)$ of (X, Y), where $x_i = (x_{i1}, \dots, x_{ip})^\top$, and
  - the inputs $x_i \in \mathbb{R}^p$ are called **predictors**
  - the outputs y_i are called **responses**
- We will use this data to **learn** the unknown relationship f between Y and X, i.e., to **train** (or **estimate**) a predictive model $\widehat{f}$.

Example : $[Y =$ house price (100k); $\quad X =$ size (m^2)]; house price $\approx \widehat{f}$(size) $= 2.977 + 0.027 \times$ size

- The training data are realizations of the random vector (X, Y):

Observation	X (Size in m²)	Y (House Price in 100k)
(x_1, y_1)	327	9.0
(x_2, y_2)	154	7.9
(x_3, y_3)	233	8.7

2.2.2 Why do we estimate f?

1. **Prediction** : Machine Learning

- A set of inputs X are readily available, but Y cannot be easily obtained.
- We can predict Y by: $\widehat{Y} = \widehat{f}(X)$ where $\widehat{f}$ is our **predictive model** for f.

- The function $\hat{f}$ is often treated as a **black box**: we are not concerned with the exact form of $\hat{f}$ provided it yields good prediction for Y.

2. **Inference**: Classical Multivariate Analysis

- We are often interested in understanding **how** Y is affected as $X_1, \ldots, X_p$ change.
- Cannot treat $\hat{f}$ as a black box; we need to know the exact form.
- We want to answer questions such as:
  - Which predictor is associated with the response?
  - What is the relationship of the response and each predictor (positive, negative, ...)?
  - Is the relationship linear or more complicated?
  - etc.

2.3 What is a good $\hat{f}$?

2.3.1 Understanding the optimal prediction

- For a fixed value of X, say $X = 3$, we want to know what would be an **optimal prediction** $\hat{f}(X)$ for the response Y.
- Since the relationship between X and Y is stochastic,[2] there might be **many different observed values** of Y corresponding to the same $X = 3$. A **good** $\hat{f}$ is the function that **minimises the expected squared prediction error**: $\hat{f}(X) = \mathbb{E}[Y \mid X]$
- In particular, for $X = 3$: $\hat{f}(3) = \mathbb{E}[Y \mid X = 3]$. That means $\hat{f}(3)$ gives the **expected value (average)** of Y for all observations where $X = 3$.

In other words, among all possible prediction functions, the conditional expectation $\mathbb{E}[Y \mid X]$ is **optimal** in the sense that it minimises the **mean squared error (MSE)**:

$$\mathbb{E}[(Y - f(X))^2]$$

Geometrically, if you look at all points with the same $X = 3$, their Y-values vary due to noise. The best prediction $\hat{f}(3)$ is the **centre** of that distribution, i.e. the average Y value for that X.

[2] Since that relationship it involves random variation, it is not perfectly deterministic .

2.3.2 Visual interpretation

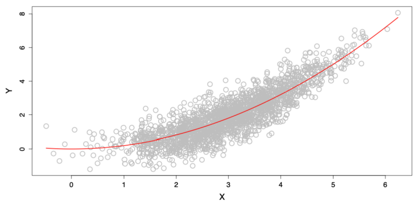


Figure 4: What is a good $\hat{f}$?

As we see in the plot on the left:

- The **red curve** represents $\hat{f}(X)$, the *conditional mean* of Y given X.
- For each value of X, $\hat{f}(X)$ passes through the *average height* of the cloud of observed points.
- This illustrates how the regression function captures the **systematic (non-random)** trend in the data, summarising the central tendency of Y as X changes.

2.4 The Loss Function

2.4.1 Defining what a "good" prediction means

To decide what makes a **good prediction**, we need a way to measure how far a predicted value $\hat{y}$ is from the true value y. This is done through a **loss function** L, which quantifies the **discrepancy** between prediction and reality. In other words, a loss function is a rule that assigns a numerical penalty to each prediction, depending on how wrong it is.

2.4.2 Squared Error Loss (pointwise)

In regression, the most common choice is the **squared error loss**:

$$L(y, \hat{y}) = (y - \hat{y})^2$$

This loss penalises large errors more heavily than small ones, because the difference is squared.

2.4.3 Expected squared prediction error (population risk)

For a predictive model $\hat{f}$, we can define the **expected loss**, that is, the expected squared prediction error across all possible pairs (X, Y):

$$\text{Err}_{\hat{f}} = \mathbb{E}\{[Y - \hat{f}(X)]^2\}$$

Here, the expectation $\mathbb{E}$ is taken over the joint distribution of (X, Y). It tells us the *average squared distance* between our predictions and the true outcomes in the population.

2.4.4 Finding the optimal predictor

Ideally, we would like to find a function f^* that **minimises** this expected error:

$$f^* = \arg\min_f \mathbb{E}\{[Y - f(X)]^2\}$$

This function f^* is called the **regression function**.

2.4.5 The optimal regression function

It can be shown (and we'll later prove) that the function minimising the expected squared error is:

$$f^*(x) = \mathbb{E}[Y \mid X = x]$$

That is, the **conditional expectation** of Y given $X = x$.

This satisfies the population model:

$$Y = f(X) + \varepsilon, \quad \text{where} \quad \mathbb{E}(\varepsilon) = 0.$$

- The loss function $L(y, \hat{y})$ defines how we measure mistakes.
- The expected loss $\mathrm{Err}_{\hat{f}}$ defines how well our model performs on average.
- The best possible function f^* (the **regression function)** predicts the *mean value* of Y given X, because this minimises the expected squared error.

2.5 How do we estimate $\widehat{f}$?

In theory, we know that the optimal prediction function is the **conditional expectation**:$f^*(x) = \mathbb{E}[Y \mid X = x]$. However, in practice, we often have **few or no observations** with exactly the same X value. For instance, there might be few or no data points where $X = 3$. Therefore, we **cannot compute** $\mathbb{E}[Y \mid X = 3]$ directly from the data.

What we can do to estimate $\widehat{f}(x)$ in practice is to approximate the expectation ($\mathbb{E}[Y \mid X = x]$) using nearby observations to approximate. One simple and intuitive method is **k-Nearest Neighbours (k-NN) regression**.

2.5.1 The k-Nearest Neighbours approach

The **k-Nearest Neighbours (k-NN)** method estimates $\hat{f}(x_0)$ by averaging the Y values of the k data points whose X_i values are closest to x_0:

$$\hat{f}(x_0) = \text{Ave}(Y \mid X \in N_k(x_0))$$

where:

- $N_k(x_0)$ is the set of the k observations (x_i, y_i) whose x_i values are closest to x_0.
- The **number** k is a **tuning parameter**, i.e. it controls how "local" or "smooth" the estimate is.
  - If k is **small**, $\hat{f}(x_0)$ is based on very few nearby points → the estimate is **very local** and can capture detail (low bias, high variance).
  - If k is **large**, $\hat{f}(x_0)$ averages over more distant points → the estimate is **smoother** and more stable (high bias, low variance).

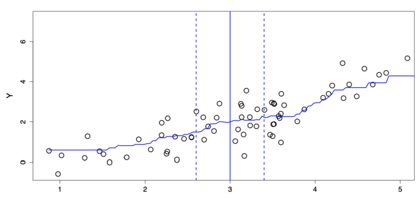


Figure 5: Estimating $\hat{f}$ with kNN

As we see in the plot on the left:

- The **vertical solid line** marks the target point $x_0 = 3$.
- The **dashed vertical lines** indicate the neighbourhood $N_k(x_0)$, which contains the k nearest observations around x_0.
- The **average of the** Y **values** within this neighbourhood provides $\hat{f}(3)$, the *estimated regression value* at $X = 3$.

2.5.2 Limitations of k-Nearest Neighbours

Nearest neighbour averaging works well when the number of predictors (dimension) p is **small**, and the training dataset size n is **large**.

However, when the number of predictors $X_1, \ldots, X_p$ increases, **kNN does not perform well**. This problem is known as the **curse of dimensionality**.

2.5.3 The curse of dimensionality

As the number of predictors p grows an data points become **sparser** in high-dimensional space. Therefore, the notion of "closeness" loses meaning since even the nearest neighbours are far away. As a result, the **averages become unreliable**, and the kNN estimate $\hat{f}(x)$ becomes noisy.

2.5.4 Towards structured methods

We will therefore study **structured methods** that:

- need **less data** to fit,
- may be more **interpretable**,
- but come at the cost of **lower flexibility**.

The key challenge is to find a good **compromise** between:

- **flexibility** (capturing complex relationships), and
- **parsimony** (simplicity and interpretability).

2.6 The empirical loss: model evaluation

2.6.1 Why the true error is unknown

The **expected prediction error** is a *theoretical quantity*:

$$\mathrm{Err}_{\hat{f}} = \mathbb{E}\{[Y - \hat{f}(X)]^2\}$$

It represents the **average squared difference** between the true Y values and the model's predictions $\hat{f}(X)$ across the entire population.

However, since we only have a finite dataset, we do **not know** the true joint distribution $P(X, Y)$. That means we cannot compute this expectation exactly, because we do not know *all possible* pairs (x, y) in the population.

Therefore, to approximate $\mathrm{Err}_{\hat{f}}$, we use a new, independent sample, called the **test dataset**:

$$\mathcal{T}_m = \{(\tilde{x}_1, \tilde{y}_1), \dots, (\tilde{x}_m, \tilde{y}_m)\}.$$

- For each input $\tilde{x}_i$, we **predict** the response using our model:

$$\hat{y}_i = \hat{f}(\tilde{x}_i)$$

- We then **compare** these predictions $\hat{y}_i$ with the observed outputs $\tilde{y}_i$.

The **empirical prediction error** (also called the **Mean Squared Error**, MSE) is:

$$\mathrm{err}_{\hat{f}} = \frac{1}{m} \sum_{i=1}^{m} [\tilde{y}_i - \hat{f}(\tilde{x}_i)]^2$$

This quantity estimates the expected squared prediction error using the available data.

We square the error for three main reasons:

1. **To penalise large errors more strongly:**
   Squaring makes large deviations more costly. For example, an error of 4 leads to a squared loss of 16, whereas an error of 2 leads to a squared loss of 4, so the larger error is penalised four times as much.
2. **To avoid cancellation of positive and negative errors:**
   Without squaring (or taking absolute values), overestimates and underestimates would cancel each other out.
3. **For mathematical and statistical convenience:**
   The squared loss is smooth and convex, making optimisation easier.
   Moreover, minimising the expected squared loss leads directly to the **regression function**:

$$f^*(X) = \mathbb{E}[Y \mid X]$$

2.6.2 Practical use

- In practice, we **split** our data into **training** and **test** sets.
- The training data are used to **fit** $\widehat{f}$, while the test data are used to **evaluate** its performance.
- When data are limited, we use **cross-validation**, which rotates the training and test roles to get a reliable estimate.
- To obtain a more reliable measure of performance, we use **cross-validation**, which consists of:
  - Dividing the dataset into K subsets (called "folds");
  - Training the model on $K-1$ folds and testing it on the remaining one;
  - Repeating this process K times and averaging the test errors.

This ensures that every observation is used for both training and testing, giving a more robust estimate of the model's predictive accuracy.

Concept	Definition	Interpretation
$\mathrm{Err}_{\hat{f}}$	$\mathbb{E}\{[Y - \widehat{f}(X)]^2\}$	True (theoretical) expected error
$\mathrm{err}_{\hat{f}}$	$\frac{1}{m}\sum_{i=1}^{m}[\tilde{y}_i - \widehat{f}(\tilde{x}_i)]^2$	Empirical estimate (MSE) from test data

The squared loss penalises large errors and ensures all deviations are positive, giving a smooth, interpretable measure of how well $\widehat{f}$ generalises to unseen data.

3 Linear Regression

3.1 Simple Linear regression

Recall the stochastic model: we start from the general regression framework: $Y = f(X) + \varepsilon$ where ε is a random error term with $\mathbb{E}(\varepsilon) = 0$ and $\mathrm{Var}(\varepsilon) = \sigma^2$.

The simple linear model

In **simple linear regression**, we assume that the relationship between Y and X is **linear**: $f(X) = \beta_0 + \beta_1 X$, where:

- β_0 is the **intercept**, i.e., the predicted value of Y when $X = 0$;
- β_1 is the **slope**: the expected change in Y for a one-unit increase in X.

Together, β_0 and β_1 are called the **model coefficients** or **parameters**.

In order to apply linear regression in practice, we first **split the data** into **training** and **test** sets. For the training data, suppose we observe n data points: $\{(x_i, y_i)\}_{i=1}^n$. We then use these observations to estimate the parameters (β_0, β_1), obtaining their fitted (sample) estimates: $(\hat{\beta}_0, \hat{\beta}_1)$ For each observation i:

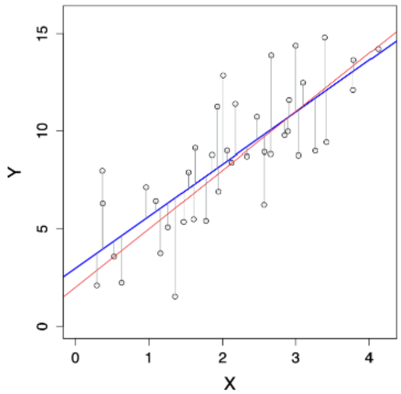
- The **fitted value** (predicted response) is
$$\hat{y}_i = \hat{f}(x_i) = \hat{\beta}_0 + \hat{\beta}_1 x_i$$
- The **residual** (prediction error) is
$$e_i = y_i - \hat{y}_i$$

Each residual measures how far the model's prediction is from the actual observed value of Y.

Finally, to measure the total prediction error across all observations, we compute the **Residual Sum of Squares (RSS):**
$$\mathrm{RSS}(\hat{\beta}_0, \hat{\beta}_1) = e_1^2 + e_2^2 + \cdots + e_n^2$$

or equivalently,
$$\mathrm{RSS}(\hat{\beta}_0, \hat{\beta}_1) = \sum_{i=1}^{n} (y_i - \hat{f}(x_i))^2 = \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

Minimising this RSS gives the **least squares estimates** $\hat{\beta}_0$ and $\hat{\beta}_1$, in other words, the line that best fits the data in the least-squares sense.



Simple Linear Regression: true function vs. estimated function

As we see in the plot on the left:

- The **red line** represents the *true* function $f(X) = \beta_0 + \beta_1 X$, which is generally unknown.
- The **blue line** represents the *estimated* regression line $\hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$.
- The **vertical grey lines** show the residuals $e_i = y_i - \hat{y}_i$, i.e., the differences between the observed and fitted values.

The goal of linear regression is to make the estimated line $\hat{f}(X)$ as close as possible to the true underlying relationship $f(X)$ by minimising the RSS.

3.2 Estimation of parameters by least squares

To estimate the coefficients (β_0, β_1) of the simple linear model, we choose the values that **minimise the Residual Sum of Squares (RSS):**

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{(\beta_0, \beta_1) \in \mathbb{R}^2} \text{RSS}(\beta_0, \beta_1)$$

where

$$\text{RSS}(\beta_0, \beta_1) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

This means we are finding the intercept and slope that make the predicted values $\hat{y}_i = \beta_0 + \beta_1 x_i$ as close as possible to the actual y_i, in the least-squares sense.

So, **how do we solve the optimisation problem?**

There are two main approaches to find $(\hat{\beta}_0, \hat{\beta}_1)$:

1. **Numerically**, using computer algorithms such as **gradient descent**, which iteratively adjusts the coefficients to reduce the RSS.
2. **Analytically**, by **differentiation**, i.e., setting the partial derivatives of the RSS with respect to β_0 and β_1 to zero and solving the system of equations.

The analytical (closed-form) solutions are:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n} (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$ and $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$ are the **sample means** of X and Y respectively.

Therefore, $\hat{\beta}_1$ measures the **average change in** Y associated with a one-unit increase in X, whereas $\hat{\beta}_0$ is the **predicted value of** Y when $X = 0$. Both coefficients are calculated so that the **sum of squared residuals** is as small as possible.

3.2.1 Randomness of the estimates

The estimates $(\hat{\beta}_0, \hat{\beta}_1)$ are **random variables**, not fixed numbers. This is because if we collected a new training dataset (another random sample from the same population) we would obtain slightly different estimates.

In the plot on the right:

- The **red line** represents the true model $f(X) = \beta_0 + \beta_1 X$.
- The **blue line** shows the estimated regression line $\hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X$.
- The **grey lines** correspond to regression lines obtained from other random samples, illustrating how $(\hat{\beta}_0, \hat{\beta}_1)$ vary due to sampling variability.

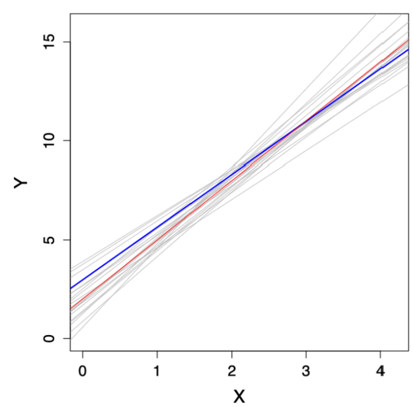


Figure 6: Sampling variability in least squares estimates

3.3 Advertising data

3.4 Multiple Linear Regression

We now consider p predictors $X = (X_1, \dots, X_p)$ for the response variable Y. The model becomes:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p + \varepsilon$$

where β_0 is the intercept (the predicted value of Y when all predictors are 0), β_j for $j = 1, \dots, p$ are the coefficients representing the partial effect of each predictor X_j, and ε is the random error term.

Intuition:
This is the natural extension of simple linear regression to multiple predictors. In simple regression, Y varies with a single X, forming a **line** in 2D space.

With multiple predictors, the model defines a **hyperplane** in $(p+1)$-dimensional space, since each predictor adds one dimension. Moreover, each coefficient β_j tells us how Y changes *on average* when X_j increases by one unit, while all other predictors are held constant.

Example: The Advertising dataset

For instance, suppose we study how different types of media advertising affect sales. Then our model could be:

$$\text{sales} = \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times \text{newspaper} + \varepsilon$$

Here: Y = sales, X_1 = TV advertising budget, X_2 = radio budget, X_3 = newspaper budget.

Each predictor contributes independently to explaining Y. The regression finds the "best-fitting" hyperplane that minimises the total squared distance between the observed and predicted sales values.

3.4.1 Matrix form of the training data

We observe n data points $\{(x_i, y_i)\}_{i=1}^n$, where each $x_i = (x_{i1}, \dots, x_{ip})^\top$ contains the p predictor values for observation i. All predictors can be represented compactly as an $n \times p$ matrix:

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

Intuition: each **row** of **X** corresponds to one observation, and each **column** to one predictor (feature). This compact matrix notation allows us to handle all data points and predictors simultaneously, which is essential when we later express the model in vector–matrix form. In other words, x_{11} would be the value for the observation $i = 1$ coupled with the feature $x = 1$, whereas x_{32} would be the value for observation $i = 3$ coupled with the feature $x = 2$.

3.4.2 Dot product notation

For notational convenience, we sometimes write the model using a **dot product** between vectors $x, y \in \mathbb{R}^p$:

$$x^\top y = \sum_{i=1}^{p} x_i y_i = x_1 y_1 + x_2 y_2 + \cdots + x_p y_p$$

This lets us express the regression model compactly as:

$$Y = \beta_0 + x^\top \beta + \varepsilon$$

Intuition: the dot product $x^\top \beta$ is just a concise way of combining all predictors and their coefficients: $\beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_p X_p$. This compact form simplifies derivations and computations, it's the same as saying we take a weighted average of predictors, where the weights are the regression coefficients. In other words, the term $x^\top \beta$ includes all coefficients except β_0, because β_0 (the intercept) is handled separately.

3.5 Estimation and Prediction in Multiple Linear Regression

As before, we estimate the parameters $(\beta_0, \beta_1, \dots, \beta_p)$ by minimising the **residual sum of squares (RSS)**:

$$\text{RSS}(\beta_0, \dots, \beta_p) = \sum_{i=1}^{n} (y_i - \beta_0 - x_i^\top \beta)^2$$

Expanding the dot product $x_i^\top \beta$ gives

$$\text{RSS}(\beta_0, \dots, \beta_p) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_{i1} - \cdots - \beta_p x_{ip})^2$$

This expression measures how far the model's predictions are from the true observed values, across all n observations. The parameter estimates $(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p)$ are the values that **minimise** this total squared error.

Intuition: this generalises the simple linear regression idea, in other words, we're fitting a *hyperplane* rather than a line. Each coefficient β_j tells us how Y changes when predictor X_j increases by one unit, **holding all other predictors constant**. The minimisation process finds the hyperplane that best fits all the data points in p-dimensional space by minimising the vertical distances (residuals).

3.5.1 Making Predictions

Once the parameters are estimated, we can predict the response for a new observation $x_0 = (x_{01}, \dots, x_{0p})$ using

$$\hat{y}_0 = \hat{\beta}_0 + x_0^\top \hat{\beta}$$

or equivalently

$$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_{01} + \cdots + \hat{\beta}_p x_{0p}$$

where $\hat{\beta} = (\hat{\beta}_1, \dots, \hat{\beta}_p)$.

Here, the new point x_0 represents a new combination of predictor values (e.g. TV, radio, and newspaper budgets). By plugging these into the model, we obtain the **predicted response** $\hat{y}_0$, which lies on the estimated regression surface.

In the 3D plot on the right:

- The **green–blue surface** represents the fitted regression plane (the model $\hat{f}(X)$).
- The **red points** are the observed data points (x_{i1}, x_{i2}, y_i).
- The **vertical grey lines** show residuals, i.e., the distances between the actual values and the fitted surface.

Intuition: when we move from simple to multiple regression, we replace the best-fitting line with the best-fitting *plane* (or *hyperplane* in higher dimensions). The same least-squares principle applies: the plane is positioned to minimise the overall squared vertical distances between the observed points and the regression surface.

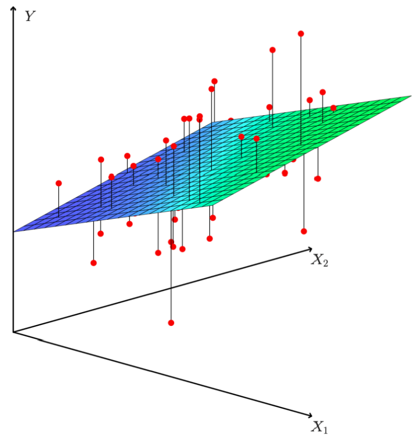


Figure 7: Fitted regression plane in multiple linear regression

3.6 Advertising Data

The **Advertising** dataset is a classic example of multiple linear regression, where we try to predict **sales** (in thousands of units) from the amount spent on different media types. We fit the model

$$\text{sales} = \beta_0 + \beta_1 \times \text{TV} + \beta_2 \times \text{radio} + \beta_3 \times \text{newspaper} + \varepsilon$$

Coefficient	Estimate	Std. Error	t value	p-value
Intercept	2.939	0.3119	9.42	< 2e−16
TV	0.046	0.0014	32.81	< 2e−16
radio	0.189	0.0086	21.89	< 2e−16
newspaper	−0.001	0.0059	−0.18	0.8599

The **intercept** (2.939) represents the expected sales when all advertising budgets are zero. The **TV** and **radio** coefficients are positive and highly significant ($p < 2e - 16$), meaning they have a clear effect on sales. The **newspaper** coefficient is near zero and not statistically significant ($p = 0.8599$), suggesting newspaper advertising has little to no predictive power in this model.

Intuition: each coefficient tells us how much sales are expected to change, on average, for a one-unit increase in that advertising budget, **holding other media constant**.

3.6.1 Correlations Among Predictors

	TV	radio	newspaper
TV	1	0.0548	0.0567
radio		1	0.354
newspaper			1

Intuition: correlations among predictors help us detect **multicollinearity**, in other words, when predictors are correlated with each other. Here, the correlations are low (except a moderate 0.354 between radio and newspaper), so multicollinearity is not a major issue.

3.7 Example: The Auto Data Set

The **Auto** dataset contains measurements for $n = 397$ cars, such as **mpg** (miles per gallon), **horsepower**, **year**, etc. We can fit the model

$$\text{mpg} = \beta_0 + \beta_1 \times \text{horsepower} + \beta_2 \times \text{horsepower}^2 + \varepsilon$$

Even though this model includes horsepower^2, it is still a **linear model** in the parameters $(\beta_0, \beta_1, \beta_2)$, since no coefficients are raised to a power.

Intuition: here, $X_1 = $ horsepower and $X_2 = $ horsepower^2. By including the squared term, the model captures curvature since the relationship between horsepower and mpg is not strictly linear but decreases more steeply at higher horsepower levels.

In this case, both β_1 and β_2 are statistically significant, meaning the quadratic relationship is meaningful. Even higher-order polynomials (degree 3, 4, 5, etc.) can also be fitted to capture more complex patterns, but doing so increases the risk of **overfitting**.

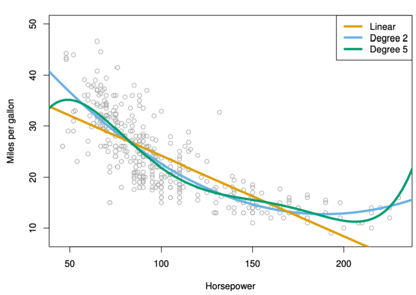


Figure 8: Different degrees Polynomial Regressions

- The yellow line represents a simple linear fit (degree 1).
- The **blue curve** (degree 2) captures curvature and fits the data better.
- The **green curve** (degree 5) fits the data even more closely but may overfit small variations.

The goal is to strike a balance between **bias** (oversimplification) and **variance** (overfitting) when choosing model complexity.

3.8 Beyond Linearity: Linear Basis Functions

Most often, the true function $f(X)$ is not linear in the predictors $X_1, \dots, X_p$. However, we can adjust the linear model by transforming the predictors.

Instead of using $X = (X_1, \dots, X_p)$ directly, we consider transformations of them. Let $h_m(X) : \mathbb{R}^p \to \mathbb{R}$ be the mth transformation, for $m = 1, \dots, M$. The model becomes

$$f(X) = \sum_{m=1}^{M} \beta_m h_m(X)$$

This model is still **linear in the parameters** $\beta_1, \dots, \beta_M$, but **nonlinear in the predictors** X. The functions $h_1(X), \dots, h_M(X)$ are called **basis functions**, and estimation or prediction can be applied as before using least squares.

By using basis functions, we can bend or reshape the linear model to capture more complex relationships between X and Y, i.e., without losing the simplicity of linear estimation. The model is "linear" in β but flexible in X.

3.8.1 Possible Choices for the Basis Functions $h_m(X)$

- $h_m(X) = X_m$, for $m = 1, \dots, p$: essentially recovers the original linear model.
- $h_m(X) = X_j^2$ or $h_m(X) = X_j X_k$: allows higher-order polynomial terms or interactions between predictors.
- $h_m(X) = \log(X_j)$ or $h_m(X) = \sqrt{X_j}$: captures other nonlinear transformations.
- etc.

Intuition: each $h_m(X)$ acts as a new transformed version of the original predictors, expanding the feature space to better approximate the true relationship $f(X)$. Linear regression then finds the best linear combination of these transformed predictors.

4 Gradient Descent

In least squares regression, we often minimise the residual sum of squares (RSS) to estimate the coefficients. Here, for simplicity, we fix β_1 to its true value β_1^* and only estimate β_0 by least squares.

In gradient descent, the **cost function** (often denoted as $J(\theta)$ or $L(\theta)$) **is the mathematical expression that measures how wrong the model is**, that is, **how far the model's predictions are from the true values**. It tells us how costly a particular choice of parameters θ (like β_0, β_1, …) is.

The goal of gradient descent is to find the parameters θ that **minimise the cost function**, i.e., make the model as accurate as possible. In this case, to find the value of β_0 that minimises

$$\hat{\beta}_0 = \arg\min_{\beta_0 \in \mathbb{R}} \text{RSS}(\beta_0) = \arg\min_{\beta_0 \in \mathbb{R}} J(\beta_0)$$

- $\hat{\beta}_0$: this is the **estimated intercept**. The hat indicates it's an estimate obtained from the data, not the true parameter.
- arg min: this stands for *"argument of the minimum."* It means: find the value of β_0 that minimises the expression following it. For example, if you think of a function $f(x)$, $\arg\min_x f(x)$ gives the x that produces the lowest possible value of $f(x)$.
- $\beta_0 \in \mathbb{R}$: the parameter β_0 is a real number.
- $\text{RSS}(\beta_0)$: this is the **Residual Sum of Squares** as a function of β_0. It measures the total squared difference between the observed y_i and the predicted values given β_0:

  $$\text{RSS}(\beta_0) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

  Since β_1 is fixed here, RSS only changes with β_0.
- $J(\beta_0)$: this is a **generic name for the cost function**. In optimisation problems, we often denote the function to be minimised as $J(\cdot)$. Here, $J(\beta_0)$ is simply another name for $\text{RSS}(\beta_0)$.

4.1 The Gradient Descent Algorithm

- Start with an arbitrary initial value $\beta_0^{(0)} \in \mathbb{R}$.
- In the ith step:
  - Compute the **gradient** (or derivative) of the cost function J at the current position $\beta_0^{(i-1)}$:

    $$\nabla J(\beta_0^{(i-1)}) = \frac{\partial J(\beta_0)}{\partial \beta_0}\Big|_{\beta_0 = \beta_0^{(i-1)}}$$

  - Update your estimate using:

    $$\beta_0^{(i)} = \beta_0^{(i-1)} - \alpha \nabla J(\beta_0^{(i-1)})$$

    where α is a **tuning parameter** known as the **learning rate**.
- Repeat until convergence, i.e. when $\beta_0^{(i)}$ changes very little or not at all.

Gradient descent is an iterative algorithm that adjusts parameters step by step in the direction that most rapidly decreases the cost function (here, RSS). The learning rate α controls how large each step is:

- If α is too small $\rightarrow$ convergence is very slow.
- If α is too large $\rightarrow$ the algorithm may overshoot and fail to converge.

This process continues until the algorithm reaches the minimum of $J(\beta_0)$, i.e., the point where its slope is zero.

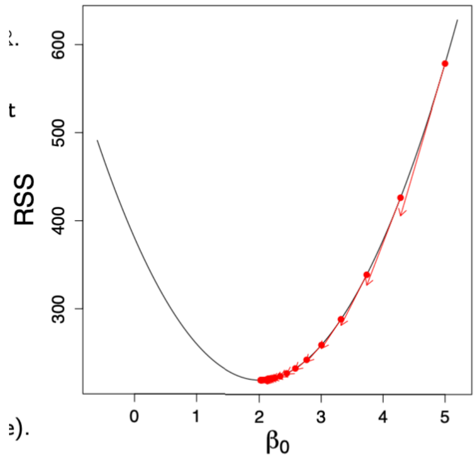


Figure 9: Gradient descent minimising the cost function $J(\beta_0)$

4.2 Gradient Descent: Higher Dimensions

In multiple linear regression, we estimate both β_0 and β_1 (and, in general, all β_j for $j = 1, \ldots, p$) by minimising the cost function:

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{(\beta_0,\beta_1)\in\mathbb{R}^2} \text{RSS}(\beta_0, \beta_1) = \arg \min_{\theta\in\mathbb{R}^d} J(\theta)$$

Here $\theta = (\beta_0, \beta_1)^\top$ denotes the parameter vector, and $J(\theta)$ (or RSS) measures the total squared error between predictions and observed values.

4.3 The Gradient Descent Algorithm (in multiple dimensions)

- Start with an arbitrary initial value $\theta^{(0)} \in \mathbb{R}^d$.
- In the ith step:
  - Compute the **gradient** of the cost function J at the current position $\theta^{(i-1)}$:

    $$\nabla J(\theta^{(i-1)}) = \frac{\partial J(\theta)}{\partial \theta}\Big|_{\theta=\theta^{(i-1)}} \in \mathbb{R}^d$$

  - Update your estimate using:

    $$\theta^{(i)} = \theta^{(i-1)} - \alpha \nabla J(\theta^{(i-1)}) \in \mathbb{R}^d$$

  where α is the **learning rate** (a tuning parameter controlling the step size).

- Stop when the algorithm **converges**, i.e. when $\theta^{(i)}$ changes very little or not at all.

- The plot shows contour lines of the cost surface $J(\beta_0, \beta_1)$, where each line connects points with equal RSS.
- The red dot marks the minimum: the optimal pair $(\hat{\beta}_0, \hat{\beta}_1)$.
- Gradient descent iteratively updates (β_0, β_1), moving step by step along the steepest downward direction until it reaches the centre.

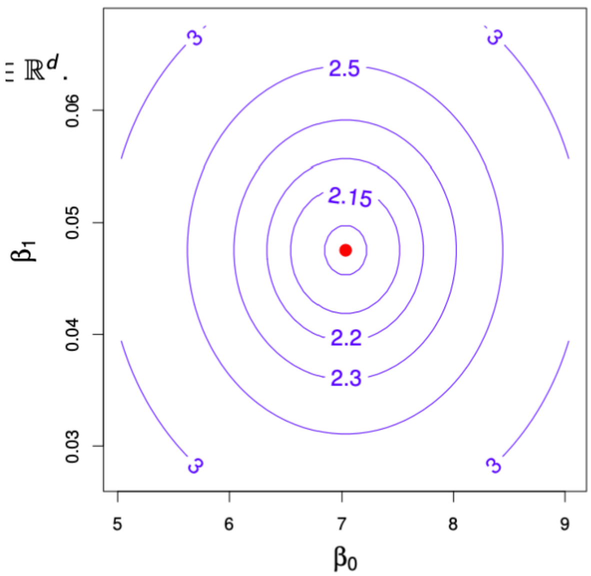


Figure 10: Gradient descent in higher dimensions: contour plot of the cost function $J(\beta_0, \beta_1)$.

4.4 Gradient Descent: Remarks

Gradient descent is a general optimisation algorithm used to find the parameter values that minimise a cost (or objective) function $J(\theta)$. Formally, it solves the problem $\arg\min_{\theta \in \mathbb{R}^d} J(\theta)$.

- The **essential components** are:
  - The **objective function** $J(\theta)$, which measures model error or loss.
  - Its **derivatives** (gradients) in all directions, denoted ∇J.

- The **gradient** $\nabla J(\theta)$ is the collection of **derivatives** of J with respect to each parameter in θ. Intuitively, a derivative measures how much the cost function J changes when one parameter changes slightly.
  In one dimension, it's just the slope $\frac{dJ}{d\theta}$; in multiple dimensions, the gradient vector
  $\nabla J(\theta) = \left[\frac{\partial J}{\partial \beta_0}, \frac{\partial J}{\partial \beta_1}, \dots, \frac{\partial J}{\partial \beta_p}\right]^\top$
  points in the direction of the **steepest increase** of J at point θ. Therefore, gradient descent moves **in the opposite direction** (the direction of the **steepest decrease**) to minimise $J(\theta)$.

- **Important restriction:** the function J must be **differentiable**. Otherwise, gradients cannot be computed.
- Gradient descent is a **generic numerical method** that typically finds a **local minimum** of $J(\theta)$.
- If $J(\theta)$ is **convex**, then the local minimum is also the **global minimum**, and convergence is guaranteed.
- The **tuning parameter** α, also called the **learning rate**, determines how large each update step is. It must be carefully chosen or adapted. If too large, the algorithm overshoots the minimum; if too small, it converges very slowly.
- In complex or high-dimensional models, gradient descent is often the **only feasible way** to find even a local minimum.
- Depending on the model, there exist several **variants and approximations**, such as:
  - **Stochastic Gradient Descent (SGD)**: uses a subset (batch) of data per iteration.
  - **Backpropagation**: applies gradient descent efficiently in neural networks.

Think of gradient descent as walking downhill on a complex landscape. You can't always see the entire surface, but by feeling the slope beneath your feet (the gradient), you can take small steps in the direction that lowers your elevation (the loss) until you reach a valley, ideally, the lowest one.

5 Training versus Test Errors

5.0.1 Training Data

- We fit our model $\hat{f}$ on the **training data** $\{(x_1, y_1), \dots, (x_n, y_n)\}$, representing n observations of (X, Y).
- The model learns patterns from this dataset by adjusting its parameters (e.g. β_0, β_1, …) to minimise the error between predictions and true values.
- The **training error** measures how well the model fits this known data:

$$\text{MSE}_{\text{Tr}} = \frac{1}{n}\sum_{i=1}^{n}\{y_i - \hat{f}(x_i)\}^2$$

- Because the model has already seen these points, increasing its complexity (e.g. more parameters or higher-order terms) almost always reduces this error, sometimes to zero.
- However, this makes the training error **overly optimistic**, since it does not reflect how the model performs on unseen data.

5.0.2 Test Data

- To evaluate **generalisation**, we use a separate **test dataset** $\{(\tilde{x}_1, \tilde{y}_1), \dots, (\tilde{x}_m, \tilde{y}_m)\}$ containing m new observations not seen during training.
- The model $\hat{f}$, trained earlier, is now applied to these new points to produce predictions $\tilde{y}_i = \hat{f}(\tilde{x}_i)$.
- The **test (or generalisation) error** is then computed as:

$$\mathrm{MSE}_{\mathrm{Te}} = \frac{1}{m}\sum_{i=1}^{m}\{\tilde{y}_i - \hat{f}(\tilde{x}_i)\}^2$$

- This test error provides an approximately unbiased estimate of how well the model is expected to perform on future data.
- Formally, it estimates the **expected prediction error:**

$$\mathrm{Err}_{\hat{f}} = \mathbb{E}[(Y - \hat{f}(X))^2]$$

The training error measures how well the model explains the data it has already seen, while the test error shows how well it generalises to unseen examples. As model flexibility increases, the training error steadily decreases, but the test error often follows a U-shaped curve, first improving, then worsening, because of **overfitting**.

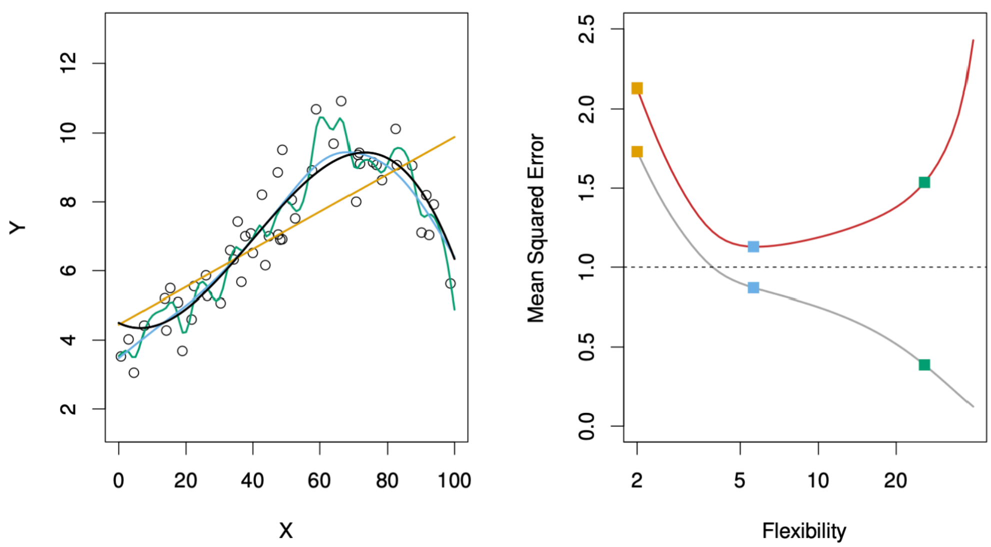


Figure 11: Training versus test errors

Figure 11 illustrates how **training** and **test errors** evolve as model **flexibility** increases.

- On the **left**, the data are generated from the true function f (black line) and fitted using three models:

- a **linear model** (orange line),
- a **moderately flexible polynomial** (blue line),
- and a **highly flexible polynomial** (green line).

- On the **right**, the curves show:
  - **Training error** MSE_{Tr} in grey, which decreases steadily as flexibility increases.
  - **Test error** MSE_{Te} in red, which decreases at first but eventually rises due to overfitting.

When model flexibility is low, both training and test errors are high, in other words, the model underfits and fails to capture the structure of the data. As flexibility increases, the model fits the training data better and generalises more effectively, reaching a point of **optimal complexity** where test error is minimised. Beyond this point, further flexibility causes the model to memorise noise, leading to rising test error, i.e., the hallmark of **overfitting**. This U-shaped behaviour of the test error can be formally explained by the bias–variance decomposition of the expected prediction error.

5.1 The Bias–Variance Decomposition

- We assume a simple model where the outcome is generated as

$$Y = f(X) + \varepsilon, \quad \text{with } \mathbb{E}(\varepsilon) = 0 \text{ and } \text{Var}(\varepsilon) = \sigma_\varepsilon^2.$$

- For a regression fit $\hat{f}$, the **expected prediction error** at an input $X = x_0$ is

$$\text{Err}_{\hat{f}}(x_0) = \mathbb{E}[(Y - \hat{f}(X))^2 \mid X = x_0].$$

- This can be decomposed into three components:

$$\begin{aligned}\text{Err}_{\hat{f}}(x_0) &= \sigma_\varepsilon^2 + \text{Bias}^2(\hat{f}(x_0)) + \text{Var}(\hat{f}(x_0)) \\ &= \text{Irreducible Error} + \text{Bias}^2 + \text{Variance}.\end{aligned}$$

- The **Irreducible Error** is the error due to the noise variable ε and cannot be improved.
- The **Bias** measures how far the model's average prediction $\mathbb{E}[\hat{f}(x_0)]$ is from the true function $f(x_0)$.
- The **Variance** captures how much $\hat{f}(x_0)$ fluctuates across different training sets.
- **A good model finds a balance between low bias and low variance to minimise total error**.

5.1.0.1 Example: Bias–Variance Trade-off in k-Nearest Neighbours

The k-Nearest Neighbours (k-NN) algorithm predicts $\hat{f}_k(x_0)$ as the average of the k closest training points to x_0:

$$\hat{f}_k(x_0) = \frac{1}{k} \sum_{x_i \in N_k(x_0)} y_i.$$

The expected prediction error is

$$\mathrm{Err}_{\hat{f}}(x_0) = \sigma_\varepsilon^2 + \Big[f(x_0) - \frac{1}{k} \sum_{x_i \in N_k(x_0)} f(x_i)\Big]^2 + \frac{\sigma_\varepsilon^2}{k}.$$

When k is small, the model is very flexible → low bias, high variance. As k increases, predictions become smoother → higher bias, lower variance. The total prediction error reaches its minimum when bias and variance are optimally balanced.

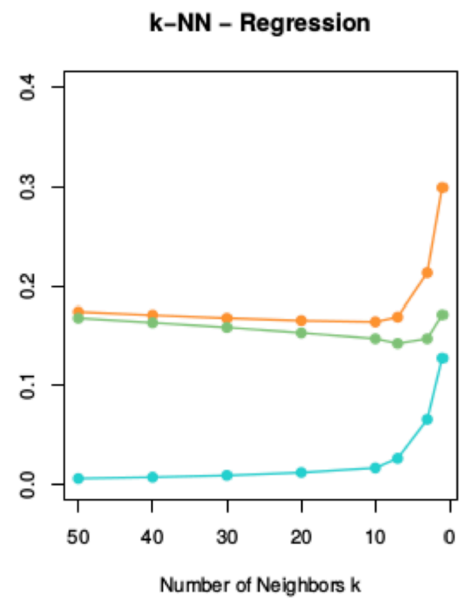


Figure 12: Bias–variance decomposition in k-NN

Intuition:
- When k is **small**, the model is very flexible → **low bias** but **high variance**.
- As k increases, predictions become smoother → **higher bias** but **lower variance**.
- The **orange curve** in the figure shows the **total prediction error**, which is minimised at an intermediate k, where bias and variance are optimally balanced.
- The **green curve** represents the **squared bias**, and the **blue curve** shows the **variance**, both plotted as functions of k to illustrate their trade-off.

5.2 The Bias–Variance Trade-off

Simple parametric models such as **linear regression** or **logistic regression**:

- Have **low flexibility** → **high bias**, but require less data to fit.
- Are **stable** across different datasets → **low variance**.
- Fit well when the data satisfy model assumptions.
- Are suitable for **inference** and **interpretation**, since model parameters have clear meaning.

Complex (often non-parametric) models such as **k-Nearest Neighbours** or **decision trees**:

- Can be **highly flexible** → **low bias**, but require more data to train.
- Are more prone to **overfitting** and can vary strongly across datasets → **high variance**.
- Perform poorly in **high-dimensional** settings (large p) due to the **curse of dimensionality**.
- Are often better suited for **prediction**, where flexibility outweighs interpretability.

In other words, simple models generalise better when data are limited or assumptions hold, while complex models can capture richer patterns but risk memorising noise.

Choosing the right model involves balancing **bias**, **variance**, and the **purpose** of analysis, i.e., explanation or prediction.

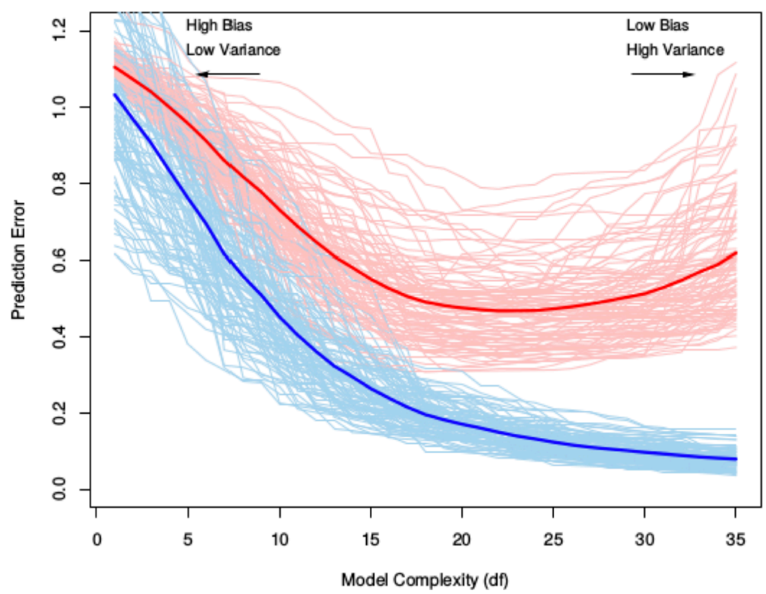


Figure 13: Training versus test error as model complexity increases.

Blue = training error, falls as complexity increases Red = test error, falls at first, then rises (U-shape)
Faint curves show variability across training sets; thick curves show the typical pattern.

Left (low complexity): underfitting — the model is too rigid $\Rightarrow$ **high bias, low variance**; both training and test error are relatively high because important structure is missed.
Middle (intermediate complexity): best generalisation — as flexibility increases, bias drops and the model captures the real signal present in both train and test data $\Rightarrow$ the test error decreases and reaches its minimum.
Right (high complexity): overfitting — beyond a point, the model becomes too sensitive and starts fitting noise $\Rightarrow$ **variance dominates, low bias**; the training error keeps decreasing, but the test error rises again because fitted noise does not generalise well to new data.

The U-shaped test error reflects the sum of two competing effects:

$$\text{Test Error} = \text{Bias}^2 + \text{Variance} + \text{Irreducible Error}$$

As model complexity increases, Bias^2 decreases (better fit to the signal), while variance increases (greater sensitivity to the particular training set). Initially, bias reduction dominates so the test error falls; eventually, variance dominates so the test error rises.

6 Cross-Validation

- The **training error** $\mathrm{MSE}_{\mathrm{Tr}}$[3] is **not reliable** for model selection, since a model that overfits the training data can achieve a very small $\mathrm{MSE}_{\mathrm{Tr}}$ but perform poorly on unseen data.
- A good model should minimise the **expected prediction error** $\mathrm{Err}_{\hat{f}}$, that is, it should generalise well to new data. In other words, $\mathrm{Err}_{\hat{f}}$ measures how well will our model predict outcomes for new data that it has never seen before.
- However, in practice, we only have **one dataset**, so we must find a way to **estimate** $\mathrm{Err}_{\hat{f}}$ using it.

Two main approaches:

1. **Theoretical approximations** of $\mathrm{Err}_{\hat{f}}$
   - Based on analytical criteria such as the **Akaike Information Criterion (AIC)** or the **Bayesian Information Criterion (BIC)**.
   - These rely on assumptions about the model class (e.g. linear models) and penalise excessive complexity to prevent overfitting.
2. **Empirical estimation** of $\mathrm{Err}_{\hat{f}}$
   - Based on the **test error** $\mathrm{MSE}_{\mathrm{Te}}$ estimated from resampling methods such as **Bootstrap** or **Cross-Validation**.
   - These methods approximate the model's performance on unseen data by dividing the available dataset into subsets used for training and testing.

Intuition: Cross-validation provides a systematic way to simulate how the model would perform on new, unseen data. By repeatedly splitting the dataset into training and validation parts, we can estimate the expected prediction error $\mathrm{Err}_{\hat{f}}$ and choose the model complexity that minimises it.

6.1 K-Fold Cross-Validation

- We only have **one dataset** $\{(x_1, y_1), \dots, (x_n, y_n)\}$. The question is: *how can we estimate how the model will perform on new, unseen data*?
- In **K-Fold Cross-Validation**, we randomly divide the dataset into K equally sized subsets, called **folds**, denoted $C_1, C_2, \dots, C_K$.
- Each fold acts once as a **validation set**, while the remaining $K-1$ folds are used for training.

[3] The **Mean Squared Error (MSE)** measures the average squared difference between the predicted values $\hat{y}_i$ and the true values y_i: $\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$. It quantifies how close the model's predictions are to the actual outcomes. In cross-validation, MSE is computed on the validation folds to estimate the model's **generalisation ability**. **Lower MSE values indicate better predictive performance on unseen data, making it the most common metric for model comparison and selection in machine learning**.

- We then compute the **validation MSE** for each fold:

$$\text{MSE}_k = \frac{K}{n} \sum_{i \in C_k} (y_i - \hat{f}_{-k}(x_i))^2$$

where $\hat{f}_{-k}$ is the model trained on all data *except* fold C_k.

- Finally, the **K-Fold Cross-Validation estimate** of the prediction error is the **average validation error** across all folds:

$$\text{CV}_{(K)} = \frac{1}{K} \sum_{k=1}^{K} \text{MSE}_k$$

Since we cannot collect new data to test our model, we simulate this process by holding out different subsets of the data in turn. Each fold provides an estimate of how well the model generalises, and averaging these results gives a more stable and less biased estimate of the **expected prediction error** $\text{Err}_{\hat{f}}$.

Choosing the number of folds:
The parameter K determines how the data are divided between training and validation, directly influencing the **bias**, **variance**, and **computational cost** of the cross-validation estimate.

- With a **small** K (e.g. 5): each **training set** is smaller because a larger portion of the data is held out for validation in each iteration. As a result, the model is fitted on less data and may **underfit slightly**, leading to a **higher bias** in the estimated prediction error. However, the **validation sets are larger**, which makes the error estimates across folds more stable and results in **lower variance**. Training fewer models also makes the procedure **computationally faster**.
- With a **large** K (e.g. 10, or $K = n$ in Leave-One-Out Cross-Validation): each **training set** is larger, and the **validation set** becomes smaller. This reduces bias since each model is trained on nearly the full dataset, but increases variance because the validation results depend more on specific observations. It also requires fitting more models, which increases **computational cost**.

In practice, choosing $K = 5$ or $K = 10$ usually provides the best balance: bias remains reasonably low, variance manageable, and computation time practical for most applications.

6.2 Interpreting the K-Fold Cross-Validation Estimate

- The **K-Fold Cross-Validation estimate** is a widely used and general method that can be applied to almost any model.
- This estimate allows us to **select the best model** among several candidates and provides an empirical approximation of the **expected prediction error** $\text{Err}_{\hat{f}}$ for the final chosen model.
- The appropriate choice of K depends on the dataset size n, but in practice, $K = 5$ or $K = 10$ are commonly used as they provide a good balance between bias, variance, and computational efficiency.

Interpreting the cross-validation estimate:
K-Fold Cross-Validation not only estimates how well a model generalises, but also gives an indication of the **uncertainty** associated with this estimate, that is, how much the result would vary if we repeated the procedure with a different random split of the data. This variability is measured through the **standard error** of the cross-validation estimate:

$$\widehat{\text{SE}}(\text{CV}_{(K)}) = \frac{1}{\sqrt{K}} \sqrt{\frac{\sum_{k=1}^{K} (\text{MSE}_k - \text{CV}_{(K)})^2}{K-1}}$$

A **small standard error** means that the model performs consistently across folds. Its generalisation is stable and reliable.
A **large standard error**, on the other hand, indicates that the model's performance varies substantially depending on which data points are used for training and validation, suggesting possible instability or overfitting.

Thus, cross-validation helps assess not only *which model performs best on average*, but also *which model performs most consistently* across different subsets of the data.

6.3 Leave-One-Out Cross-Validation

- One limitation of **K-Fold Cross-Validation** is that it requires training K separate models, which can become computationally expensive for large datasets.
- A useful special case of K-Fold CV is **Leave-One-Out Cross-Validation (LOOCV)**, where each observation is treated as its own validation fold. In this case, $K = n$ and $C_k = \{k\}$ for $k = 1, \ldots, n$.
- The **LOOCV estimate** of the prediction error is:

  $$\text{CV}_{(n)} = \frac{1}{n} \sum_{k=1}^{n} (y_k - \hat{f}_{-k}(x_k))^2$$

  where $\hat{f}_{-k}$ denotes the model fitted on all data except the k-th observation.
- In general, this requires fitting n separate models (one for each observation) which can be computationally intensive. However, for **linear models**, a convenient approximation exists:

  $$\text{CV}_{(n)} \approx \frac{1}{n} \sum_{k=1}^{n} \left(\frac{y_k - \hat{f}(x_k)}{1 - S_{kk}} \right)^2$$

  where S_{kk} are leverage values (diagonal elements of the hat matrix), automatically available from the linear model fit.
- Overall, $\text{CV}_{(K)}$ is a reliable estimate of the **expected prediction error** $\text{Err}_{\hat{f}}$. The choice of K reflects a **bias–variance trade-off**:
  - Large K (e.g. LOOCV) → smaller bias but higher variance.

– Small K (e.g. 5-Fold CV) → higher bias but lower variance.

In practical terms, LOOCV makes almost full use of the data for training in each iteration, which **reduces bias** because each model is trained on nearly all available observations. However, since each validation fold contains only one observation, the resulting errors vary more from case to case, leading to **higher variance**. This makes LOOCV less stable for complex models or large datasets, but highly effective and appealing for small samples or simple models such as linear regression.

6.4 Cross-validation applied to the body fat dataset

The body fat dataset is used to illustrate how **10-fold cross-validation (CV)** can guide **model selection** when we have many possible predictors. For each model size k (number of predictors), we consider the **best subset model** of that size, and then we estimate its **test performance** using CV.

- The **y-axis (CV)** is the average cross-validated error (MSE) across the 10 folds.
- Each point corresponds to the CV error for the **best model with k predictors**.
- The **grey bars** are ± 1 **standard error** around the CV estimate (computed from the fold-to-fold variability).
- The **black dot** marks the model size with the **lowest CV error** in this particular random split.

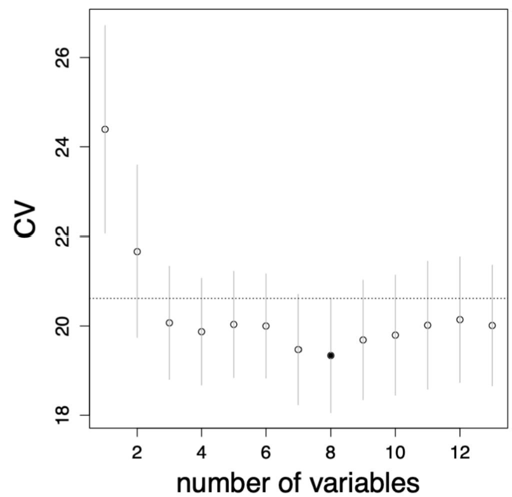


Figure 14: 10-fold CV

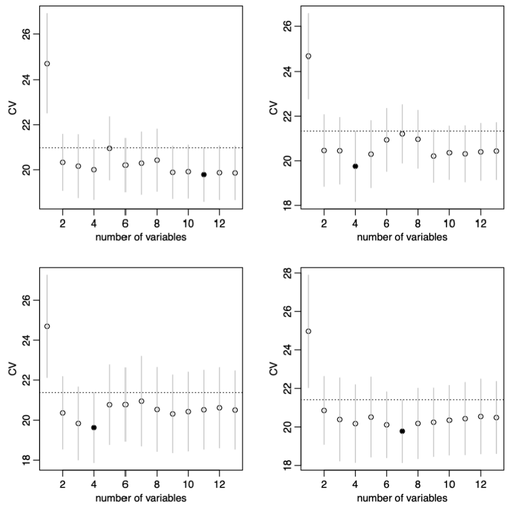


Figure 15: Four repetitions of 10-fold CV

Random folds imply randomness in the CV curve: because the 10 folds are formed **at random**, repeating 10-fold CV can give a **slightly different CV curve** and the **minimum** may occur at a **different** k from run to run. The overall shape is typically similar (often decreasing then increasing), but the exact minimiser can change.

- Repeating **10-fold CV** can lead to **different** selected model sizes.
- The **argmin of CV** is therefore not always a stable choice.
- This instability comes from the fact that **CV depends on the random partitioning** of observations into folds.

6.4.1 Stability and the one-standard-error rule

Because the CV estimate has uncertainty, a practical way to stabilise model selection is the **one-standard-error rule**:

1. Find the **minimum** CV error across model sizes.
2. Add **one standard error**.
3. Choose the **simplest** model whose CV error is **no more than one standard error above** the minimum.
   This rule treats models within that band as essentially "good enough" given estimation noise and favours the most parsimonious option.

To illustrate this, the 10-fold CV procedure was repeated **100 times**. Table Table 6 shows how often each model size (k) was selected under (i) direct minimisation of CV, and (ii) the one-standard-error rule.

Table 6

Model sizes (k) selected across 100 repetitions of 10-fold cross-validation.

k	1	2	3	4	5	6	7	8	9	10	11	12	13
arg min of CV	0	5	8	11	0	1	11	20	11	12	18	3	0
one-sd-error rule	0	83	15	2	0	0	0	0	0	0	0	0	0

6.5 Final model selection

Cross-validation identified a model with **three predictors** ($\hat{k} = 3$): **weight**, **abdomen**, and **wrist**. As shown in Figure 14, this specification achieved the **lowest cross-validation error** in the *best-subset selection* analysis, striking an effective balance between predictive accuracy and simplicity. It explains most of the variation in body fat percentage without overfitting.

The corresponding model $\hat{f}_3$ is then refitted on **all 252 observations** to obtain the final parameter estimates based on the full dataset.

Overall, cross-validation proved useful not only for identifying the best-performing model ($p = 3$) but also for ensuring that the final specification remains **interpretable, stable, and generalisable**.

7 Linear Model Selection and Regularization

7.1 Credit data set and data standardisation

- The **Credit data set** from the package *ISLR* contains the response *balance* (average credit card debt) and quantitative predictors such as *age*, number of credit *cards*, *income*, and *education*.
- It also includes four qualitative predictors: *gender*, *student*, *status*, and *ethnicity*.
- Such variables often exhibit **different numeric scales** or units of measurement, which can affect the behaviour of many algorithms.
- When predictors $X_1, \dots, X_p$ have **different scales**, the relative magnitude of each variable can distort distance-based methods (e.g. k-Nearest Neighbour) or affect regularisation penalties.
- To ensure comparability, it is good practice to **standardise the predictors** in the training data $\mathbf{x}_i$, $i = 1, \dots, n$, for each margin $j = 1, \dots, p$:

$$x_{ij}^{\text{scaled}} = \frac{x_{ij} - \bar{x}_j}{\hat{\sigma}_j}$$

  where $\bar{x}_j$ and $\hat{\sigma}_j$ are the sample mean and standard deviation of predictor j
- In Python, this can be done with the transformer `StandardScaler()`.
- For a new input $x_0 \in \mathbb{R}^p$, standardise as $x_{0j}^{\text{scaled}} = (x_{0j} - \bar{x}_j)/\hat{\sigma}_j$ before applying the predictive model.

Standardisation ensures that all predictors contribute equally to the model by removing scale effects. Without it, a variable with large numerical values (such as *income* measured in thousands) can dominate distance-based or penalty-based calculations, not because it is more informative, but because it stretches the geometry of the feature space.

As shown in Figure 16, unscaled predictors distort distances and decision boundaries in the original data, whereas standardisation recentres each variable and rescales it to unit variance. After this transformation, classification and regularisation methods operate on a balanced feature space, where each predictor influences the model according to its information content rather than its measurement scale.

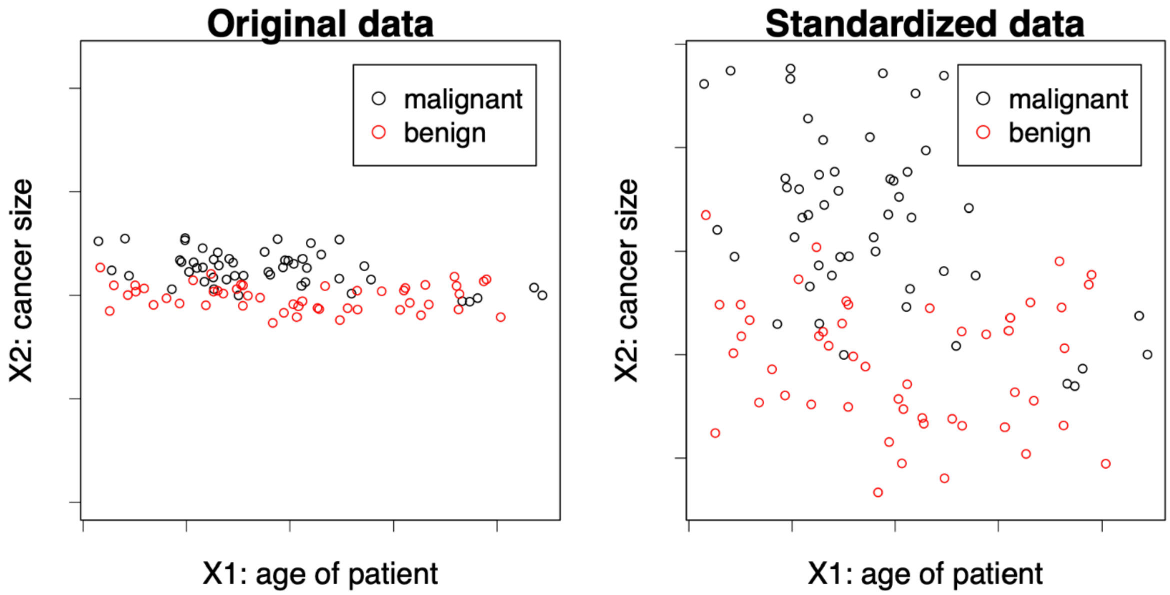


Figure 16: Effect of standardisation on feature scale

7.2 Back to linear models

- In the **regression setting**, a multiple linear model

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p$$

  is fitted to the data (x_i, y_i), $i = 1, \ldots, n$.[4]

- The estimated vector $\widehat{\beta} = (\widehat{\beta}_0, \ldots, \widehat{\beta}_p)$ minimises the residual sum of squares (RSS):

$$\text{RSS} = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} \beta_j x_{ij} \right)^2$$

- Why might we want to use a **different estimation method** than least squares?

(1) **Prediction accuracy:** If n is not much larger than the number of predictors p, or even $p \gg n$ (a *high-dimensional setting*), the least squares estimate $\widehat{\beta}$ may have high variance, leading to overfitting. By **shrinking** or **regularising** the coefficients, we can reduce variance and improve prediction accuracy on new test data.

[4] X_1 refers to the vector of values for the first predictor variable across all observations in your dataset. This means it contains all individual observations for the predictor associated with the coefficient β_1.

(2) **Model interpretability:** Some predictors may have weak or no relationship with the response, yet least squares rarely estimates their coefficients as exactly zero. Excluding such **irrelevant predictors** simplifies the model and improves interpretability.

As we saw in Figure 7, the fitted regression plane represents the least-squares solution that minimises the vertical distances between the observed data points and the estimated surface. However, when the number of predictors grows or when predictors are correlated, the position and slope of this plane become unstable. Regularisation methods modify this fitting process by penalising large coefficient values, thereby stabilising the plane and improving both prediction accuracy and interpretability.

7.3 Alternatives to least squares

Several approaches extend or modify least squares estimation to improve prediction accuracy and interpretability when dealing with many or correlated predictors.

(1) **Subset selection:** Identify a subset of the p predictors and then fit a linear model by least squares.

- *Best subset selection* evaluates all possible combinations of predictors.
- *Stepwise selection* (forward or backward) adds or removes predictors sequentially.
  Details in [ISL], Chapter 6.

(2) **Dimension reduction:** Project the p predictors into an M-dimensional subspace ($M < p$) using techniques such as PCA, then fit a linear regression to the projected variables by least squares. Details in [ISL], Chapter 6.

(3) **Shrinkage (regularisation):** Fit a linear regression to all p predictors, but shrink the estimated coefficients towards zero relative to the least squares fit. Depending on the regularisation method (e.g. Ridge, LASSO), some coefficients may even become exactly zero. This penalisation both stabilises estimation and can perform **automatic variable selection**.

These alternative methods adjust the standard least-squares fitting process by reducing the number of predictors, transforming them into a lower-dimensional representation, or constraining their estimated effects to achieve models that generalise better and are easier to interpret.

7.4 Ridge regression

- Instead of minimising the RSS, the **ridge regression** coefficient estimates $\hat{\beta}_\lambda^R$ minimise the penalised version

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}\beta_j^2 = \text{RSS} + \lambda\sum_{j=1}^{p}\beta_j^2$$

  where $\lambda \geq 0$ is the **tuning parameter**.

- The penalty term $\lambda \sum_j \beta_j^2$ is called a **shrinkage penalty** and becomes small when all $\beta_1, \dots, \beta_p$ are close to zero.
- For each $\lambda \geq 0$, ridge regression produces a different set of coefficient estimates $\hat{\beta}_\lambda^R$.
  - If $\lambda = 0$, they coincide with the least-squares estimates.
  - As $\lambda \to \infty$, they shrink towards zero.
- Choosing an appropriate λ is crucial and is typically done through **cross-validation**.

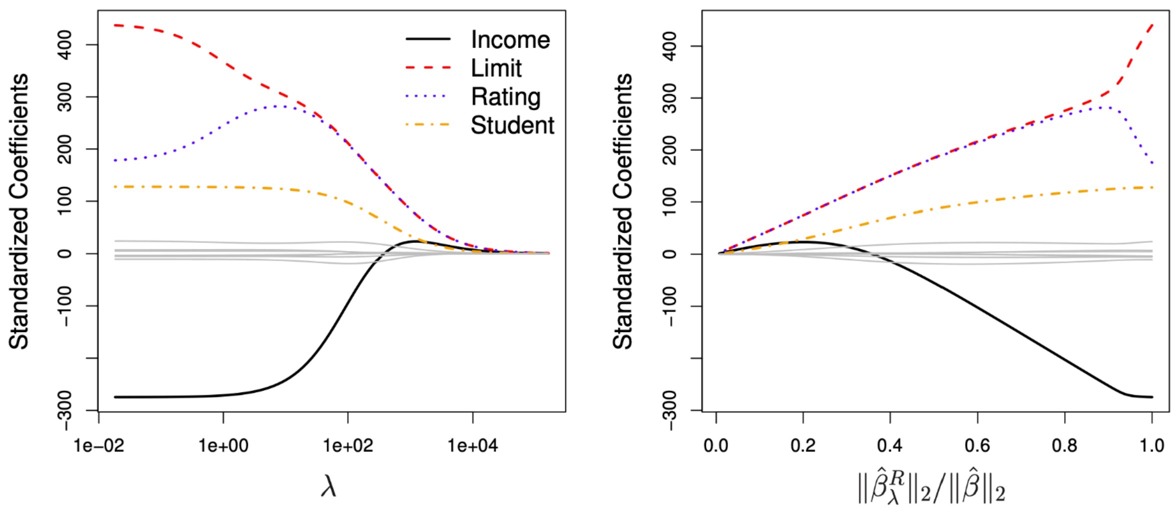


Figure 17: Ridge regression coefficient paths

As shown in Figure 17, ridge regression progressively shrinks coefficient magnitudes as the regularisation parameter λ increases.

In the **left panel**, λ is displayed on a logarithmic scale. Each coloured curve represents the standardised coefficient for one predictor. The **black curve** corresponds to *Income* and starts around –200, indicating that higher income initially reduces the predicted credit balance. As regularisation strengthens, its magnitude decreases smoothly, meaning the model assigns progressively less weight to that predictor. The **red curve** (*Limit*), **blue curve** (*Rating*), and **orange/yellow curve** (*Student*) exhibit the same continuous shrinkage towards zero.

In the **right panel**, the same coefficient paths are plotted against the ratio $\|\hat{\beta}_\lambda^R\|_2/\|\hat{\beta}\|_2$, which measures the remaining overall coefficient magnitude relative to the unpenalised least-squares solution.

Together, these plots illustrate the **bias–variance trade-off**: increasing λ reduces variance by stabilising the estimates but introduces bias by shrinking coefficients towards zero. Ridge regression is therefore particularly effective in the presence of multicollinearity or high-dimensional data, with the optimal λ typically selected via cross-validation.

7.5 Ridge Regression: Closed-Form Solution

- Ridge regression has a closed-form solution:

$$\hat{\beta}_\lambda^R = (X^\top X + \lambda I)^{-1} X^\top y$$

  where $(X \in \mathbb{R}^{n\times p})$ is the data matrix, $(I \in \mathbb{R}^{p\times p})$ is the identity matrix, and $(y = (y_1, \dots, y_n))$ is the vector of responses.

- This can be compared to the least squares solution:

$$\hat{\beta} = (X^\top X)^{-1} X^\top y$$

- The ridge estimator modifies the least-squares expression by adding (λI) before inversion. This term prevents instability when predictors are highly correlated or when $(p > n)$, and demonstrates how ridge regression shrinks the least squares estimates towards zero.

7.6 Lasso: least absolute shrinkage and selection operator

- Ridge regression has the disadvantage that it includes **all** p predictors in the final model.
- The **lasso** overcomes this limitation. The lasso coefficients $\hat{\beta}_\lambda^L$ minimise the penalised RSS

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 + \lambda\sum_{j=1}^{p}|\beta_j| = \text{RSS} + \lambda\sum_{j=1}^{p}|\beta_j|$$

  where $\lambda \geq 0$ is again a **tuning parameter**.
- The key difference from ridge regression is that the **lasso penalty** uses the ℓ_1-norm of β.
- This penalty allows some coefficient estimates to become **exactly zero** when λ is large enough, meaning the lasso performs **automatic variable selection**.
- As with ridge regression, a suitable value of λ is critical and typically chosen via **cross-validation**.

As shown in Figure 18, lasso regression behaves similarly to ridge regression for small values of λ, but differs in how coefficients shrink as λ increases.
- In the **left panel**, the coefficient paths are plotted against λ on a logarithmic scale. As λ grows, some coefficients are driven exactly to zero, effectively removing those predictors from the model.

- In the **right panel**, the horizontal axis is the ratio $\|\hat{\beta}_\lambda^L\|_1/\|\hat{\beta}\|_1$, which measures how much total coefficient magnitude remains compared with least squares. Values near 1 mean weak regularisation; values near 0 mean strong regularisation, where many coefficients have been driven to zero. Unlike λ, this ratio is a normalised "model size" scale: moving left corresponds to a sparser model with fewer active predictors.

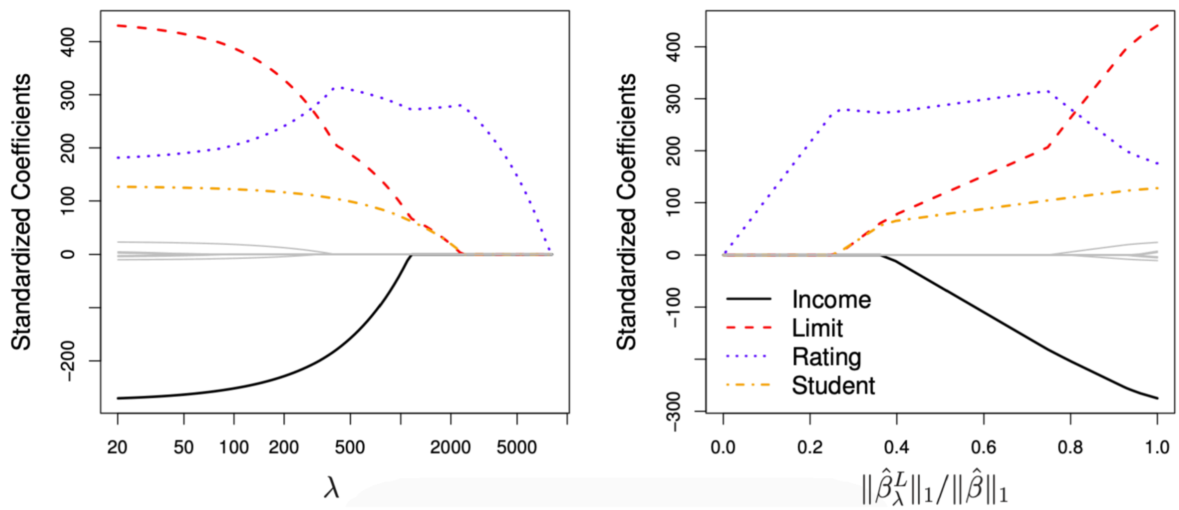


Figure 18: Lasso regression coefficient paths

This sparsity property distinguishes the lasso from ridge regression: while ridge continuously shrinks all coefficients, lasso can produce a simpler, more interpretable model by selecting only a subset of relevant predictors.

7.7 Regularisation: geometric interpretation

- **Equivalent formulation:** ridge and lasso regression can each be expressed as constrained optimisation problems. The estimated coefficients $\hat{\beta}^R_\lambda$ and $\hat{\beta}^L_\lambda$ minimise the residual sum of squares subject to different types of constraints:

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 \quad \text{subject to} \quad \sum_{j=1}^{p}\beta_j^2 \le s,$$

$$\sum_{i=1}^{n}\left(y_i - \beta_0 - \sum_{j=1}^{p}\beta_j x_{ij}\right)^2 \quad \text{subject to} \quad \sum_{j=1}^{p}|\beta_j| \le s.$$

- The first corresponds to **ridge regression** (a quadratic constraint, L_2-norm), and the second to **lasso** (a diamond-shaped constraint, L_1-norm).
- This formulation provides a **geometric interpretation** of penalisation: the constraint defines a region of feasible coefficient values, and the solution occurs where this region first touches the elliptical contours of the RSS.

As shown in Figure 19, the picture can be read as "fit versus complexity" in the (β_1, β_2)-plane. The **red ellipses** are RSS level sets: moving to ellipses closer to the origin means a smaller training error. The

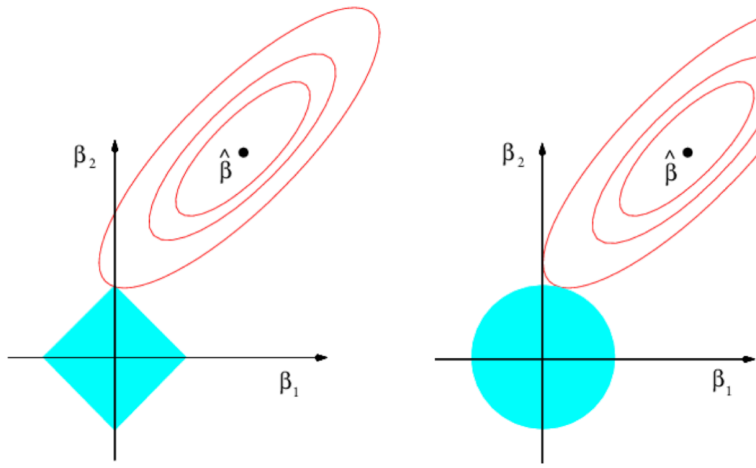


Figure 19: Geometric interpretation of lasso and ridge

blue shape is the constraint imposed by regularisation, i.e. the set of coefficient values we are allowed to choose.

The estimate is the point where the smallest (best) RSS ellipse first touches the constraint region. In other words, we take the best-fitting solution we can achieve without leaving the allowed region.

- With **ridge regression** (right panel), the constraint $\sum_{j=1}^{p} \beta_j^2 \leq s$ is round (an L_2 ball). Because its boundary is smooth, the tangency point typically occurs away from the axes, so coefficients are **shrunk smoothly** towards zero but are **rarely exactly zero**.

- With **lasso** (left panel), the constraint $ ∑_{j=1}^{p}|□_j| ≤s$ has corners (an L_1 ball). Since the corners lie on the coordinate axes, the first contact often happens at a corner, which forces at least one coefficient to be **exactly zero**. This is why lasso performs **variable selection**, whereas ridge mainly performs **shrinkage**.

Exam takeaway: ridge uses a round constraint $\Rightarrow$ continuous shrinkage; lasso uses a diamond constraint with corners on the axes $\Rightarrow$ coefficients can hit zero.

7.8 Comments

- For large p, best subset selection is computationally infeasible. In such cases, the **lasso** can be used to perform **variable selection** efficiently.
- While ridge coefficients $\hat{\beta}_\lambda^R$ have a closed-form solution, lasso coefficients $\hat{\beta}_\lambda^L$ do not. They are obtained through **numerical optimisation**, which remains tractable since both problems are **convex**.

- In practice, ridge and lasso often achieve similar **predictive performance** (for well-chosen λ), but lasso models tend to be **more interpretable** because of their sparsity.
- The **elastic net** (Zou & Hastie, 2005) combines ridge and lasso penalties into

$$\lambda \sum_{j=1}^{p} \left((1-\alpha)\beta_j^2 + \alpha|\beta_j| \right),$$

  where $\alpha \in [0,1]$ controls the relative weight of each penalty ($\alpha = 0$ gives ridge, $\alpha = 1$ gives lasso).
- In applied settings, lasso is often used to **pre-select** a manageable subset of relevant predictors, which can then be refined in more flexible predictive models.
- **Regularisation** extends beyond linear regression and can also be applied to models such as **logistic regression** or **generalised linear models**.

7.9 Example: Hitters data set

The **Hitters** data set from the *ISLR* package contains the annual **Salary** of 263 baseball players and $p = 19$ predictors such as the number of **Years** played in the major leagues and the number of **Hits**. The colour scale encodes salary, from low (yellow–red) to high (blue–green).
This dataset illustrates how regularisation handles models with many correlated predictors. Salary tends to increase with both *Years* and *Hits*, but these variables are also strongly correlated, leading to **multicollinearity**, an ideal case for ridge or lasso regression.

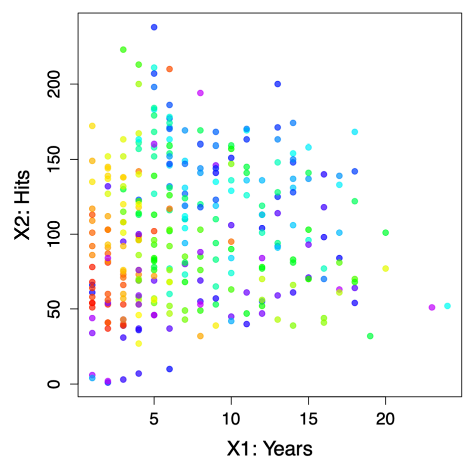


Figure 20: Hitters data set

7.9.1 Ridge regression on the Hitters data set

As shown in Figure 21, the **left panel** displays the *ridge coefficient paths* as functions of $\log(\lambda)$.
- Each coloured curve corresponds to one of the 19 predictors.
- As λ increases, coefficients shrink continuously towards zero but never reach it, meaning all variables remain in the model.

The **right panel** shows the **mean squared error (MSE)** estimated by cross-validation for different λ values.

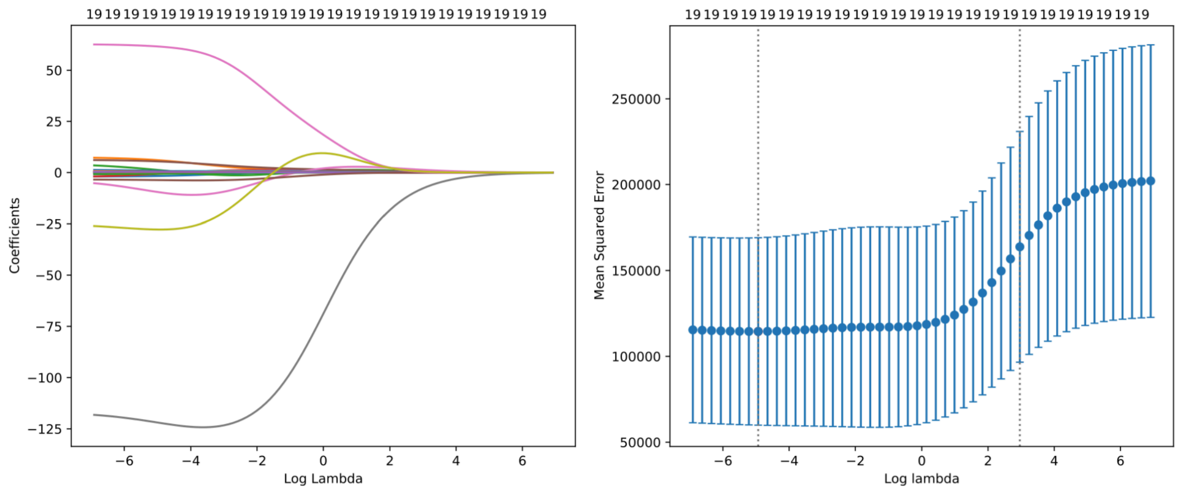


Figure 21: Ridge regression: coefficient paths and cross-validation error

- The dotted vertical line indicates the value of λ that minimises prediction error.
- For small λ, the model overfits; for large λ, coefficients are overly constrained and bias dominates.
- The optimal λ achieves a balance between bias and variance, providing stable but slightly biased predictions.

7.9.2 Lasso regression on the Hitters data set

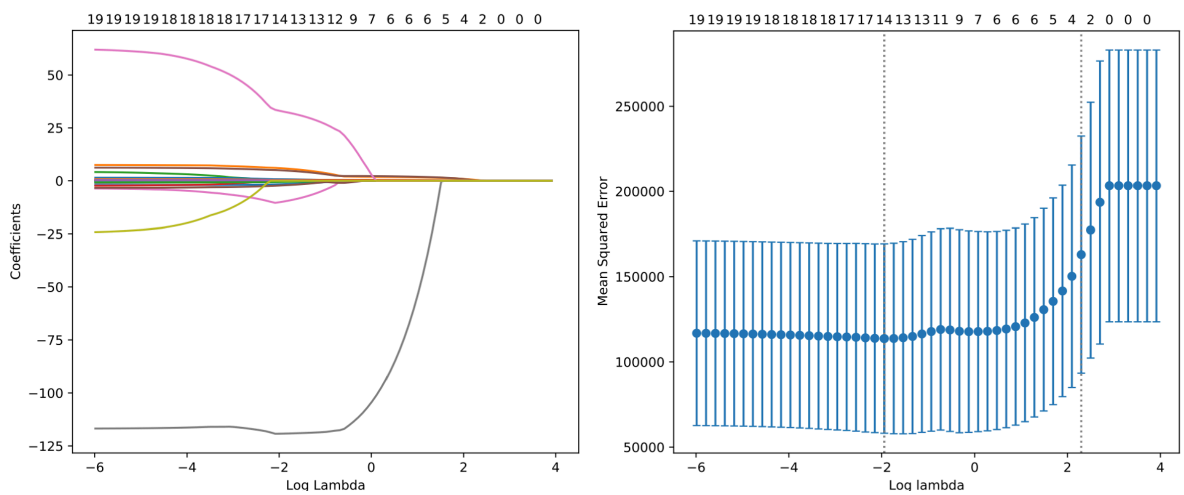


Figure 22: Lasso regression: coefficient paths and cross-validation error

In Figure 22, the **left panel** shows the *lasso coefficient paths*.

- As λ increases, some coefficients drop exactly to zero, removing the corresponding predictors from the model.
- This reflects lasso's ability to perform **automatic variable selection** by enforcing sparsity.

The **right panel** again shows the cross-validation MSE across values of λ.

- The optimal λ (dotted line) identifies a simpler model that excludes less relevant predictors while maintaining strong predictive performance.
- Compared with ridge, lasso yields more interpretable models by retaining only a subset of significant predictors.

Together, these results demonstrate how both ridge and lasso manage the bias–variance trade-off, but differ in **model complexity**: ridge smooths coefficients continuously, while lasso introduces sparsity by setting some exactly to zero.

8 Classification Framework

8.1 Classification: Introduction

Classification, also called *pattern recognition*, aims to predict **qualitative** response variables Y that take values in an unordered set $\mathcal{G} = \{G_1, \dots, G_q\}$, for example: infected $\in \{\text{Yes}, \text{No}\}$ and eye colour $\in \{\text{brown}, \text{blue}, \text{green}\}$.

Given the vector of predictors $X = (X_1, \dots, X_p)$, the goal of classification is to build a function G, called a *classifier*, that takes X and predicts the value of Y, i.e. $G(X) \in \mathcal{G}$.

Often, one is more interested in the **probability** that Y belongs to each class $G_1, \dots, G_q$, for instance:

- the probability that a bank client will default
- the probability that a tumour is malignant
- etc.

We typically identify the q classes with the numbers 1 to q, such that $\mathcal{G} = \{1, \dots, q\}$, or if there are only two classes (*binary data*), then $\mathcal{G} = \{0, 1\}$.

Example. Classification of cancer with $p = 2$ and $q = 2$. We aim to find a classifier

$$G(age, size) \in \mathcal{G} = \{\text{benign}, \text{malignant}\}$$

that predicts whether a cancer is benign or malignant, based on the predictors *age* and *size*.
Each observation corresponds to a patient, represented by their age (X_1) and tumour size (X_2). The classifier will learn a decision boundary that best separates the two classes.

Figure 23: Scatter plot of tumour size versus patient age, coloured by diagnosis.

This example illustrates a **binary classification** task where each observation is assigned to one of two possible categories. The goal is to construct a decision function $G(X)$ that correctly predicts the label Y for new, unseen patients.

Let us clarify that the classifier G maps the two predictor variables $(age, size)$ to one of the possible outcomes in $\mathcal{G}$, either *benign* or *malignant*. In other words, G is a decision rule that assigns each patient to a diagnostic category based on these predictors.

8.2 The 0–1 Loss Function

How do we measure the quality of a prediction? For classification, the squared error loss is not ideal. Therefore, to measure the **misclassification error** for a prediction $\hat{y}$, we usually use the **0–1 loss function**

$$L(y, \hat{y}) = 1\{y \neq \hat{y}\} = \begin{cases} 0 & \text{if } y = \hat{y} \\ 1 & \text{if } y \neq \hat{y} \end{cases}.$$

- For a classifier $\hat{G}$, the **expected misclassification rate** is

  $$\text{Err}_{\hat{G}} = \mathbb{E}[\,1\{Y \neq \hat{G}(X)\}\,] = \Pr\{Y \neq \hat{G}(X)\},$$

  where the expectation is taken over (X, Y). The expression essentially states that the error rate of the classifier $\hat{G}$ is equivalent to the expected value of the indicator function, which tracks the instances where the prediction does not equal the true classification. This is also equal to the probability of making an incorrect classification.

- **Note:** Other loss functions for classification tasks are also common, since:
  - In medical or safety-critical settings, false negatives and false positives are not equally severe, so one may wish to penalise one type of error more than the other.
  - The 0–1 loss is not smooth, which can lead to optimisation difficulties when fitting models.

8.3 The Bayes Classifier

What is a **good classifier**? Ideally, we would like to find the optimal G^* that minimises

$$\text{Err}_{\hat{G}} = \Pr\{Y \neq \hat{G}(X)\}.$$

For simplicity, consider **two-class classification**, i.e. $\mathcal{G} = \{0, 1\}$.
The optimal G^* is called the **Bayes classifier**, which assigns each observation to the class with the highest conditional probability $\Pr(Y = j \mid X = x)$.

Formally, the Bayes classifier assigns to a predictor x the most likely class:

$$G^*(x) = \underset{j=0,1}{\arg\max}\ \Pr(Y = j \mid X = x) = \begin{cases} 0 & \text{if } \Pr(Y = 0 \mid X = \mathbf{x}) > 1/2, \\ 1 & \text{otherwise.} \end{cases}$$

- The **Bayes decision boundary** is the curve (in $\mathbb{R}^p$) where

$$\Pr(Y = 0 \mid X = x) = 1/2.$$

- The Bayes classifier achieves the **lowest possible 0–1 loss**; this minimum achievable error is called the **Bayes error rate**, given by

$$\text{Err}_{G^*} = \Pr\{Y \neq G^*(X)\}.$$

The Bayes classifier represents the theoretical ideal: it predicts the class with the highest conditional probability given the input $X = \mathbf{x}$.
In practice, the true probabilities $\Pr(Y \mid X)$ are unknown, so learning algorithms approximate this rule from data.
The **decision boundary** (white curve in Figure 24) separates regions where each class is more likely; in other words, it marks $\Pr(Y = 0 \mid X) = 0.5$.

Example: Bayes classifier for a two-class problem. The **Bayes classifier** is the theoretical ideal: it assigns each observation to the class with the highest conditional probability $\Pr(Y = j \mid X = \mathbf{x})$. Since the true probabilities $\Pr(Y \mid X)$ are unknown, learning algorithms approximate this rule from sample data. The **decision boundary** (white curve in the plot) separates regions where one class is more likely than the other, corresponding to $\Pr(Y = 0 \mid X) = 0.5$.

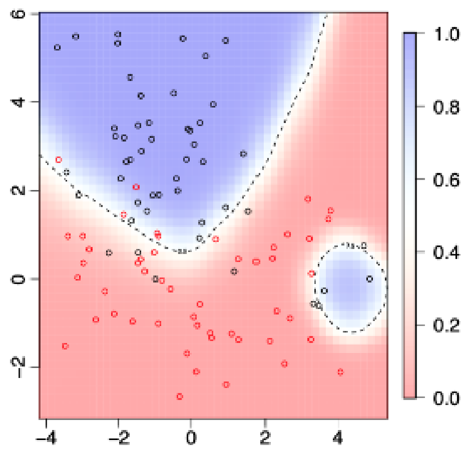


Figure 24: Bayes classifier and decision boundary for a two-class problem.

8.4 The Bayes Decision Rule

The **Bayes classifier** provides the theoretical optimum but requires knowing the true probabilities $\Pr(Y \mid X)$, which are unknown in practice. To approximate it, we estimate the **posterior class probabilities**

$$\widehat{\Pr}(Y = j \mid X = x), \quad j = 0, 1,$$

and then apply the **plug-in decision rule**:

$$\hat{G}(x) = \begin{cases} 0 & \text{if } \widehat{\Pr}(Y = 0 \mid X = x) > 1/2, \\ 1 & \text{otherwise.} \end{cases}$$

The **decision boundary** is the line (or surface in higher dimensions) where both posterior probabilities are equal:

$$\widehat{\Pr}(Y = 0 \mid X = x) = \widehat{\Pr}(Y = 1 \mid X = x) = 1/2.$$

Example. Estimated and true Bayes decision boundaries. This figure compares the **true** and **estimated decision boundaries** for a two-class classification problem. The blue solid line represents the **estimated boundary** obtained from sample data, while the dashed line corresponds to the **theoretical Bayes boundary**.
In practice, the plug-in rule approximates the Bayes classifier by estimating $\Pr(Y \mid X)$ from observed data rather than knowing it exactly. This illustrates how statistical learning methods attempt to reproduce the optimal (but unattainable) Bayes boundary using empirical information.

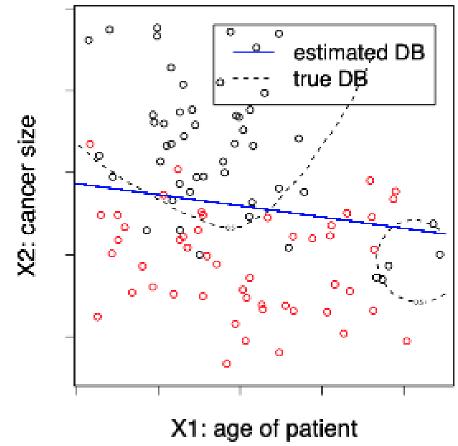


Figure 25: Estimated and true Bayes decision boundaries for a two-class problem.

8.5 Statistical Modelling of $\hat{G}$

Depending on how we model the relationship between Y and X, we can distinguish three main approaches to classification:

(a) Generative models estimate the **joint distribution** of (Y, X) and then deduce the posterior class probabilities for $j = 0, 1$:

$$\widehat{\Pr}(Y = j \mid X = x) = \frac{\widehat{\Pr}(Y = j, X = x)}{\widehat{\Pr}(X = x)}.$$

- More complex and data-demanding, as they require modelling both Y and X.
- **Examples:** Linear/Quadratic Discriminant Analysis, Naive Bayes.

(b) Discriminative models estimate only the **conditional distribution** $\Pr(Y \mid X)$. That is, they directly model the posterior class probabilities $\widehat{\Pr}(Y = j \mid X = \mathbf{x})$.
- Usually preferred, since they provide posterior probabilities and are less complex than generative models.
- **Examples:** k-Nearest Neighbours (kNN), Logistic Regression, Random Forest, Neural Networks.

(c) Geometric models directly estimate a **decision boundary function** that predicts the class (0 or 1) from $\mathbf{x}$ without modelling probabilities.
- Simpler, since no probability distribution is assumed, but they do not provide estimates of class probabilities.
- **Examples:** Support Vector Machines (SVM).

8.6 The Empirical Misclassification Rate

Given a **training data set** $(x_1, y_1), \dots, (x_n, y_n)$, with $x_i \in \mathbb{R}^p$ and $y_i \in \{1, \dots, q\}$, how can we estimate the model $\hat{G}$?

In regression, we minimise the **empirical mean squared error** on the training set. Similarly, for classification, we can minimise the **empirical misclassification rate**:

$$\text{err}_{\hat{G}} = \frac{1}{n} \sum_{i=1}^{n} 1\{y_i \neq \hat{G}(x_i)\}.$$

However, the **0–1 loss function** $1\{y \neq \hat{y}\}$ is **discontinuous**, which makes **numerical optimisation** difficult.

Different methods such as **maximum likelihood estimation** are therefore used for **parameter estimation**, providing differentiable alternatives that approximate the 0–1 loss.

The **0–1 loss** can still be used for **model selection** or for comparing models fitted with different methods.

Example. Empirical misclassification rate. The plot on the right shows how the empirical misclassification rate changes with different values of β_0. Each point represents the average error across the training data for a given parameter value. The goal is to minimise this error to improve classification performance.

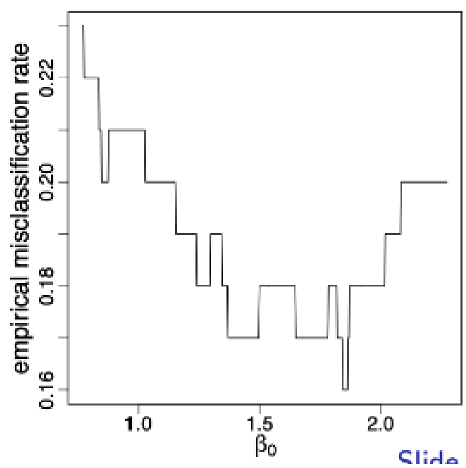


Figure 26: Empirical misclassification rate as a function of β_0.

8.7 K-Fold Cross-Validation

Given a **data set** $\{(x_1, y_1), \dots, (x_n, y_n)\}$, we perform **K-Fold Cross-Validation (CV)** to estimate how well a classifier generalises to unseen data.

For each fold C_k, $k = 1, \dots, K$, we treat C_k as the **validation/test set** and fit the classification model $\hat{G}_{-k}$ on the remaining $K - 1$ folds.

The **empirical misclassification rate** for fold k is given by

$$\text{err}_k = \frac{1}{|C_k|} \sum_{i \in C_k} L(y_i, \hat{G}_{-k}(x_i)) = \frac{1}{n/K} \sum_{i \in C_k} 1\{y_i \neq \hat{G}_{-k}(x_i)\}.$$

The **K-Fold CV estimate** is then the average of these values:

$$\text{CV}(K) = \frac{1}{K} \sum_{k=1}^{K} \text{err}_k.$$

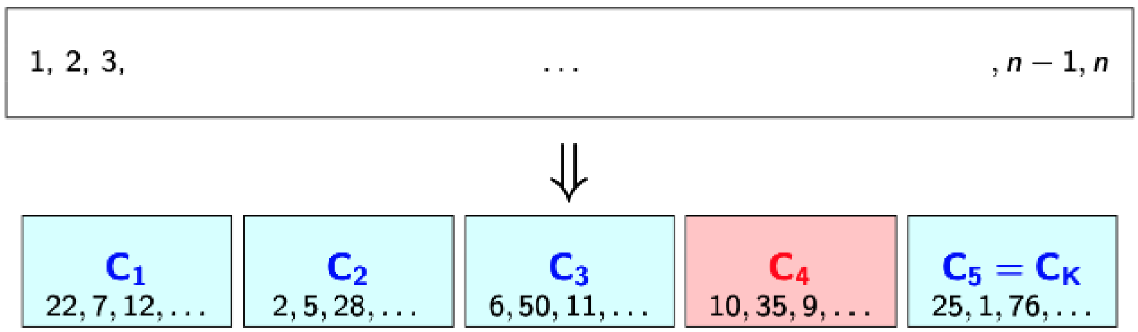


Figure 27: K-Fold Cross-Validation: the dataset is divided into K subsets (folds). Each fold acts once as a validation set (in red) while the remaining folds (in blue) are used for training. The process is repeated K times, and the average error provides an overall estimate of model performance.

As shown in Figure 27, each block $C_1, C_2, \dots, C_K$ represents a different subset of the data. The algorithm cycles through these folds, training on $K - 1$ parts and validating on the remaining one. This rotation ensures that every observation is used both for training and validation exactly once, producing a balanced and robust estimate of model accuracy.

9 Linear and Quadratic Discriminant Analysis (LDA & QDA)

9.1 Linear Decision Boundaries: Hyperplanes

Linear classification methods assign classes according to the sign of a linear function. Suppose we have two classes $G = 0, 1$. Each observation x is evaluated through a linear combination of its predictors, and the points where this expression equals zero define the decision boundary (or hyperplane).

In a p-dimensional space, a **hyperplane** is a flat affine subspace of dimension $p-1$. It is defined by parameters $b_0, b_1, \ldots, b_p \in \mathbb{R}$ as the set of points $x \in \mathbb{R}^p$ satisfying

$$b_0 + b_1 x_1 + \cdots + b_p x_p = 0.$$

If a point x does not satisfy this equality, then

$$b_0 + b_1 x_1 + \cdots + b_p x_p > 0$$

indicates that it lies on one side of the hyperplane, while a negative value places it on the opposite side.

Plugging any point x into this expression tells us on which side of the boundary it lies: a positive value corresponds to one class, a negative value to the other, and zero means the point lies exactly on the hyperplane.

Example $p = 1$. In one dimension, the hyperplane degenerates into a single point separating two classes along a line:

$$b_0 + b_1 x = 0 \iff x = -\frac{b_0}{b_1}.$$

The scalar $x = -b_0/b_1$ is where the linear function crosses zero, i.e. the *decision boundary*. Plugging in values to the left or right yields positive or negative outputs, determining the class. The function's sign alone decides the classification: points to the left or right of this threshold yield opposite signs of $b_0 + b_1 x$, with positive values corresponding to one class and negative values to the other.

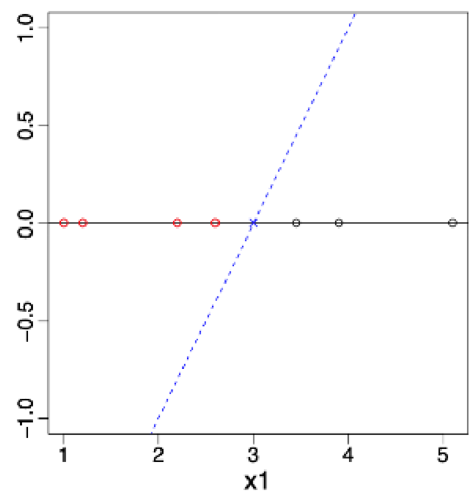


Figure 28: Linear decision boundary in $\mathbb{R}^1$ (a point dividing two classes).

Example $p = 2$. In two dimensions, the hyperplane becomes a line dividing the plane into two half-spaces:

$$(x_1, x_2) \in \mathbb{R}^2 : \; b_0 + b_1 x_1 + b_2 x_2 = 0$$

or equivalently,

$$x_2 = -\frac{b_0}{b_2} - \frac{b_1}{b_2} x_1.$$

The slope $-\frac{b_1}{b_2}$ defines the orientation of the boundary, and the intercept $-\frac{b_0}{b_2}$ determines its position.
Any point (x_1, x_2) that satisfies this equation lies on the decision boundary; points above make the linear function positive and belong to one class, those below negative and to the other. In general, a hyperplane in $\mathbb{R}^p$ always has one dimension fewer than the input space.

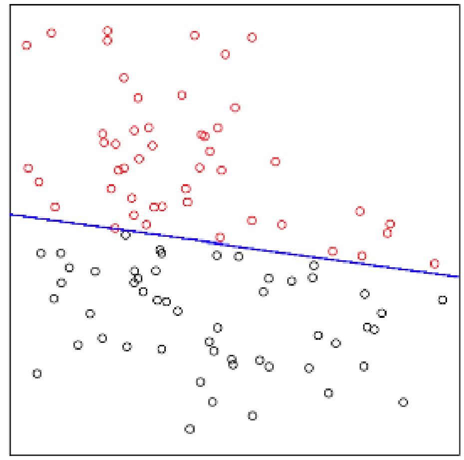


Figure 29: Linear decision boundary in $\mathbb{R}^2$ separating two classes.

Figures 28 and 29 illustrate that a hyperplane in $\mathbb{R}^p$ is an affine subspace of dimension $p - 1$. Classification decisions depend solely on the sign of $b_0 + b^\top x$—that is, on which side of the boundary each observation lies. We next show how *Linear Discriminant Analysis (LDA)* uses probabilistic modelling to generate these linear decision boundaries.

Example $p = 3$

When we move to three dimensions, the hyperplane becomes a two-dimensional *plane* in $\mathbb{R}^3$:

$$(x_1, x_2, x_3) \in \mathbb{R}^3 : \; b_0 + b_1 x_1 + b_2 x_2 + b_3 x_3 = 0.$$

This plane divides the three-dimensional feature space into two half-spaces, one for each class.
Each additional predictor adds a new dimension to the feature space, while the decision boundary always remains one dimension lower.
The geometry is analogous to the regression surface shown earlier in Figure 7, except that here the plane represents a *separating boundary* rather than a fitted surface.

9.2 Linear Methods for Classification

A direct but naive approach to classification is to apply **linear regression** to a binary response, e.g. $G = \{0, 1\}$. By fitting a regression line to class labels treated as numerical values, we can obtain a simple linear decision boundary, but this is usually **not a good idea**, as linear regression is not designed for categorical outcomes.

Many other classification methods, however, also produce **linear decision boundaries**. These boundaries are linear **in the feature space**, corresponding to hyperplanes of the form

$$b_0 + b_1 x_1 + \cdots + b_p x_p = 0.$$

By augmenting the feature space with transformations such as squares, interactions, or higher powers (e.g. $x_1^2, x_1 x_2, x_2^3$), the model remains linear in this *expanded space*, but its projection onto the original predictor space becomes **non-linear**.
Thus, linear classification in a rich feature space can yield flexible, curved decision boundaries in the original variables.

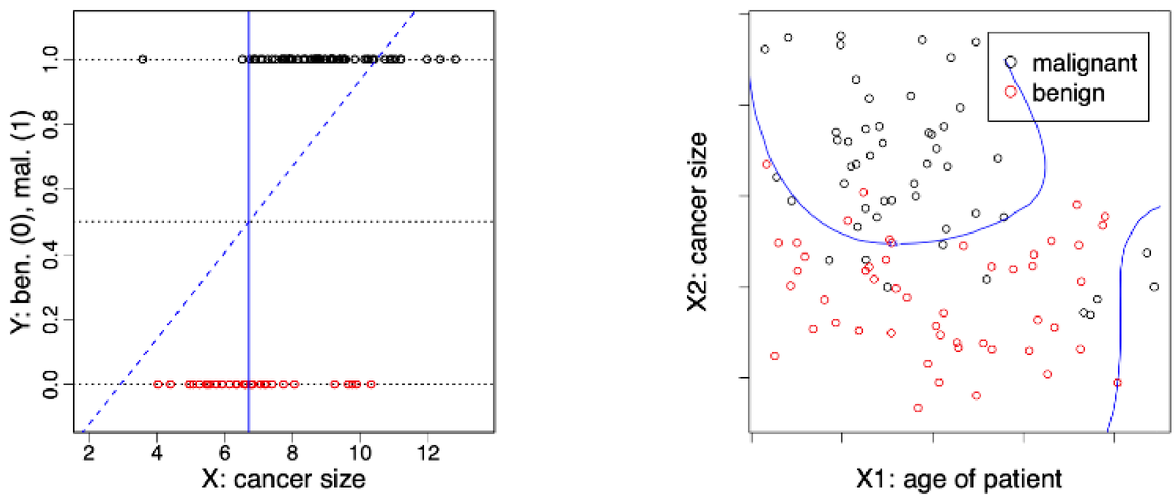


Figure 30: Linear and piecewise-linear decision boundaries in feature and original spaces.

In problems with more than two classes, linear methods produce **piecewise linear decision boundaries**, i.e. separate hyperplanes that divide the feature space into multiple regions, one for each class.

9.3 Linear Discriminant Analysis

9.3.1 Setup

We aim to classify an observation $Y \in \mathcal{G} = \{1, \ldots, q\}$ given predictors $X = (X_1, \ldots, X_p)$.
The idea is to model the **class-conditional densities** of X given the class $Y = j$, denoted $g_j(x)$, and the **marginal class probabilities** $\pi_j = \Pr(Y = j)$ for $j = 1, \ldots, q$.
Using Bayes' theorem, we can express the posterior probability of belonging to class j as

$$\Pr(Y = j \mid X = x) = \frac{\Pr(X = x \mid Y = j)\Pr(Y = j)}{\Pr(X = x)} \propto g_j(x)\pi_j.$$

This proportionality means that, for classification, we only need to compare the quantities $g_j(x)\pi_j$ across classes.

9.3.2 Modelling Assumptions

In **Linear Discriminant Analysis (LDA)**:

1. Each class-conditional density $g_j(x)$ is assumed to follow a **multivariate normal distribution**:

$$g_j(x) = \frac{1}{(2\pi)^{p/2}(\det \Sigma_j)^{1/2}} \exp\left\{-\frac{1}{2}(x-\mu_j)^\top \Sigma_j^{-1}(x-\mu_j)\right\}.$$

2. All classes share the **same covariance matrix**, that is, $\Sigma_j = \Sigma$ for all j.

Under these assumptions, the decision boundary between two classes is **linear** in x, because the logarithm of the posterior probabilities yields a linear function of x.

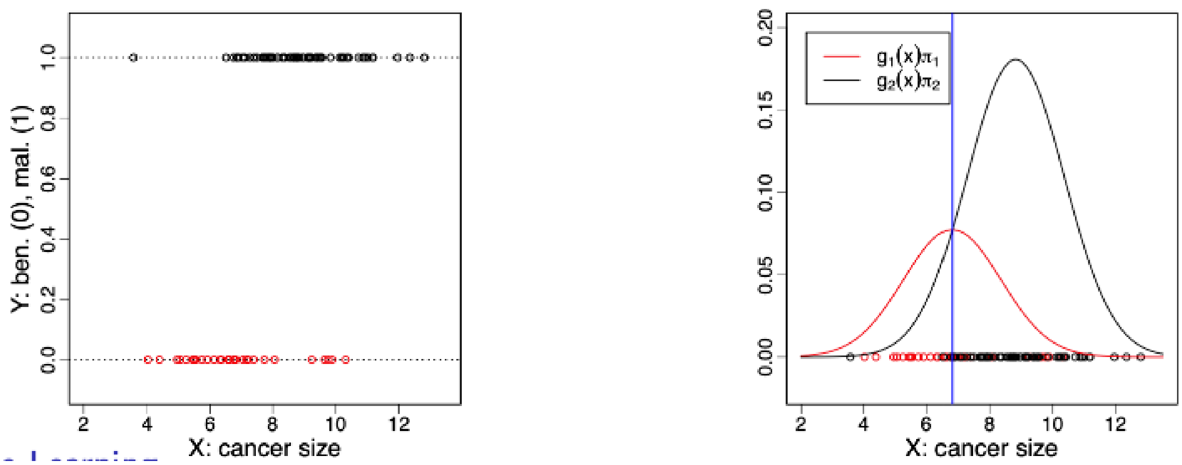


Figure 31: Class-conditional normal densities $g_1(x)\pi_1$ and $g_2(x)\pi_2$ under equal covariance assumption.

Figure 31 shows a one-dimensional example with two classes. Each class distribution is modelled by a Gaussian density with its own mean but identical variance.
The intersection point between the scaled densities $g_1(x)\pi_1$ and $g_2(x)\pi_2$ corresponds to the **decision boundary**, i.e., the value of x where the posterior probabilities are equal.

9.4 Estimation

In practice, the parameters of LDA are estimated from a training sample $(y_1, x_1), \ldots, (y_n, x_n)$.
The model includes:

- the **prior class probabilities** $\pi_1, \dots, \pi_q \in (0, 1)$
- the **class mean vectors** $\mu_1, \dots, \mu_q \in \mathbb{R}^p$
- the **common covariance matrix** $\Sigma \in \mathbb{R}^{p \times p}$

We estimate these by the **maximum likelihood estimators**:

$$\hat{\pi}_j = \frac{1}{n} \sum_{i=1}^{n} 1\{y_i = j\}, \quad \hat{\mu}_j = \frac{\sum_{i=1}^{n} x_i 1\{y_i = j\}}{\sum_{i=1}^{n} 1\{y_i = j\}},$$

$$\hat{\Sigma} = \frac{\sum_{j=1}^{q} \sum_{i=1}^{n} (x_i - \hat{\mu}_j)(\mathbf{x}_i - \hat{\mu}_j)^\top 1\{y_i = j\}}{n - q}.$$

These estimators are simple closed forms, i.e., no iterative optimisation is needed — which makes LDA **computationally very fast**.
Unlike linear regression, we do not use the numerical values of the y_i's, only their class membership.

9.5 Prediction

Given a new observation $x_0 \in \mathbb{R}^p$, we predict its class by maximising the posterior probability estimated from the training data:

$$\hat{y}_0 = \arg \max_{j=1,\dots,q} \widehat{\Pr}(Y = j \mid X = x_0) = \arg \max_{j=1,\dots,q} \log \widehat{\Pr}(Y = j \mid X = x_0).$$

From the Gaussian model, we have:

$$\log \widehat{\Pr}(Y = j \mid X = x_0) = \log \hat{g}_j(x_0) + \log \hat{\pi}_j + \text{constant},$$

which expands to

$$-\tfrac{1}{2}(x_0 - \hat{\mu}_j)^\top \hat{\Sigma}^{-1}(x_0 - \hat{\mu}_j) + \log \hat{\pi}_j + \text{constant}.$$

Ignoring terms independent of j, this simplifies to a **linear discriminant function**:

$$\delta_j(x_0) = x_0^\top \hat{\Sigma}^{-1} \hat{\mu}_j - \tfrac{1}{2} \hat{\mu}_j^\top \hat{\Sigma}^{-1} \hat{\mu}_j + \log \hat{\pi}_j.$$

We assign x_0 to the class with the largest $\delta_j(x_0)$.

9.5.1 Geometric interpretation

LDA classifies points based on their **Mahalanobis distance** to each class mean, adjusted by the prior probabilities:

$$d_{\Sigma^{-1}}(x, \mu_j) = (x - \mu_j)^\top \Sigma^{-1} (x - \mu_j).$$

In geometric terms, the decision boundary lies at points where two discriminant functions are equal, i.e. where the posterior probabilities of the classes coincide.
These boundaries are **linear** because Σ is common to all classes.

9.6 Decision Boundaries

The **decision boundary** separating two classes $j, m \in \{1, \dots, q\}$ is the set of all $x \in \mathbb{R}^p$ such that

$$\Pr(Y = j \mid X = x) = \Pr(Y = m \mid X = x).$$

This is equivalent to

$$\log \Pr(Y = j \mid X = x) - \log \Pr(Y = m \mid X = x) = 0.$$

Replacing these probabilities by their Gaussian expressions and the prior class probabilities gives the **decision boundary equation**:

$$\log \frac{\pi_j}{\pi_m} - \tfrac{1}{2}(\mu_j + \mu_m)^\top \Sigma^{-1} (\mu_j - \mu_m) + x^\top \Sigma^{-1} (\mu_j - \mu_m) = 0.$$

This is a **linear equation in x**.
Hence, the boundaries separating any two classes j and m are **hyperplanes** in $\mathbb{R}^p$ of dimension $p - 1$.
For $q > 2$ classes, these hyperplanes together form a **piecewise linear decision boundary**.

9.7 Iris Data Set

To illustrate LDA in practice, we use the classic **Iris data set**, which contains 150 observations of three flower species: *setosa*, *versicolor*, and *virginica*. Each plant is described by $p = 4$ predictors (sepal length, sepal width, petal length, and petal width), and the response variable $G = \{\textit{setosa}, \textit{versicolor}, \textit{virginica}\}$ indicates the species.

Here, we restrict the analysis to only **two predictors** (sepal length and sepal width) to visualise the classification boundary in two dimensions.
After fitting LDA to these two features, we obtain three **piecewise linear decision boundaries**, each separating one species from the others.

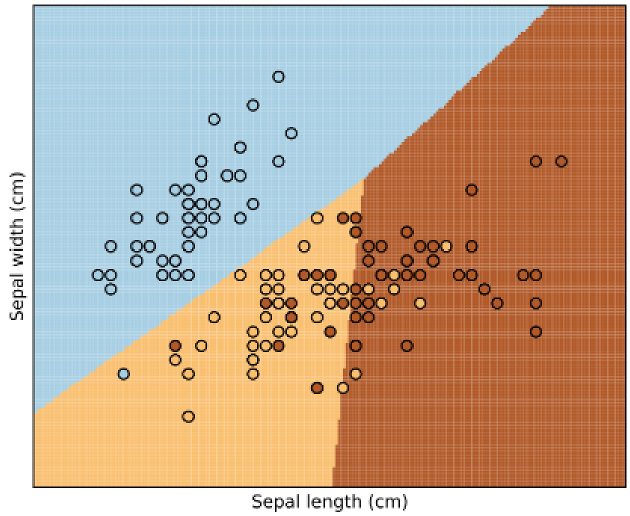


Figure 32: LDA decision regions for the Iris data using two predictors (sepal length and sepal width).

In Figure 32, each coloured region represents the set of points assigned to one of the three classes. The black circles correspond to the training observations. As expected from LDA, the boundaries between classes are straight lines, i.e., consistent with the linear decision boundaries derived theoretically. Despite using only two features, LDA achieves clear separation between *setosa* and the other two species, while *versicolor* and *virginica* overlap slightly, reflecting their natural similarity in the data.

9.8 Extensions of LDA

There are many ways to extend LDA. The Bayes' formulation

$$\Pr(Y = j \mid X = x) = \frac{\Pr(X = x \mid Y = j)\Pr(Y = j)}{\Pr(X = x)} = \frac{g_j(x)\pi_j}{\sum_{k=1}^{q} g_k(x)\pi_k}$$

can be used with different forms of the class-conditional densities $g_j(x)$.

Quadratic Discriminant Analysis (QDA) models each g_j as a multivariate normal distribution but allows for **different covariance matrices** Σ_j, leading to **quadratic decision boundaries**.

Mixture models use combinations of Gaussian components to represent more complex class-conditional densities $g_j(x)$, providing greater flexibility when class distributions are not unimodal.

Naive Bayes assumes **conditional independence** between predictors given the class. In this case,

$$g_j(x) = \prod_{\ell=1}^{p} g_{j,\ell}(x_\ell),$$

where each feature contributes independently to the overall class likelihood.

When some predictors are **categorical**, such as X_1, we can write

$$g_j(x) = \Pr(X_1 = x_1 \mid Y = j) g_j(x_2, \dots, x_p \mid X_1 = x_1, Y = j),$$

where $g_j(x_2, \dots, x_p \mid X_1 = x_1, Y = j)$ represents the density of the remaining continuous variables.

Finally, **non-parametric approaches** estimate $g_j(x)$ directly from data without assuming any specific parametric form, offering the most flexibility at the cost of greater computational effort.

9.9 Iris Data Set: QDA

We again use the **Iris data set** to compare QDA with LDA. The same three classes (*setosa*, *versicolor*, and *virginica*) are modelled using only two predictors: sepal length and sepal width.

Fitting QDA allows each class to have its own covariance structure, producing **curved boundaries** that better adapt to class shapes in the feature space.

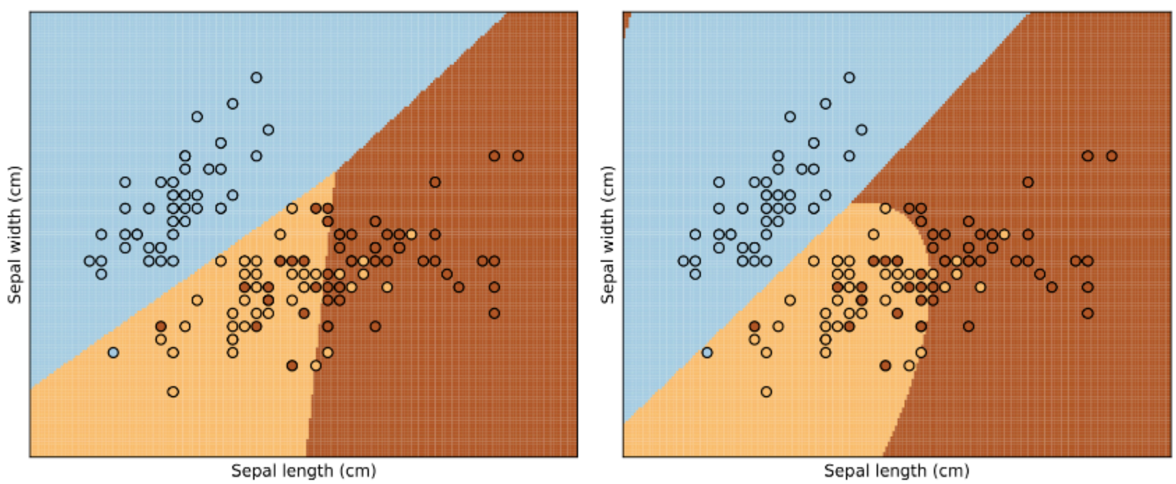


Figure 33: Comparison between LDA (left) and QDA (right) decision regions on the Iris data set.

Figure 33 compares the classification regions obtained with LDA (left) and QDA (right).
LDA yields **linear boundaries**, while QDA produces **quadratic, more flexible** ones that better capture non-linear class separation.
Although QDA can fit the data more closely, it also increases the risk of overfitting when the number of predictors or classes grows.

9.10 Comments on LDA

LDA is **easy** and **fast to compute**.
It also provides estimates of the **posterior class probabilities**:

$$\widehat{\Pr}(Y = j \mid X = x_0) = \frac{\hat{g}_j(x_0)\hat{\pi}_j}{\sum_{k=1}^{q} \hat{g}_k(x_0)\hat{\pi}_k}.$$

In practice, LDA usually performs well and can compete with more complex methods.
This is not because the normality assumption is realistic, but because a **linear decision boundary** is often the best the data can support.
Moreover, LDA is a **stable estimation method**, meaning that even if the assumed Gaussian model is imperfect, the resulting decision boundary remains robust.

We do not need highly accurate g_j to obtain a good posterior; Figure 34 illustrates this idea.

On the **left**, the two class-conditional densities $g_1(x)$ and $g_2(x)$ are clearly **non-Gaussian**, as we see, the blue density is even bimodal.
On the **right**, we see the corresponding **posterior probability** of belonging to the red class.
Despite the complex shape of the densities, the posterior remains **smooth and monotonic**, producing a clear linear-like separation.
This shows that even if the normality assumption fails, LDA still yields reliable classification boundaries.

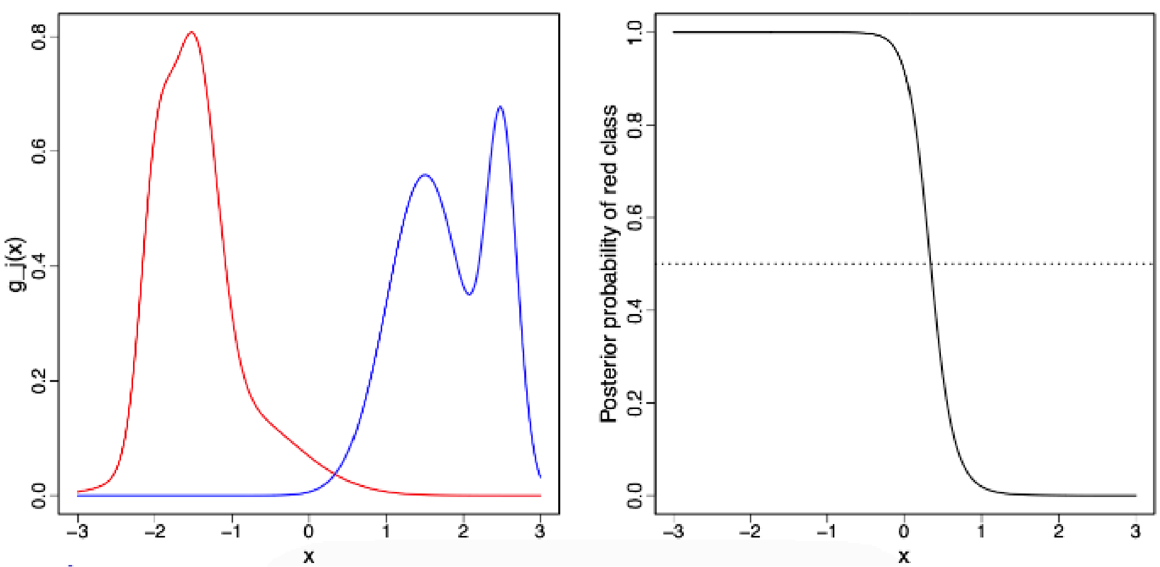


Figure 34: Even with non-Gaussian class densities (left), LDA yields smooth, stable posterior probabilities (right).

10 Logistic Regression

10.1 Logistic Regression

We begin with binary classification, where the response variable takes values $Y \in \{0, 1\}$. To simplify notation, we include the intercept directly in the feature vector by setting $X_1 = 1$.
With this convention, the linear predictor can be written compactly as $x^\top \beta$ instead of $\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$.

Logistic regression models the conditional probability of class 1 as a smooth function of this linear predictor:

$$\Pr(Y = 1 \mid X = x) = \sigma(x^\top \beta),$$

where the function σ is defined by

$$\sigma(t) = \frac{\exp(t)}{1 + \exp(t)}.$$

The function σ is called the *logistic function*. Its role is to map any real-valued input $x^\top \beta \in \mathbb{R}$ into the interval $(0, 1)$, making it suitable for modelling probabilities.
For large negative values of t, $\sigma(t)$ is close to 0; for large positive values, it is close to 1. Around $t = 0$, the function increases most rapidly, so observations near the decision boundary exhibit the greatest uncertainty.

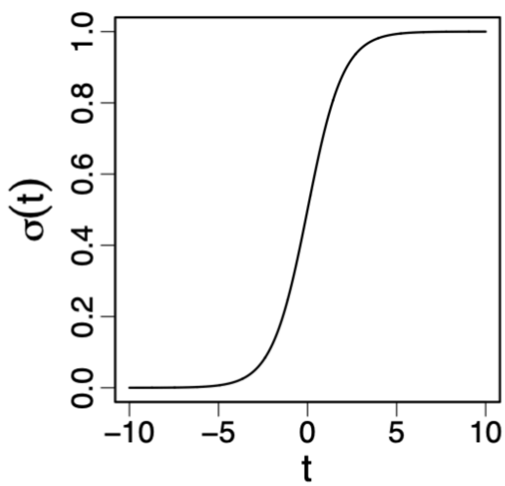


Figure 35: The logistic function $\sigma(t)$ maps real values to probabilities in $(0, 1)$.

As shown in Figure 35, logistic regression replaces a hard threshold by a smooth probabilistic transition, yielding well-defined class probabilities.

10.2 Logistic Regression: Interpretation

A key property of logistic regression follows directly from the model definition.
Since

$$\Pr(Y = 1 \mid X = x) = \sigma(x^\top \beta),$$

applying the inverse of the logistic function gives

$$\sigma^{-1}(\Pr(Y = 1 \mid X = x)) = x^\top \beta,$$

where

$$\sigma^{-1}(u) = \log\Big(\frac{u}{1-u}\Big)$$

is called the *logit function*.

This shows that logistic regression assumes linearity of the log-odds:

$$\log\left(\frac{\Pr(Y = 1 \mid X = x)}{\Pr(Y = 0 \mid X = x)}\right) = x^\top \beta.$$

Each coefficient β_j represents the change in the log-odds associated with a one-unit increase in the corresponding predictor, holding all other variables fixed.

An important consequence is that the decision boundary of logistic regression is linear. The boundary is defined by $\Pr(Y = 1 \mid X = x) = 0.5$, which is equivalent to $x^\top \beta = 0$.
Although the probability model is non-linear, the resulting class separation in feature space remains a hyperplane. This contrasts with linear regression applied to a binary response, which produces unbounded predictions and poorly calibrated probabilities.

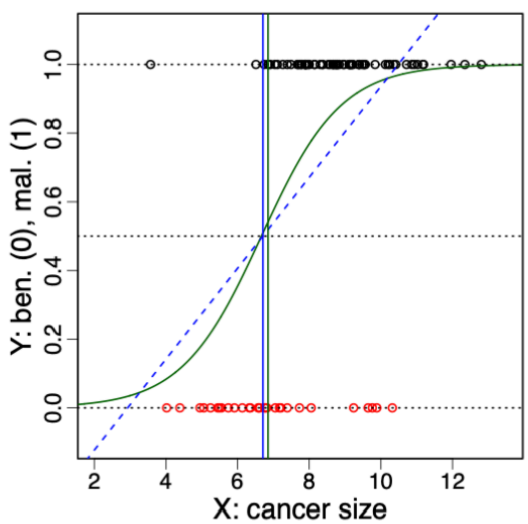


Figure 36: Logistic regression compared with linear regression for a binary response.

As illustrated in Figure 36, logistic regression combines a linear decision boundary with a probabilistic output that is appropriate for classification.

10.3 Logistic Regression: Estimation

The parameters β of logistic regression are estimated by maximum likelihood. Given a training sample $(x_1, y_1), \dots, (x_n, y_n)$ with $y_i \in \{0, 1\}$, the logistic model assumes that, conditionally on x_i, the response Y_i follows a Bernoulli distribution with success probability $\sigma(x_i^\top \beta)$.

Under this assumption, the conditional log-likelihood of the sample is

$$\ell(\beta) = \sum_{i=1}^{n} \log \Pr(Y = y_i \mid X = x_i) = \sum_{i=1}^{n} \left(y_i x_i^\top \beta - \log\left(1 + e^{x_i^\top \beta}\right) \right).$$

The maximum likelihood estimator $\hat{\beta}$ is defined as the value that maximises this log-likelihood:

$$\hat{\beta} = \arg\max_{\beta} \ell(\beta).$$

Unlike linear regression, this optimisation problem has no closed-form solution. The log-likelihood is concave, but its maximiser must be found numerically using iterative methods such as Newton–Raphson or gradient-based algorithms.

The figure illustrates why likelihood-based estimation is preferable to directly minimising classification error. The blue curve shows the negative log-likelihood as a smooth, well-behaved function of the parameter, with a clear global minimum. In contrast, the black curve represents the 0–1 classification error, which is irregular and non-smooth, making optimisation difficult.
Hence, logistic regression is estimated by maximising likelihood rather than minimising misclassification error.

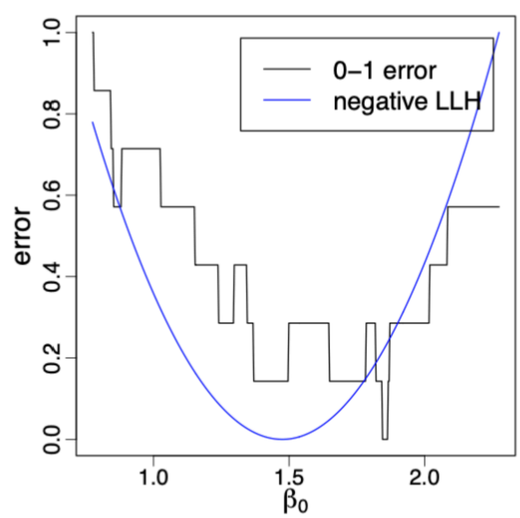


Figure 37: Negative log-likelihood and 0–1 classification error as functions of a regression parameter.

Figure 37 highlights that, even when classification error is minimised at a similar parameter value, likelihood-based estimation yields a smoother and more reliable optimisation landscape.

10.4 Logistic Regression: Prediction and Perfect Separation

Given an estimated parameter vector $\hat{\beta}$ and a new input $x_0 \in \mathbb{R}^p$, logistic regression predicts the probability of class 1 as

$$\widehat{\Pr}(Y = 1 \mid X = x_0) = \sigma(x_0^\top \hat{\beta}).$$

To obtain a class label, we apply the Bayes classifier with equal misclassification costs:

$$\hat{y}_0 = \begin{cases} 1 & \text{if } \sigma(x_0^\top \hat{\beta}) > 1/2, \\ 0 & \text{otherwise.} \end{cases}$$

Equivalently, since σ is increasing, this rule can be written as $\hat{y}_0 = 1$ if $x_0^\top \hat{\beta} > 0$ and $\hat{y}_0 = 0$ otherwise.

A practical caveat is that the maximum likelihood estimator $\hat{\beta}$ may fail to exist when the classes are perfectly separable. If there is a hyperplane that classifies all training points correctly, the likelihood can be increased indefinitely by making the logistic curve steeper and steeper. In the one-dimensional setting this corresponds to letting a coefficient (here β_1) grow without bound, $\beta_1 \to \infty$, so that $\sigma(x^\top \beta)$ approaches a step function.

This phenomenon explains why numerical optimisation may appear to diverge in perfectly separable data sets: the fitted probabilities push towards 0 and 1, and the parameter norm increases rather than stabilising.

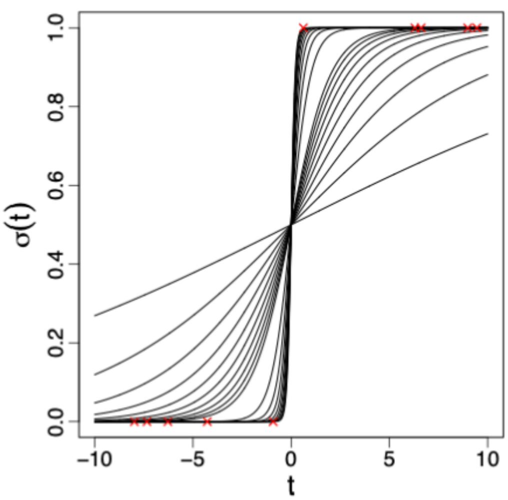


Figure 38: Perfect separation: increasing the slope makes the logistic curve approach a step, preventing a finite MLE.

Figure 38 shows that as the slope increases, the transition around $1/2$ becomes arbitrarily sharp while still classifying the red observations perfectly, so the likelihood keeps improving without reaching a finite optimum.

10.5 Multiclass Logistic Regression (Multinomial Regression)

Logistic regression extends naturally to the multiclass setting, where the response variable takes values $Y \in \{1, \ldots, q\}$.

Instead of modelling a single probability, we directly model the conditional probabilities of each class given the input x.

To ensure that all class probabilities are positive and sum to one, the model is specified using a multinomial (or softmax-type) formulation.
For classes $j = 1, \dots, q-1$, we define

$$\Pr(Y = j \mid X = x) = \frac{\exp(x^\top \beta^{(j)})}{1 + \sum_{\ell=1}^{q-1} \exp(x^\top \beta^{(\ell)})},$$

and for the remaining class q we set

$$\Pr(Y = q \mid X = x) = \frac{1}{1 + \sum_{\ell=1}^{q-1} \exp(x^\top \beta^{(\ell)})}.$$

Here, $\beta^{(1)}, \dots, \beta^{(q-1)} \in \mathbb{R}^p$ are parameter vectors to be estimated. Only $q-1$ parameter vectors are needed for q classes because the probabilities must sum to one; the last class acts as a reference class.

As in the binary case, classification is performed by assigning x to the class with the largest posterior probability $\Pr(Y = j \mid X = x)$.

10.5.1 Estimation

From a probabilistic perspective, the conditional distribution $Y \mid X = x$ is multinomial.
Given a training sample $(x_1, y_1), \dots, (x_n, y_n)$, the parameters are estimated by maximising the log-likelihood

$$\ell(\beta) = \sum_{i=1}^{n} \log \Pr(Y = y_i \mid X = x_i).$$

As in binary logistic regression, this optimisation problem has no closed-form solution and must be solved numerically using iterative methods.
Nevertheless, the objective function remains smooth and concave, which makes maximum likelihood estimation tractable in practice.

10.6 Iris Data Set: Multinomial Logistic Regression

We illustrate multiclass classification using the **Iris data set**, which contains 150 observations from three species: *setosa*, *versicolor*, and *virginica*. Each plant is described by $p = 4$ predictors, but for visualisation we restrict attention to the first two: sepal length and sepal width.
With three classes, multinomial logistic regression models the full set of posterior probabilities $\Pr(Y = j \mid X = x)$ and predicts the class with the largest posterior probability.

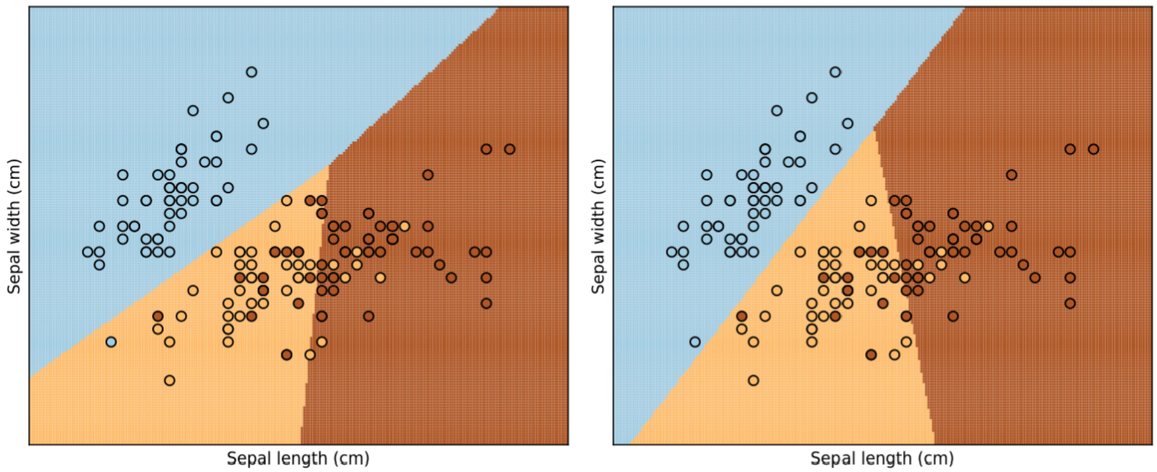


Figure 39: Decision regions on the Iris data set (sepal length and sepal width). Left: LDA. Right: multinomial logistic regression.

In Figure 39, each coloured region corresponds to the predicted class $\hat{y}(x) = \arg\max_j \hat{p}_j(x)$, and the points are the observed flowers.
The boundaries are the locations where two classes have the same (largest) posterior probability, for example $\hat{p}_a(x) = \hat{p}_b(x)$.

Because both LDA and multinomial logistic regression are **linear classifiers**, their decision boundaries in this two-dimensional feature space are **straight lines**. In the multiclass setting, several straight boundaries combine to create a **piecewise linear partition** of the plane.

The two panels look very similar, and that is expected in this simple two-predictor example:
- *setosa* is cleanly separated in the upper-left part of the plane, so both methods agree almost everywhere for that class.
- The main ambiguity lies in the overlap between *versicolor* and *virginica*, so any differences between methods appear only in that overlap zone.

The differences between the two fits are therefore subtle and local: they mainly affect (i) the precise position of the boundary that separates *versicolor* from *virginica* through the central cloud of points, and (ii) the location of the "junction" where all three regions meet.
In other words, the global structure is the same (linear separation with three regions), but a few borderline observations in the overlap area can be assigned differently depending on whether we fit a generative model (LDA) or a discriminative model (multinomial logistic regression).

This illustrates an important exam point: even when two methods produce linear decision boundaries, they need not yield identical boundaries on the same data because they rely on different modelling assumptions and estimation criteria.

10.7 Comments

Logistic regression is defined by the choice of a link function that maps the linear predictor $x^\top\beta$ to a probability. While the logistic function is the most common choice, other link functions can be used.

One important alternative is probit regression, which models the probability of class 1 as

$$\Pr(Y = 1 \mid X = x) = \Phi(x^\top\beta),$$

where Φ denotes the cumulative distribution function of a standard normal random variable.
In practice, logistic regression is usually preferred because it is more robust to outliers, while yielding very similar classification boundaries.

More generally, logistic regression belongs to the broader family of generalised linear models (GLMs). In this framework, we model the conditional distribution of $Y \mid X$ through:

- a linear predictor $x^\top\beta$,
- a link function connecting this predictor to the mean response,
- and a likelihood appropriate for the response type (Bernoulli for binary outcomes).

10.7.1 Logistic Regression vs. LDA

Logistic regression and Linear Discriminant Analysis share an important structural property: for both models, the log-odds between any two classes are linear functions of x.

The key difference lies in how the models are estimated.

- LDA is a generative model: it specifies a joint model for (Y, X) by assuming Gaussian class-conditional densities for $X \mid Y$.
- Logistic regression, by contrast, is a discriminative model: it models only the conditional distribution $Y \mid X$, without making assumptions about the distribution of X itself.

As a result:

- when the Gaussian assumptions underlying LDA are reasonable, LDA can outperform logistic regression;
- when these assumptions are clearly violated, logistic regression typically performs better.
- In many practical situations, however, the two methods exhibit very similar performance, since both ultimately produce linear decision boundaries.

11 Image Recognition

11.1 Image Recognition: introduction

Image recognition aims to identify or classify objects within digital images by learning the relationship between pixel intensities and their corresponding labels. Each image is represented as a grid of pixels, and each pixel encodes a greyscale intensity (often rescaled or normalised for modelling). In this example, every handwritten digit image consists of $16 \times 16 = 256$ pixels, so we have $p = 256$ predictors per observation.

Each observation is represented by a feature vector $x_i = (x_{i1}, x_{i2}, \ldots, x_{i256}) \in \mathbb{R}^{256}$, and the response is the digit label $y_i \in \{0, \ldots, 9\}$. A classifier learns a mapping $\hat{f}(x_i)$ from pixel intensities to digit labels (e.g. multinomial logistic regression, LDA/QDA, neural networks, CNNs).

This approach converts visual information into numerical data, allowing statistical or machine learning methods to identify complex patterns of brightness and shape that define each digit. The more pixels that are used as predictors, the higher the potential resolution and detail the model can learn, although this also increases computational cost and the need for regularisation.

Exam takeaway: Images become standard supervised learning data by treating pixel intensities as predictors. As p grows, models can learn richer patterns, but overfitting risk increases and regularisation becomes important.

11.2 Two-class classification: one predictor

Figure 40 illustrates how an image is converted into numerical predictors. Each digit is a 16×16 greyscale image, so every pixel corresponds to one intensity value. In this two-class example (zero vs. one), the bottom row is shown in red to visually distinguish the class, while the blue cross marks the single pixel (Pixel 1) whose value is extracted as the predictor X_1 (normalised to $[-1, 1]$).

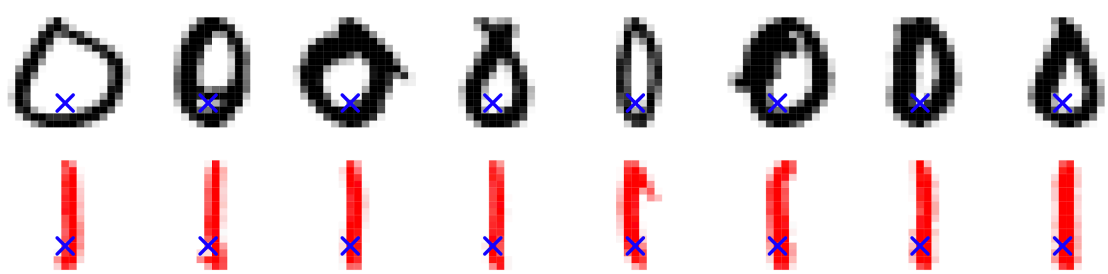


Figure 40: Handwritten zeros (top) and ones (bottom, shown in red for class illustration). The blue cross marks Pixel 1, whose intensity is used as a single predictor.

This highlighted pixel is taken from every image and used as a single predictor X_1. Darker pixels correspond to larger values of X_1, while lighter pixels correspond to smaller values. Although one pixel captures only a tiny fraction of the image information, it can still carry some discriminative signal between the two classes.

Two-class example (one predictor): We classify only two digits (zero and one) using the value at Pixel 1 as predictor X_1. Logistic regression yields a one-dimensional decision rule (a threshold on X_1).

Setup:

- Predictor: X_1 = value at Pixel 1, $X_1 \in [-1, 1]$.
- Class: $Y \in \{0, 1\} = \{\text{zero}, \text{one}\}$.
- Classifier:

$$\hat{G}(x) = \begin{cases} \text{zero} & \text{if } X_1 < 0.45, \\ \text{one} & \text{otherwise.} \end{cases}$$

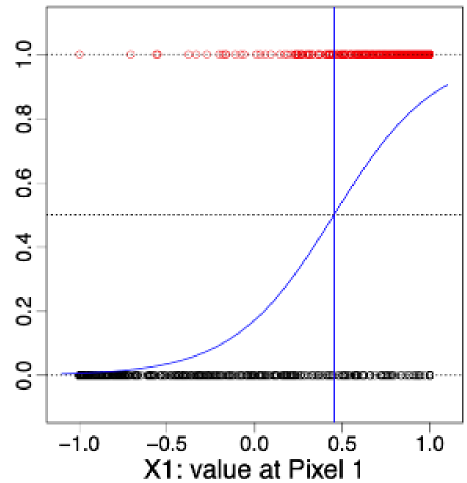


Figure 41: Two-class classification using a single pixel as predictor

This shows that even a single pixel may carry some information about class membership, but meaningful recognition requires combining all 256 pixel predictors to capture the full spatial structure of each digit. As seen in Figure 41, logistic regression produces a smooth probability curve separating the two classes based on Pixel 1 intensity.

Exam takeaway: With one predictor, logistic regression produces a threshold rule in 1D. This is useful for intuition but insufficient for accurate digit recognition.

11.3 Two-class classification: two predictors

Figure 42 shows the extension to two pixel predictors. The blue crosses mark two fixed pixel locations whose intensities are extracted to form the predictor vector $(X_1, X_2) \in [-1, 1]^2$.

The red colouring is used only to visually distinguish the classes; the images should still be interpreted as greyscale. The two pixels are treated simply as numerical predictors, without using spatial neighbourhood information. Considering two pixels jointly already improves class separation compared to a single pixel.

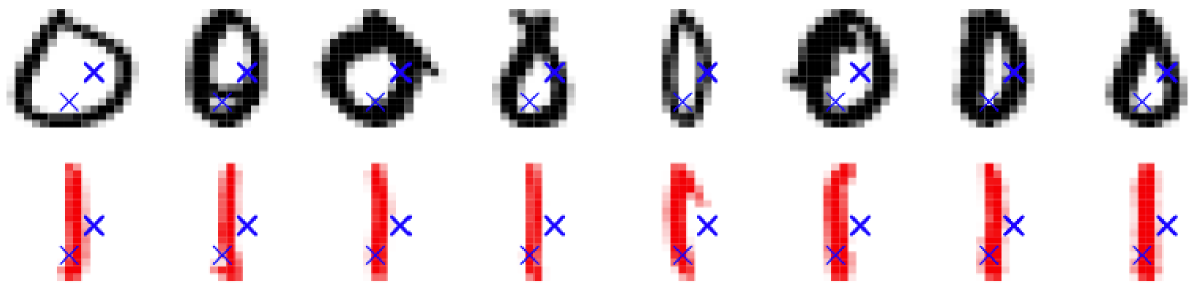


Figure 42: Two-predictor example for handwritten digits. The blue crosses mark the two pixel locations used as predictors; red colouring distinguishes the two classes (zero and one).

Two predictors: Increasing the number of predictors from one to two expands the predictor space from one to two dimensions. Logistic regression now uses the pair (X_1, X_2) to estimate a decision boundary in the two-dimensional pixel space. The setup is summarised as follows:

- $p = 2$: Predictors are X_1 = "value at Pixel 1" $\in [-1, 1]$, X_2 = "value at Pixel 2" $\in [-1, 1]$.
- The class is $Y \in \{0, 1\} = \{\text{zero}, \text{one}\}$.
- Logistic regression is applied using only X_1 and X_2.

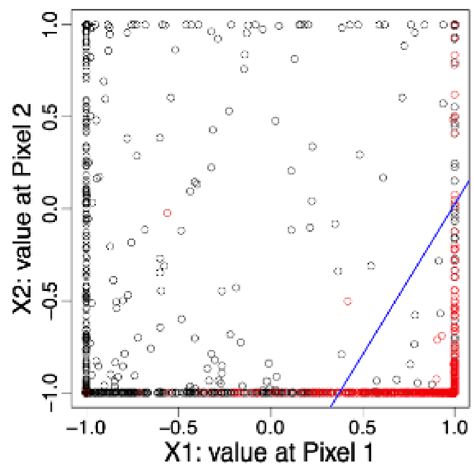
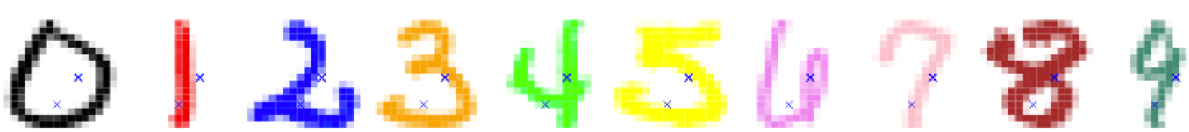


Figure 43: Two-class classification using two pixel predictors

Exam takeaway: Increasing predictors increases model flexibility; in 2D, logistic regression yields a linear separating line.

11.4 10-class classification: two predictors

We now classify all ten digits $Y \in \{0, \dots, 9\}$ using only two predictors X_1 and X_2. This illustrates the limitations of low-dimensional feature representations.



The blue crosses mark the selected pixel locations. With only two predictors, class overlap is substantial and accurate classification is impossible.

Ten classes with two predictors: Extending from binary to multiclass classification requires the model to output a probability distribution across ten possible classes. The logistic regression generalises to a *multinomial regression* model. The main elements are:

- $p = 2$: Predictors are X_1 = "value at Pixel 1" $\in [-1, 1]$, X_2 = "value at Pixel 2" $\in [-1, 1]$.
- The class is $Y \in \{0, 1, \dots, 9\} = \{\text{zero, one, } \dots \text{, nine}\}$.
- We apply *multinomial regression* based only on X_1 and X_2, though with only two pixels this remains a very difficult task.

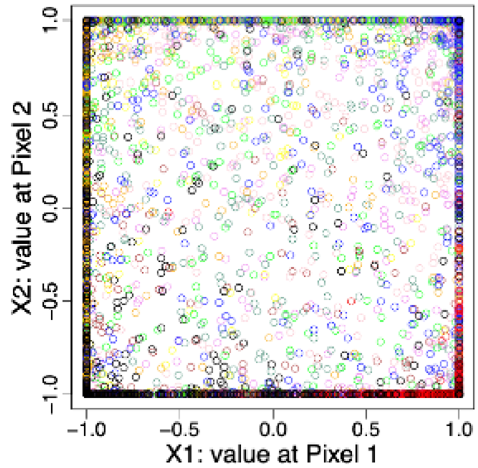


Figure 44: Ten-class classification using two pixel predictors

As seen in Figure 44, the classifier attempts to separate the ten classes within the (X_1, X_2) plane, but the overlap between classes is substantial. With only two predictors, the model cannot capture the high-dimensional structure of the digits, showing why richer feature sets are essential for effective image recognition. Luckily, we have $p = 256$ predictors (pixel values) to perform this task.

Exam takeaway: Severe class overlap in low dimensions motivates using all $p = 256$ predictors.

11.5 Digit classification (all $p = 256$ pixels)

Each image is now a point in $\mathbb{R}^{256}$, and the task is to predict the digit label.

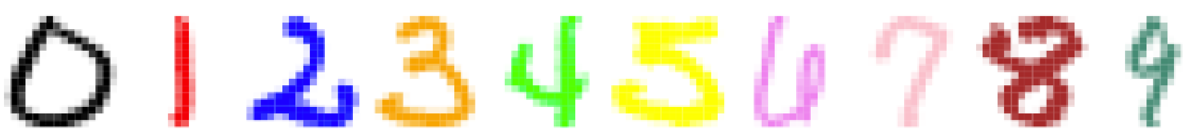


Figure 45

Training and test data:

- Training data $(x_1, y_1), \dots, (x_n, y_n)$, $n = 7290$, are used for model fitting and selection.
  - Predictors x_i = i-th image $\in \mathbb{R}^p$ are the greyscale pixel values.
  - Outcomes y_i = i-th digit $\in \{\text{zero}, \text{one}, \dots, \text{nine}\}$.
- Test data: $(\hat{x}_1, \hat{y}_1), \dots, (\hat{x}_m, \hat{y}_m)$, $m = 2006$, are not used for fitting or selection, only for evaluating performance.

Training and test errors (%)

Method	Training	Test
linreg	7.6	13.1
LDA1	6.2	11.5
LDA2	3.9	10.2
QDA	1.8	13.5
log	0.01	11.1

Models compared:

- **linreg:** direct linear regression.
- **LDA1:** Linear Discriminant Analysis (LDA) based on the values x_i.
- **LDA2:** LDA based on x_i and x_i^2.
- **QDA:** Quadratic Discriminant Analysis with a distinct covariance matrix for each class.
- **log:** logistic (multinomial) regression.

QDA yields the lowest training error but performs worst on the test data due to high variance and overfitting. In contrast, multinomial logistic regression achieves a balanced trade-off between bias and variance, maintaining strong generalisation performance across unseen digits.

Key observation: Low training error does not imply good generalisation.

11.6 Digit classification: visualisation of LDA1

Figure 46: Estimated class centres from LDA1

Visualising class centres: The figure on the left shows the estimated class means $\hat{\mu}_1, \dots, \hat{\mu}_q$ obtained from the LDA1 model. Each image corresponds to the *average digit* for one class, formed by taking the mean pixel intensity across all training examples belonging to that class.

This visualisation provides an intuitive impression of the class centroids, illustrating how the model perceives the typical form of each handwritten digit. In other words, it shows how an "average" American writer tends to shape the digits 0–9.

11.7 Example: digit classification

The table in Figure 47 presents a selection of misclassified test observations produced by the LDA1 model. Each row corresponds to a single handwritten digit from the test set, with the true class label reported in parentheses. The columns display the predicted class probabilities (in percent) assigned to each digit $k \in \{0, \dots, 9\}$.

Rather than concentrating probability mass entirely on the true class, the model often spreads it across several plausible alternatives. For example, a digit "6" may receive substantial probability for class "8" when the loop closure is faint, while a "9" can be confused with a "4" if the vertical stroke is unusually short. Such patterns illustrate how small variations in handwriting style can blur linear decision boundaries, revealing the inherent limitations of linear discriminant analysis in high-dimensional image classification problems.

Obs		0	1	2	3	4	5	6	7	8	9
	(6)	59	0	0	0	0	0	40	0	0	0
	(2)	94	0	0	6	0	0	0	0	0	0
	(3)	0	0	0	11	0	0	0	0	89	0
	(3)	0	0	0	1	0	99	0	0	0	0
	(8)	1	0	0	81	1	11	0	0	5	0
	(4)	0	0	0	0	15	0	0	0	0	85
	(9)	0	0	0	0	0	0	0	100	0	0
	(2)	100	0	0	0	0	0	0	0	0	0

Figure 47: Predicted class probabilities (in %) for selected misclassified test images under the LDA1 model.

11.8 Digit classification with regularisation

We can now refine our previous results on digit classification (see Figure 45). When estimating the multinomial logistic regression, we introduce two regularisation terms in the likelihood: a *ridge penalty* and a *lasso penalty*. These control model complexity by shrinking or selecting coefficients, helping to prevent overfitting. The optimal tuning parameters $\hat{\lambda}$ for both penalties are determined by 5-fold cross-validation, balancing the trade-off between training accuracy and test performance.

Models compared:

- **linreg:** direct linear regression.
- **LDA1:** Linear Discriminant Analysis (LDA) based on the values x_i.
- **LDA2:** LDA based on x_i and x_i^2.
- **QDA:** Quadratic Discriminant Analysis with a distinct covariance matrix for each class.
- **log:** logistic regression (multinomial).
- **log-ridge:** logistic regression (multinomial) with ridge penalty.
- **log-lasso:** logistic regression (multinomial) with lasso penalty.

Training and test errors (%)

Method	Training	Test
linreg	7.6	13.1
LDA1	6.2	11.5
LDA2	3.9	10.2
QDA	1.8	13.5
log	0.01	11.1
log-ridge	4.1	9.0
log-lasso	2.7	8.8

Regularisation improves generalisation by penalising overly complex models. Both ridge and lasso versions outperform unpenalised logistic regression, reducing the test error to below 9%. The lasso penalty performs slightly better, as it enforces sparsity by effectively selecting the most informative pixel predictors.

Why good training performance can be misleading

When the number of predictors p is large, flexible models can achieve extremely low training error, sometimes even perfect classification. In high-dimensional spaces, observations become easier to separate, and models can assign large importance to a small number of predictors that happen, by chance, to correlate with the response in the training data. As a result, methods such as QDA or unpenalised logistic regression may appear to perform very well during training.

However, these models typically have high variance. Small changes in the training data can lead to large changes in the estimated parameters, and the resulting decision boundaries depend strongly on noise and idiosyncrasies of the sample. Consequently, performance often degrades substantially on new, unseen data.

Why regularisation helps: regularisation reduces overfitting by explicitly restricting model complexity, discouraging overly flexible solutions unless strongly supported by the data.

Ridge regularisation (ℓ_2 penalty)

Ridge regularisation penalises large coefficients by adding a penalty proportional to the sum of squared coefficients. This has two important effects:

- Coefficient shrinkage: Large coefficients are discouraged, preventing the model from relying excessively on a small number of pixels.
- Stabilisation of the solution: When predictors are correlated (as is common for neighbouring pixels in images), ridge spreads influence more evenly across them. This makes the parameter estimates less sensitive to small perturbations in the data.

As a result, ridge regularisation reduces variance, produces more stable decision boundaries, and improves generalisation performance.

Lasso regularisation (ℓ_1 penalty)

Lasso regularisation also shrinks coefficients but additionally induces sparsity, setting many coefficients exactly to zero. By removing irrelevant or noisy predictors, lasso stabilises the solution by reducing the effective dimensionality of the model. This leads to simpler models that use only the most informative predictors, further reducing variance and often improving interpretability.

Why ridge and lasso outperform the unpenalised model

Both penalties enforce a bias–variance trade-off: training accuracy may decrease slightly, but test performance improves substantially. They do so by preventing extreme coefficients, reducing sensitivity to noise, and focusing the model on robust, repeatable patterns rather than accidental correlations.

12 Decision trees: regression

12.1 Tree-based methods

Tree-based methods stratify or segment the feature space recursively into simple regions, denoted $R_1, \ldots, R_J$. They can be applied to both classification and regression tasks.
For a given observation, the prediction corresponds to either:

- the *mean* of the training responses in the region (for regression), or
- the *most common class* of the training observations in the region (for classification).

The method divides the input space into rectangles that are successively refined, producing a piecewise-constant model with clear interpretability.

Tree-based approaches can be grouped as follows:

1. **Decision trees:** Single trees offering clear interpretability and simple decision rules.
2. **Bagging / Boosting:** Methods combining multiple trees to improve prediction accuracy by reducing variance or bias.
3. **Random forests:** Ensembles that decorrelate bagged trees, increasing stability and robustness.

As illustrated in Figure 48, the **left panel** displays a small regression tree built using the predictors *age* (X_1) and *cancer size* (X_2). Each internal node represents a decision rule that splits the data based on a threshold value (e.g. *size < 0.14*), dividing observations into smaller, more homogeneous groups. The **terminal nodes** (leaves) indicate the predicted response for each region. In this example, coloured as *red* and *black* to distinguish lower and higher predicted values.

The **right panel** shows how these splits translate into a partitioning of the two-dimensional predictor space. Each rectangular region corresponds to one terminal node from the tree, defining an area where all observations receive the same fitted value. The vertical and horizontal lines represent the thresholds

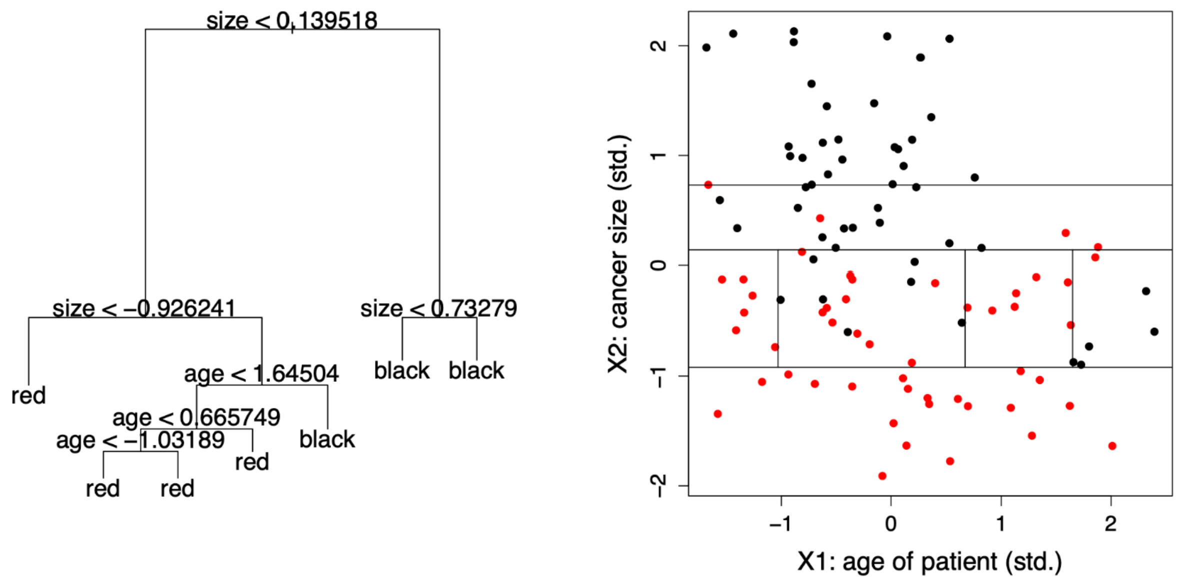


Figure 48

used in the tree's internal splits. Collectively, these partitions form a **piecewise-constant regression surface**, demonstrating how decision trees approximate complex, nonlinear relationships through a sequence of simple, interpretable divisions.

12.2 Example: Hitters data set

As introduced in Figure 20, the **Hitters** data set from the *ISLR* package contains the annual **Salary** of 263 baseball players and $p = 19$ predictors such as the number of **Years** played in the major leagues and the number of **Hits** during the previous season.

In what follows, we focus only on the variables *Years* and *Hits* to illustrate how a **regression tree** predicts the (log) salary of a player. This simplified example allows us to visualise how the tree partitions the feature space and introduces the main terminology of tree models.

12.3 Regression trees: terminology

We fit a regression tree to predict the log **Salary** of a baseball player using only **Years** and **Hits** as predictors from the **Hitters** data set.
Each split in the tree is called a **node**, and a terminal node is referred to as a **leaf**. The interior nodes connect to further branches, forming a hierarchical structure of decision rules.
The label $(X_j < t)$ indicates the *left-hand branch* from a split, while $(X_j \geq t)$ corresponds to the *right-hand branch*. Each leaf contains the mean predicted value for all observations that fall into that region.
An important advantage of trees is their strong **interpretability**: the step-by-step structure mirrors how humans make sequential decisions, making them highly intuitive in applied prediction problems.

Figure 49: Regression tree predicting log Salary using Years and Hits

12.4 Regression trees: segmentation of feature space

The regression tree partitions the feature space into three regions, each corresponding to one of the terminal nodes (leaves). Each region R_j contains the observations that satisfy the set of split conditions leading to that node.

- $R_1 = \{X \mid \text{Years} < 4.5\}$,
- $R_2 = \{X \mid \text{Years} \geq 4.5, \text{ Hits} < 117.5\}$,
- $R_3 = \{X \mid \text{Years} \geq 4.5, \text{ Hits} \geq 117.5\}$.

Interpretation:

- *Years* is the most influential variable in determining **Salary**: players with fewer than 4.5 years of experience tend to earn less.
- Among more experienced players, *Hits* becomes relevant, i.e., those with more than 117.5 hits typically have higher salaries.
- This simple tree structure highlights how regression trees create interpretable, piecewise-constant models of salary prediction.

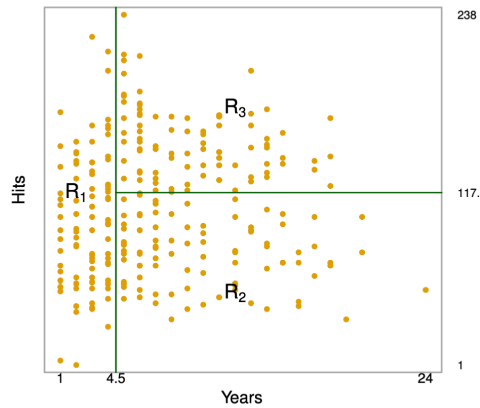


Figure 50: Segmentation of feature space by the regression tree

12.5 How to grow a regression tree?

Given a training data set $(x_1, y_1), \dots, (x_n, y_n)$ with $x_i \in \mathbb{R}^p$ and $y_i \in \mathbb{R}$, we seek to construct a regression tree that divides the predictor space into J non-overlapping regions $R_1, \dots, R_J$.

- The goal is to partition the predictor space $(X_1, \dots, X_p)$ into distinct regions that are as homogeneous as possible with respect to the response y_i.
- Although the regions could take any shape, regression trees restrict them to high-dimensional **rectangles (boxes)** for interpretability.
- For each observation falling into region R_j, we predict a constant value:

$$\hat{y}_{R_j} = \frac{1}{n_j} \sum_{i: x_i \in R_j} y_i$$

  where n_j is the number of training observations in region R_j.
- The prediction in each region is thus the **mean** of the training responses for that region.
- The optimal partitioning aims to find the regions $R_1, \dots, R_J$ that minimise the residual sum of squares (RSS):

$$\sum_{j=1}^{J} \sum_{i: x_i \in R_j} \left(y_i - \hat{y}_{R_j}\right)^2 2$$

  This recursive partitioning procedure builds the tree by successively splitting the predictor space to reduce the RSS at each step, producing a hierarchy of simple, interpretable decision rules.

12.6 Tree growing: greedy algorithm

It is computationally infeasible to test all possible ways of dividing the feature space into J regions. Instead, regression trees are built using a **top-down, greedy algorithm**, also known as **recursive binary splitting**.

- It is *top-down* because it starts at the root (the full training set) and **recursively splits** the predictor space into smaller regions.
- It is *greedy* because each split is chosen to produce the best immediate improvement in fit, without considering future splits.
- At each step, the algorithm searches over all predictors and all possible split points to find the combination (j, t) that minimises the Residual Sum of Squares (RSS).
- The process continues until a **stopping criterion** is reached. For example, when no region contains more than a minimum number of observations.

Formally, for each predictor X_j and potential cutpoint t, we define the half-planes:

$$R_1(j,t) = \{X \mid X_j < t\}, \qquad R_2(j,t) = \{X \mid X_j \geq t\}.$$

The algorithm finds the values (j^*, t^*) that minimise the sum of squared residuals:

$$\sum_{i: x_i \in R_1(j,t)} (y_i - \hat{y}_{R_1(j,t)})^2 \; + \sum_{i: x_i \in R_2(j,t)} (y_i - \hat{y}_{R_2(j,t)})^2.$$

The splitting process is then repeated **recursively** on the resulting regions, producing a hierarchical partitioning of the predictor space.

Once the regions $R_1, \dots, R_J$ are formed, predictions for a new observation x are made by assigning it to the appropriate region R_j and using the average response within that region:

$$\hat{f}(x) = \sum_{j=1}^{J} \hat{y}_{R_j} 1\{x \in R_j\}.$$

12.7 Regression tree: two-dimensional feature space

A regression tree partitions the predictor space into a series of non-overlapping rectangular regions. In a two-dimensional example with predictors X_1 and X_2, the recursive binary splits successively divide the plane into subregions $\{R_1, R_2, \dots, R_5\}$.

Each split corresponds to a threshold (cutpoint) on one of the variables, forming a hierarchical structure as shown below. The final model can be visualised both as a **tree diagram** and as a **piecewise-constant surface**, where each flat region represents the predicted response $\hat{y}_{R_j}$ for all observations in R_j.

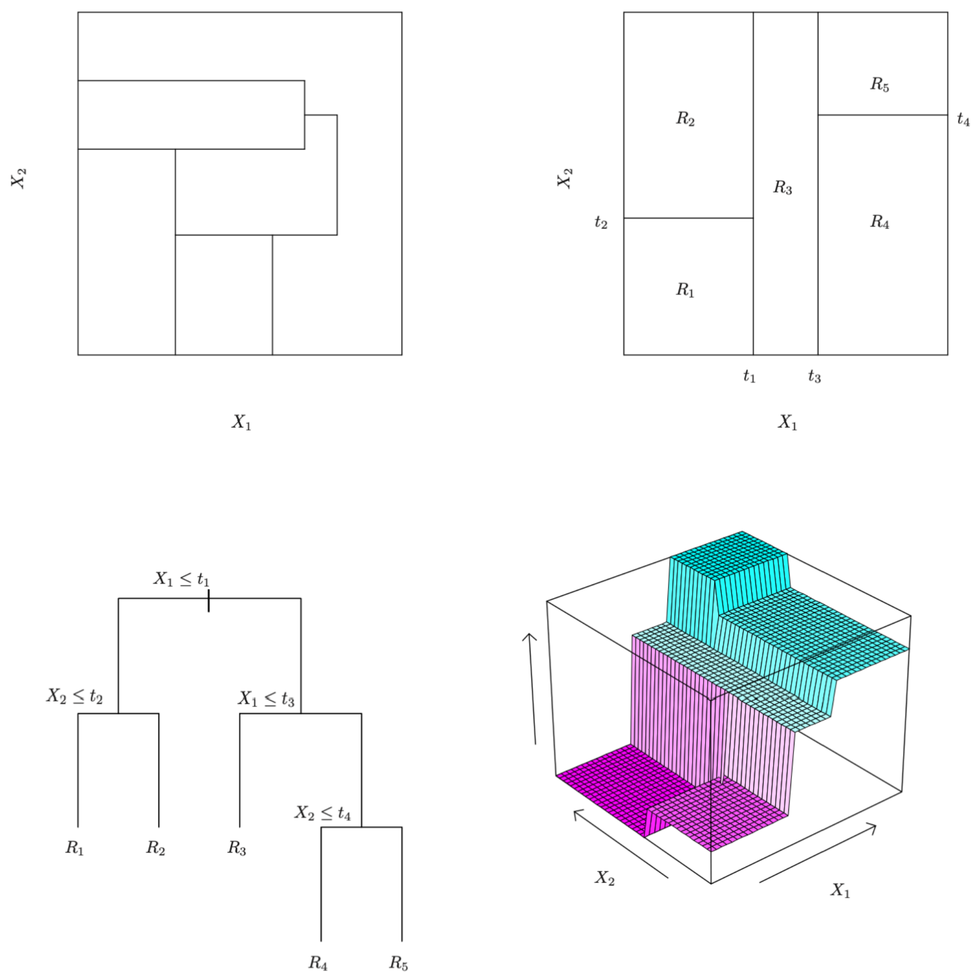


Figure 51: Two-dimensional feature space partition and corresponding regression tree

12.8 Cost-complexity pruning

When a tree is fully grown, it may overfit the training data. To improve generalisation, we prune it to a smaller subtree that balances fit and simplicity. This is achieved through **cost-complexity pruning** (or *weakest-link pruning*).

We consider a sequence of trees indexed by a tuning parameter $\alpha \geq 0$. For each α, the corresponding subtree $T_\alpha \subset T_0$ minimises:

$$\sum_{j=1}^{|T_\alpha|} \sum_{i: x_i \in R_j} (y_i - \hat{y}_{R_j})^2 + \alpha |T_\alpha|.$$

- The first term measures the **fit** of the tree (training error).
- The second term $\alpha|T_\alpha|$ penalises **model complexity**, where $|T_\alpha|$ is the number of terminal nodes.
- The tuning parameter α controls the degree of pruning:
  - Large $\alpha \rightarrow$ stronger penalty $\rightarrow$ simpler tree.
  - Small $\alpha \rightarrow$ weaker penalty $\rightarrow$ larger tree.
- α plays a similar role to λ in ridge or lasso regression.
- The optimal α is typically selected by **cross-validation**.

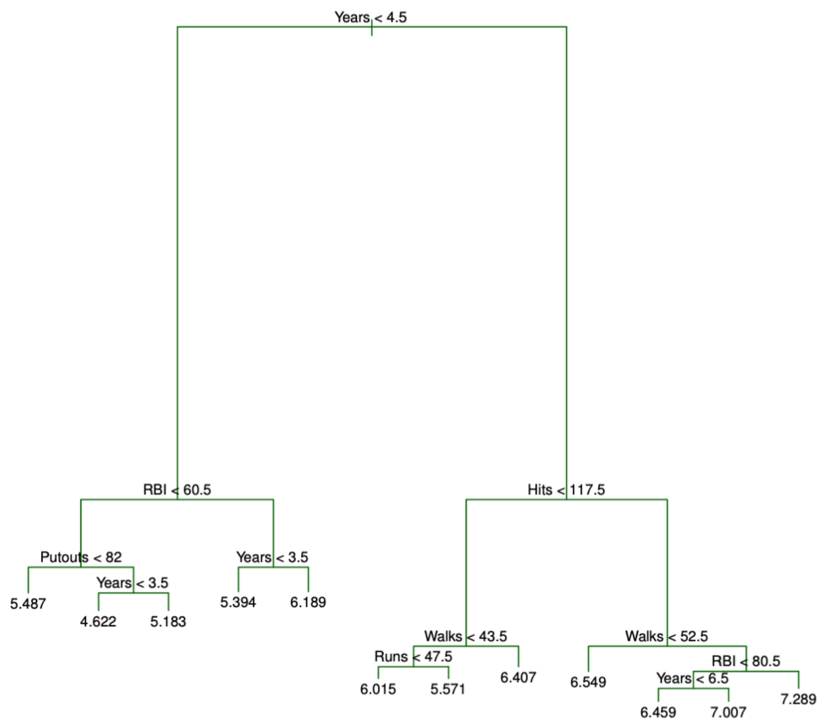


Figure 52: cost-complexity pruning applied to a regression tree

As we can see in Figure Figure 52, the **pruned tree** is a simplified version of the original full tree grown on the *Hitters* dataset.
Each internal node splits the data based on one predictor (e.g. `Years`, `Hits`, or `Walks`), while each **leaf** represents the **average log-salary** for players falling into that region.

- Players with **few years in the league** and **low hits** are predicted to have **lower salaries**.
- More experienced players with **higher hits** or **walks** tend to have **higher predicted salaries**.
- By pruning, redundant splits are removed, retaining only those that meaningfully reduce the prediction error.

The pruned tree thus strikes a **bias–variance balance**: it is simpler (less variance) but still captures the main salary patterns (low bias), leading to better test performance.

12.9 Regression example: Hitters data set

To evaluate the effect of pruning, the **Hitters** data set is split into 132 training observations and 131 test observations.
A large regression tree is first grown on the training data and then **pruned** by varying the penalty parameter α to obtain subtrees with different numbers of terminal nodes.

Six-fold cross-validation is performed on the 132 training observations to estimate the mean squared error (MSE) for each subtree.
The final test error is computed on the 131 test observations, which are **not used** for model fitting or selection.

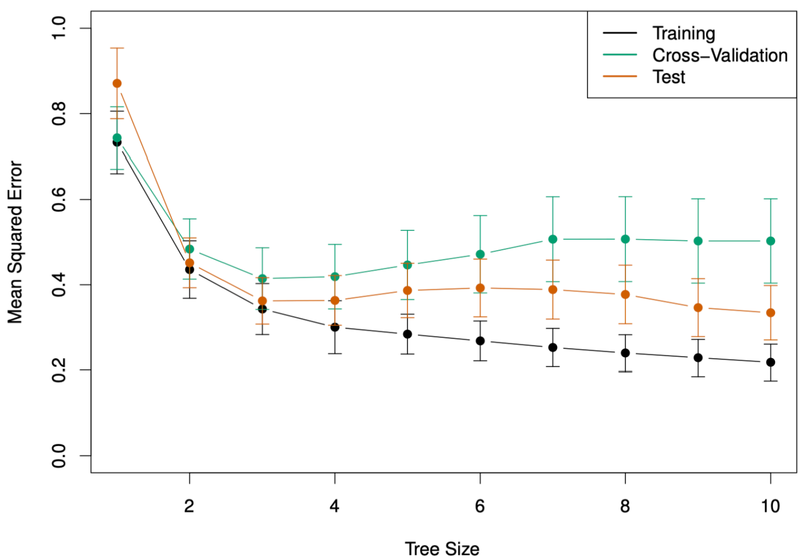


Figure 53: Cross-validation and test errors for trees of increasing size

As we see in Figure 53, **training error** (black) continues to decrease as the tree grows, while **cross-validation** (green) and **test** (orange) errors reach a minimum and then rise again, a clear sign of **overfitting** beyond the optimal tree size.

- Small trees underfit the data, yielding high bias and poor predictive accuracy.
- Large trees overfit, capturing noise and yielding high variance.
- The **optimal subtree** is chosen where the **cross-validation error is minimal**, achieving the best balance between bias and variance.

In the *Hitters* example, this corresponds to a **moderately sized tree** that retains the main predictors (Years, Hits, Walks) while discarding redundant splits, leading to improved generalisation performance.

13 Decision trees: classification

13.1 Classification trees

Classification trees are conceptually similar to regression trees, except that they predict a **qualitative** rather than a **quantitative** response. The response variable takes values in a finite set $\mathcal{G} = \{1, \dots, q\}$, representing different classes.

For each region R_j, the prediction is a constant $\hat{y}_{R_j}$ corresponding to the **most frequently occurring class** among the training observations that fall into that region. Formally, for class $g = 1, \dots, q$:

$$\hat{p}_{jg} = \frac{1}{n_j} \sum_{i: x_i \in R_j} 1\{y_i = g\}.$$

Hence, the classifier assigns each new observation to the region R_j that maximises $\hat{p}_{jg}$, producing the final prediction function:

$$\hat{G}(x) = \sum_{j=1}^{J} \hat{y}_{R_j} 1\{x \in R_j\}.$$

The construction principles (such as **tree growth**, **pruning**, and **subtree selection**) are the same as in regression trees. However, the **residual sum of squares (RSS)** cannot be used as a loss function for qualitative outcomes. Instead, classification trees rely on **measures of node impurity**, which quantify the heterogeneity of classes within a region.

Common impurity measures for a terminal node R_j include:

1. **Misclassification error:**
$$M = \frac{1}{n_j} \sum_{i: x_i \in R_j} 1\{y_i \neq \hat{y}_{R_j}\}.$$

2. **Gini index:**
$$G = \sum_{g=1}^{q} \hat{p}_{jg}(1 - \hat{p}_{jg}).$$

3. **Cross-entropy (or deviance):**
$$D = -\sum_{g=1}^{q} \hat{p}_{jg} \log \hat{p}_{jg}.$$

These impurity measures are minimised when the node is "pure", meaning that nearly all observations within it belong to the same class.

13.2 Node impurity measures

- In the case of two classes (see Figure Figure 54 below), the **misclassification error** is $1 - \max(p, 1-p)$ (black), the **Gini index** is $2p(1-p)$ (blue), and the **cross-entropy** is $-p \log p - (1-p)\log(1-p)$ (red).
- The misclassification impurity is non-differentiable and thus unsuitable for optimisation.
- The Gini index and the cross-entropy both **favour pure nodes**, where $p_{jg} \approx 0$ or $p_{jg} \approx 1$; they are numerically very similar.

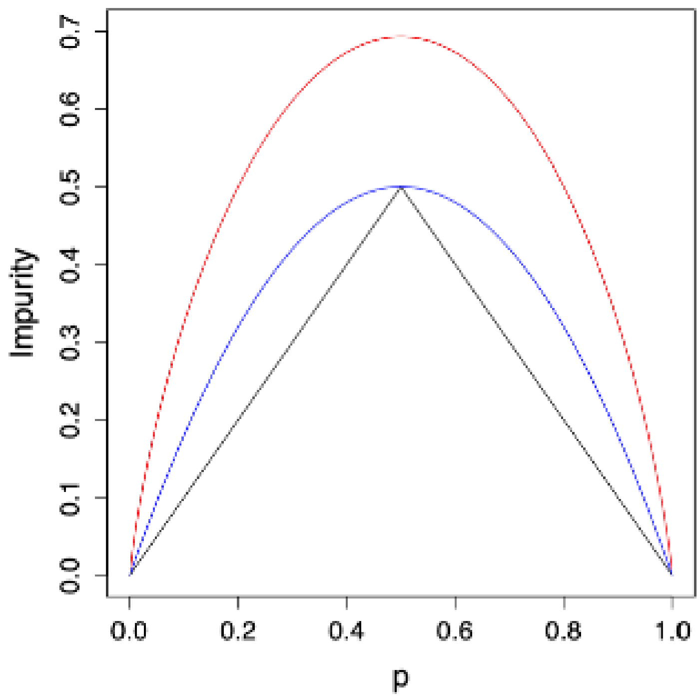


Figure 54

- As we see above in Figure 54, both the Gini and cross-entropy curves decrease smoothly toward zero as the node becomes purer.
- The cross-entropy (red) penalises uncertainty more strongly around $p = 0.5$, offering steeper gradients that make it more sensitive than the Gini index (blue).
- The misclassification error (black), by contrast, changes abruptly and lacks the smoothness required for effective optimisation.

13.3 Example: heart disease

The **heart** data set from the *ISLR* package records whether or not a patient has a heart disease (AHD), i.e. $G \in \{\text{Yes}, \text{No}\}$, for 303 patients who presented with chest pain. There are 13 predictors (e.g. *sex*, *age*, *chol*, ...). A subset of the data is shown below:

AHD	Age	Sex	ChestPain	RestBP	Chol	Fbs	RestECG	MaxHR
No	63	1	typical	145	233	1	2	150
Yes	67	1	asymptomatic	160	286	0	2	108
Yes	67	1	asymptomatic	120	229	0	2	129

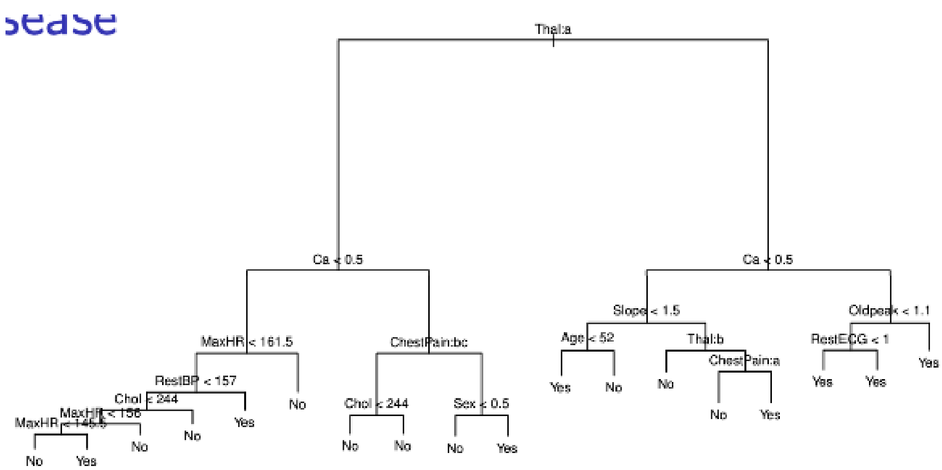


Figure 55: Unpruned tree

A **classification tree** is fitted to predict the presence of heart disease (Yes/No). The tree uses variables such as *Thal*, *Ca*, *ChestPain*, and *MaxHR* to partition patients into subgroups with different risk levels.

In Figure 55 we can see the **unpruned tree**, which is relatively complex and may overfit the training data.

In Figure 56, the **bottom-left panel** shows the training, cross-validation, and test errors for trees of increasing size. The **bottom-right panel** shows the **pruned tree** corresponding to the optimal level of complexity, selected by cross-validation. Pruning simplifies the model, improving interpretability while maintaining nearly the same predictive accuracy.

In this tree, several predictors are qualitative, for example, *Thal* and *ChestPain*. The algorithm evaluates all possible binary partitions of their categories. For instance, the split *ChestPain:bc* divides the data into two groups, one with "typical" or "asymptomatic" pain and another with "nonanginal" or "nontypical" patterns.

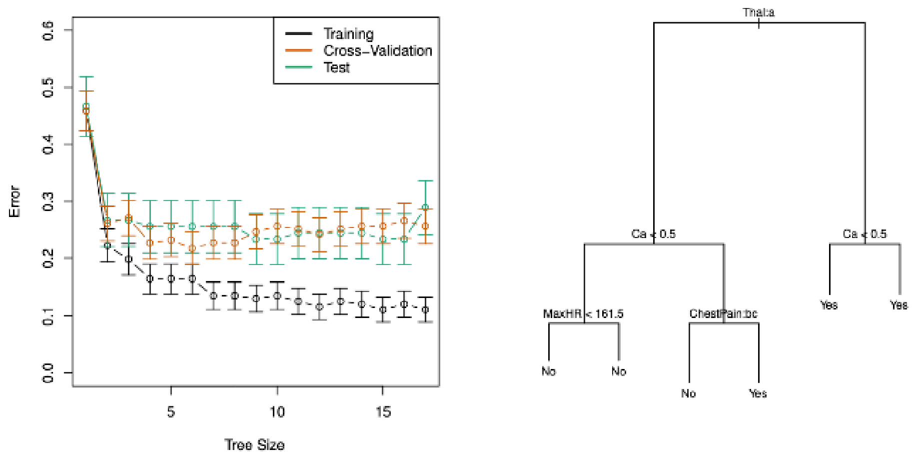


Figure 56: Training, CV and Test errors & Pruned tree

Some splits lead to terminal nodes with the same predicted value, increasing **node purity**, i.e. greater certainty in classification. Decision trees can also handle missing predictor values directly, maintaining robustness during prediction.

13.4 usTree growing in Python

The following tree, shown in Figure 57, was generated using the *Heart* dataset in Python with `DecisionTreeClassifier` from *scikit-learn*. It predicts whether a patient has heart disease (`Yes`/`No`) based on three main predictors: **Thal**, **Ca**, and **Age**.

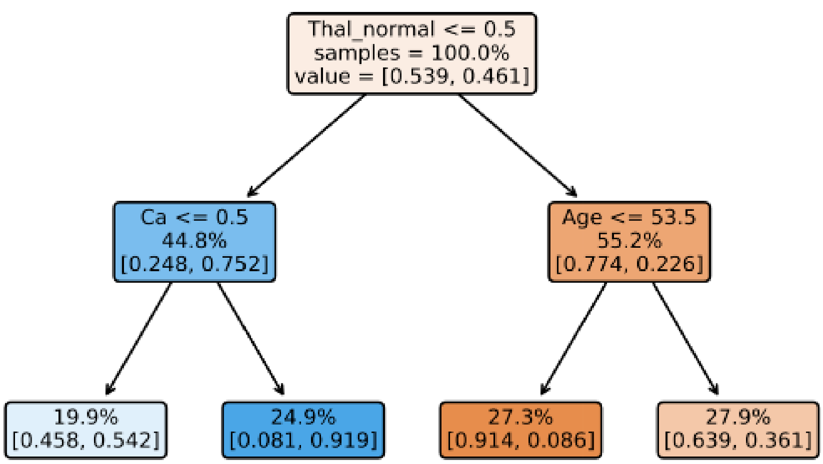


Figure 57: Decision tree for heart disease prediction

The algorithm splits the data recursively to minimise node impurity, measured by the *Gini index*. Each node in the tree contains the following information:

- **Splitting rule** (e.g. `Thal_normal <= 0.5`): the condition dividing the observations into left and right child nodes.
- **Samples**: percentage of training instances reaching that node.
- **Value**: class proportions for `No` (first element) and `Yes` (second element).
- **Colour intensity**: indicates the dominant class, with blue for `No disease` and orange for `Heart disease`.
- The **root node** splits first on `Thal_normal <= 0.5`, distinguishing patients based on the *Thal* test results, i.e., the most informative feature.
- For patients with `Thal_normal > 0.5`, the next split occurs on `Age <= 53.5`, separating younger and older individuals.
- For those with `Thal_normal <= 0.5`, the split is on `Ca <= 0.5`, reflecting calcium levels in major vessels as a key determinant.

The **leaf nodes** display the predicted class probabilities. For example:

- The leftmost leaf shows patients with *abnormal Thal* and *low calcium* having a **54% chance of not having heart disease**.
- The rightmost leaf shows older patients with *normal Thal* having a **64% chance of heart disease**.

This visualisation illustrates how a decision tree partitions the predictor space into regions of high class homogeneity. In other words, it forms a simple yet interpretable classification model.

13.5 Cost-complexity pruning in Python

The **cross-validation (CV) error plot**, shown in Figure 58, illustrates how the model's performance changes as the pruning parameter α increases.

As α grows, the penalty for tree complexity becomes stronger, leading to **simpler trees** with fewer splits. Initially, this reduces **overfitting**, and the CV error decreases. However, beyond an optimal point, excessive pruning causes **underfitting**, and the CV error rises again.

The **minimum of the CV error curve** corresponds to the value of α that achieves the best trade-off between bias and variance, in other words, the subtree with the lowest expected prediction error on unseen data.

13.6 Trees versus linear classifiers

The plots in Figure 59 compare **decision trees** and **linear classifiers** in two-dimensional feature spaces.

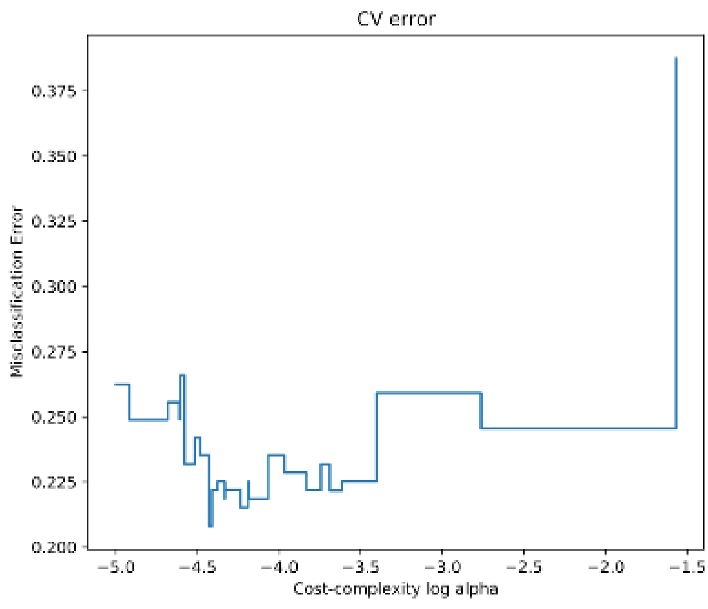


Figure 58: Cross-validation error plot

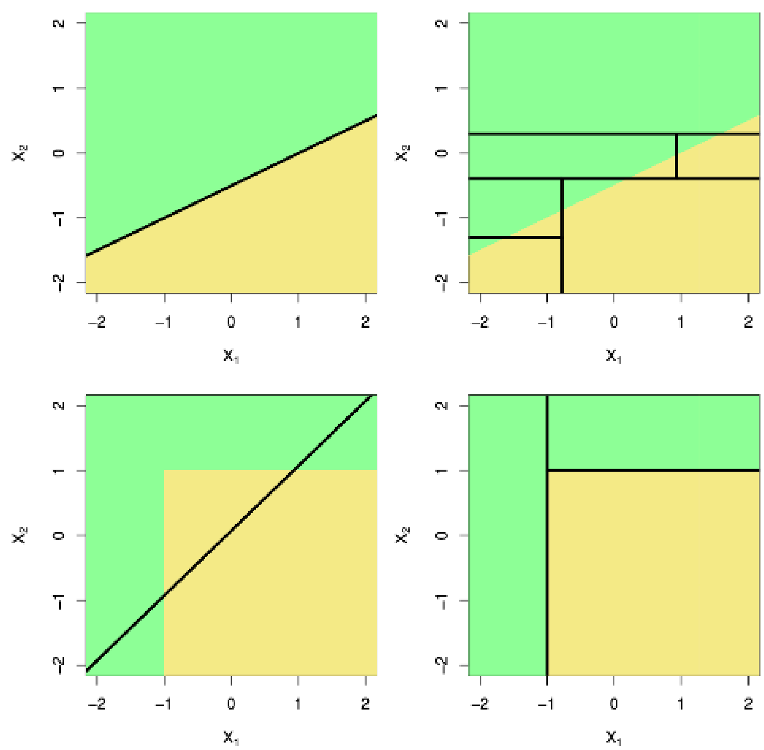


Figure 59: Comparison of decision trees and linear classifiers in two-dimensional feature spaces

- The **top-left** panel shows a linear classifier that separates the feature space using a single straight decision boundary.
- The **top-right** panel illustrates how a decision tree can approximate that same boundary by performing multiple **axis-aligned splits**, resulting in a stepwise partition of the space.
- The **bottom-left** panel shows a situation where the relationship between predictors is non-linear; here, trees can flexibly adapt their partitions to capture local structure that linear models would miss.
- The **bottom-right** panel demonstrates how trees create **rectangular regions** that align with the axes, providing a highly interpretable model at the cost of some boundary smoothness.

Overall, decision trees can approximate complex decision boundaries, while linear models are constrained to global linear relationships.

13.7 Pros and cons of trees

Pros

- Trees are very **easy to explain** and interpret, even more so than linear regression.
- They **mirror human decision-making**, following a sequential and intuitive structure.
- Results can be visualised **graphically**, making them accessible to non-experts.
- Can naturally handle **qualitative predictors** without requiring dummy variables.

Cons

- Single trees generally have **poor predictive performance** compared to ensemble methods.
- They are **highly sensitive** to small changes in the data, which can lead to very different tree structures.

Although individual trees are often unstable and limited in accuracy, **combining many trees** (through methods such as **bagging**, **random forests**, or **boosting**) leads to more powerful and reliable predictive models.

14 Bagging

15